Complete Unified Theory

— Structural Unification of Gravity, Electromagnetism, and Quantum Mechanics via the Curvature Density ρ —

A Variational Framework Based on U(1) Connections on $\mathbb{CP}^1$ and Characterization of Boundary Equations

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Abstract

This paper serves as the integrating paper of a five-part collection. Using the curvature density ρ of U(1) connections on the null Kerr screen $S^2 \simeq \mathbb{CP}^1$ as a common language, we demonstrate the structural unification of the three domains of gravity, electromagnetism, and quantum mechanics.

The argument of this paper is organized in the following three stages:

1. **Foundation building** (Part I–III): Establishing the common language, proposing the unified variational principle, and verifying GR consistency
2. **Identification of challenges**: Explicitly stating what must be demonstrated for the structural unification to be complete
3. **Resolution of challenges** (Part IV–V): Reinterpretation of the Chern number wall, presentation of observational predictions, inclusion of electromagnetic and spinor fields and the weak force, and proposal of a 4D extension

Each Part is a self-contained paper that can be read independently, and this paper integrates their results to present the overall picture of the unification argument.

Declaration on the scope of claims. The "unification" claimed by this collection of papers is not the unification of interactions in the sense of a Grand Unified Theory (GUT), but rather a **structural unification** of the connection structures appearing in boundary data. We do not derive the Einstein equations; rather, we show that the boundary equations on NEH/IH are consistently **characterized** within our framework.

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§0 Claim/Status Overview

We list all claims of this paper collection and their status.

#	Claim	Status	Part
C1	The U(1) connections in the three domains share a common curvature density representation $F = \varrho\, \omega_{FS}$	Established	I
C2	$c_1[\Omega^{(\mathrm{rot})}] = 0$ for general NEH	**Proved**	I

#	Claim	Status	Part
C3	Universality of local connection forms (isomorphism of Berry, gravitational, and electromagnetic connections)	**Proved**	I
C4	HVP reproduction of the Berry and electromagnetic sectors ($\varrho = 1$, quantization condition, $W_{\mathrm{EM}}(n{=}1) = W_{\mathrm{QM}}$)	Re-derivation (sanity check)	I
C5	Formulation of the Holonomy Variational Principle (HVP)	Axiomatic proposal (postulate)	II
C6	Physical origin of HVP (holonomy stability $\rightarrow$ local action)	Motivation	II
C7	Construction of the Ψ_2 boundary theory and characterization of Einstein boundary constraints	Characterization (not derivation)	III
C8	Effective closure A and identification of the closure problem	Effective theory (conditional)	III
C9	Determination of flux coefficients α, β, γ from conservation laws	Conditional determination	III
C10	Reinterpretation of the Chern number wall as a superselection rule	Structural interpretation	IV
C11	Horizon spectrum classification and ϱ-no-hair test	New proposal	IV
C12	Presentation of five theory-specific predictions	Qualitative predictions	IV
C13	Natural inclusion of the electromagnetic field (interpretation of Dirac quantization within $\mathcal{M}_\rho$)	Established	V

#	Claim	Status	Part
C14	Consistency with Dirac structures	Consistency verification	V
C15	Proposal of a 4D constrained unified action	Extended proposal (speculative)	V

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§1 Foundation: What Has Been Established (Results of Part I–III)

1.1 Results of Part I: Establishing the Common Language

Part I constructs the geometric language common to the three domains (gravity, electromagnetism, and quantum mechanics).

Theorem-level results:

- **Unified curvature density representation (C1).** The U(1) connections appearing in all three domains are expressed as $F = \varrho\,\omega_{FS}$ on the null Kerr screen $S^2 \simeq \mathbb{CP}^1$, and the differences reduce to the choice of the scalar density ϱ and the topological sector c_1 alone.
- **Universality of $c_1 = 0$ (C2).** For general NEH cross-sections (without assuming axial symmetry), $c_1[\Omega^{(\mathrm{rot})}] = 0$ is proved from Stokes' theorem.
- **Universality of local connection forms (C3).** It is proved that Berry connections, gravitational rotation connections, and electromagnetic connections share the same gauge transformation law and holonomy formula through the Lie algebra isomorphism $\mathfrak{u}(1) \cong \mathfrak{so}(1,1) \cong \mathbb{R}$.
- **Re-derivation of the Berry sector (C4).** From the stability condition of HVP, $\varrho = 1$ (constant curvature, the canonical connection of the Hopf bundle) is uniquely re-derived as the global minimum. The angular momentum quantization condition $j \in \frac{1}{2}\mathbb{Z}$ is also reproduced as the flux rule $c_1 = 2j \in \mathbb{Z}$. This is not a novel derivation but a verification of the internal consistency of the framework (Hello World).
- **Electromagnetic sector Hello World and three-sector comparison.** Applying the same stability argument to the electromagnetic sector ($c_1 = n$), the potential $W_{\mathrm{EM}}(\rho) = \alpha_E(\rho - n)^2/2$ is constructed. For the minimal Dirac monopole ($n = 1$), $W_{\mathrm{EM}} = W_{\mathrm{QM}}$ holds, reflecting the three-way identity of Hopf bundle = minimal Dirac monopole = Berry connection at the level of the variational principle. From the three-sector comparison of W, it is made explicit that in the quantum and electromagnetic sectors the critical points are constant configurations (W is a quadratic form), whereas in gravity alone a variable configuration (Kerr) serves as the critical point, demonstrating that **the closure problem and the complexity of W share the same root** (originating from the self-coupling structure of spin-2).

What Part I established: The three domains can be described in the same geometric grammar (curvature density and holonomy of U(1) connections on S^2), and the differences are concentrated in ϱ and c_1. In the quantum and electromagnetic sectors, the same stability argument applies, and for $n = 1$ the potentials of both sectors coincide.

1.2 Results of Part II: Proposal of the Unified Variational Principle

Part II formulates the unified variational principle on top of the common language of Part I.

- **Formulation of HVP (C5).** The Holonomy Variational Principle is formulated as an axiomatic proposal (postulate). The physics of the three domains is selected by the stationarity conditions of the holonomy functional (topological condition, stability condition, and flux rule). HVP is a postulate and is not derived from other principles.
- **Physical origin of HVP (C6).** Through a three-step logical chain—"holonomy stability" $\rightarrow$ "lifting to a surface via Stokes" $\rightarrow$ "minimization of a quadratic form of the curvature" $\rightarrow$ Maxwell/Yang–Mills type action—the connection between HVP and the action principle of local field theory is demonstrated. This does not claim HVP as a fundamental principle, but rather models local dynamics as the lowest-order gauge-invariant functional induced from small-loop holonomies.
- **Three-tier declaration.** The claim level of HVP in each sector is made explicit: (i) HVP itself is a variational ansatz, (ii) the GR side is a characterization, (iii) the quantum side is a sanity check.
- **Splitting of the ϱ symbol.** To resolve the ambiguity of the curvature density, three symbols are introduced: ϱ (abstract variable), ρ_{rot} (real rotational curvature density), and $\rho_{\mathbb{C}}$ (complex curvature density).

What Part II established: A variational principle unifying the physics of the three domains has been formulated under explicit claim levels. The common stationarity condition $\Delta_{S^2}\varrho = \partial V/\partial\varrho$ is shared across the three domains, and the differences reduce to the choice of the potential V and c_1 alone.

1.3 Results of Part III: Verification of GR Consistency

Part III verifies that the gravity sector of HVP is consistent with the Einstein boundary constraints.

- **Construction of the Ψ_2 boundary theory (C7).** A closed boundary theory with Weyl scalar Ψ_2 as the fundamental variable is constructed, and the NEH/IH Einstein boundary constraints (cross-focusing equation, structural equation of the rotation 1-form) are consistently characterized as stationarity conditions of HVP. This is a characterization, not a derivation; since the cross-focusing equation is also used as input, the argument contains a circular element.
- **EFT-style resolution of the effective closure (C8).** In the effective evolution equations from the Bianchi identities, the effective closure A ($-\eth\Psi_3 \approx c\,\Delta_{S^2}\rho_{\mathbb{C}}$) is employed. This closure is not an arbitrary assumption but is naturally derived as the minimal local effective term (leading term of the screen EFT) compatible with the screen S^2 geometry, spin-weight structure, and rotational symmetry. Consistency with the spherical harmonic expansion is also confirmed, matching the mode tower ($\ell = 0$: mass, $\ell = 1$: rotation, $\ell \geq 2$: radiation). The closure coefficient c is quantified as $c(\ell = 2) \approx 0.06M$ through the QNM consistency condition (Eq.,(6.10)), and the ℓ-dependence $c(\ell) \propto 1/\ell(\ell+1)$ originates from light-ring-dominated wave scattering. A telegrapher-type local PDE extension yields a negative result (Eq.,(6.11)), and as an alternative, non-local memory kernels (Eqs.,(6.12)–(6.13)) are constructed. The EFT second-order correction c_2 is identified as the horizon projection of the second-order Teukolsky source, with the procedure specified.

- **Conditional determination of flux coefficients (C9).** From three physical conservation laws (Bondi mass loss, area increase, angular momentum flux), the flux source term coefficients α, β, γ are conditionally determined. $\beta > 0$ is proved from the area increase theorem.
- **Minimal GR compatibility set.** The four conditions necessary and sufficient for GR consistency of this framework are identified.
- **Principled justification for all four requirements of W_{GR}.** Of the four requirements of W_{GR}, three (Kerr critical point, CR consistency, Schwarzschild stability) have principled justification from the Dirichlet energy principle. For the remaining cross-focusing consistency, it is derived as the $\delta\lambda$ Euler–Lagrange equation of the extended boundary action S_{tot} (Eq.,(7.13)), and conditions (A) (incorporation of ρ_R into the variational system), (B) (presymplectic consistency), and (C) (gauge invariance) are verified **[Closure A]**.

What Part III established: The gravity sector of HVP is consistent with the Einstein boundary constraints as an effective theory in the near-Kerr regime. It is shown that all four requirements of W_{GR} have principled justification (Requirements 1–3: Dirichlet principle; Requirement 4: extended boundary action). The closure problem is systematically described within the framework of nonlocal memory kernels and higher-order EFT terms.

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§2 Identification of Challenges: What Is Needed for the Completion of Structural Unification

Through Part I–III, the foundations of the common language, variational principle, and GR consistency have been established. However, for **the argument of structural unification to be complete**, the following challenges must be resolved.

Challenge A: Interpretation of the Chern Number Wall

The discrepancy between $c_1 = 0$ (gravity) and $c_1 = 1$ (Berry), proved in Part I (§3), places a topological barrier between the curvature density spaces $\mathcal{M}_\rho$ of the three domains. It must be discussed whether this signifies a failure of unification or constitutes a physically meaningful structure.

Challenge B: Independent Predictions from the Framework

The dictionary of Part I (C1–C3) and the HVP of Part II (C5) remain at the level of re-description and re-derivation (C4) of known structures. To demonstrate that the structural unification is not a mere formal analogy, **independent diagnostic tools and predictions** arising from the framework are needed.

Challenge C: Inclusion of Electromagnetic and Spinor Fields

The common language was constructed as a dictionary connecting gravity and quantum mechanics (Berry). Unless it is shown that electromagnetic fields and Dirac fields are **naturally** included in this framework, the unification of the three domains remains incomplete.

Challenge D: Extension to the Bulk

Part I–III are restricted to a boundary theory on the null Kerr screen S^2. For the structural unification to connect with the dynamics of 4-dimensional spacetime, an extension from the boundary

to the bulk must at least be proposed.

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§3 Resolution of Challenges (Results of Part IV–V)

3.1 Results of Part IV: Resolution of Challenges A and B

Part IV resolves Challenges A and B through the interpretation of the Chern number wall and the presentation of independent predictions.

Resolution of Challenge A: Chern number wall = superselection rule (C10).

The Chern number wall is reinterpreted not as a failure of unification but as a **superselection rule** within the same framework.

- Mathematical reason: The discreteness of $c_1 \in \mathbb{Z}$ makes it invariant under continuous deformation (topological protection)
- Physical interpretation: c_1 functions as a superselection quantum number, prohibiting continuous transitions between different sectors
- Meaning for unification: "Unification" does not mean forcing everything into the same sector, but rather **describing different topological sectors in a common language**

In exactly the same way that the different plateaux of the quantum Hall effect ($c_1 = 0, 1, 2, ...$) are different sectors of the same theory, gravity ($c_1 = 0$) and quantum mechanics ($c_1 = 1$) reside in different layers of $\mathcal{M}_\rho$ while sharing common local dynamical laws.

Resolution of Challenge B: Horizon spectrum classification and independent predictions (C11, C12).

From the ϱ language, a new diagnostic framework complementing existing BH spectroscopy is obtained.

- **Horizon spectrum classification.** NEH are classified by the support of the spherical harmonic spectrum $\{\rho_{\ell m}\}$ into Kerr type / perturbed Kerr type / non-Kerr type. The curvature power spectrum C_ℓ functions as a rotationally invariant summary statistic.
- **ϱ-no-hair test.** Statistical detection of even-ℓ ($\ell \geq 2$) components $\rho_{\ell m} \neq 0$ constitutes direct evidence of deviation from the Kerr hypothesis.
- **Five theory-specific predictions.** The universality of $c_1 = 0$, the Kerr odd-parity selection rule, the dephasing Bell threshold, area quantization, and the ϱ-no-hair test are presented as qualitative predictions.

These are not mere rewriting of the Part I dictionary but rather **payoffs of structural unification** made possible for the first time by the ϱ language.

3.2 Results of Part V: Resolution of Challenges C and D

Part V resolves Challenges C and D through the inclusion of electromagnetic and spinor fields and the proposal of a 4D extension.

Resolution of Challenge C: Inclusion of the electromagnetic field and Dirac structures (C13, C14).

- **Natural inclusion of the electromagnetic field (C13).** It is shown that the Dirac quantization condition $eg = n/2$ is equivalent to the stratification $c_1 = n$ of $\mathcal{M}_\rho$, and the Hopf bundle, the minimal Dirac monopole, and the spin-1/2 Berry connection are identified as the same geometric entity within $\mathcal{M}_\rho^{(1)}$. The nonlinear coupling of Kerr–Newman is consistent with the affine structure of $\mathcal{M}_\rho^{(0)}$.
- **Consistency with Dirac structures (C14).** It is confirmed that the null Kerr screen geometry of this framework naturally accommodates vierbein, spin connection, and the Dirac operator. This does not mean "the Dirac equation has been derived from this theory" but rather confirms compatibility with spin structures.
- **Group-theoretic analogy with the weak interaction.** From $\mathbb{CP}^1 \simeq \mathrm{SU}(2)/\mathrm{U}(1)$, a structural analogy with electroweak symmetry is pointed out, but this remains at the level of group-theoretic motivation and is not a derivation of the Standard Model.

Resolution of Challenge D: Proposal of a 4D extension (C15).

- **Constrained unified action.** A 4D unified action is proposed in which ϱ on the null Kerr screen is positioned not as an independent field but as a dependent quantity determined from the curvature of the connection, with the geometric constraint $F|_{\text{screen}} = \varrho\,\omega_{\text{screen}}$ imposed via a Lagrange multiplier.
- **Positioning.** This 4D action is not a necessary consequence of the boundary structural unification of Part I–III, and is positioned as a **proposed extension**. It is made explicit that the claim level differs from the main arena of the boundary theory (Part I–III).

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§4 The Unification Argument: Assessing Completion

4.1 Criteria for Structural Unification

For structural unification to hold, all four of the following conditions must be satisfied:

Condition	Content	Fulfillment Status
U1	A common geometric language exists for the three domains	**Fulfilled** (C1–C3, Part I)
U2	A common variational principle is formulated and selects the physics of each domain	**Fulfilled** (C5, Part II)
U3	The variational principle is consistent with known physical laws	**Conditionally fulfilled** (C4, C7–C9, Part I and III)
U4	The differences are structurally understood within the unified framework	**Fulfilled** (C10, Part IV)

4.2 Assessment

From the fulfillment status of conditions U1–U4, we make the following assessment.

Assessment: Structural unification has been achieved within the scope of effective theory.

Specifically:

1. **Common language** (U1) is established at the theorem level. The unified description $F = \varrho\,\omega_{FS}$, the universality proof of $c_1 = 0$, and the universality theorem for local connection forms are all mathematically rigorous.
2. **Variational principle** (U2) is formulated as an axiomatic proposal (postulate). HVP is not derived from other principles but is a variational ansatz based on physical motivation.
3. **Consistency with known laws** (U3) is conditionally fulfilled. On the quantum side (Berry sector), the result is completely re-derived (C4), and on the electromagnetic side, the same stability argument reproduces the result ($W_{\mathrm{EM}}(n = 1) = W_{\mathrm{QM}}$). On the gravity side, characterization as an effective theory in the near-Kerr regime is achieved (C7), and a quantitative benchmark is established through the QNM consistency condition for the closure coefficient c (Eq.,(6.10)). It is shown that all four requirements of W_{GR} have principled justification (Requirements 1–3: Dirichlet principle; Requirement 4: the $\delta\lambda$ Euler–Lagrange equation of the extended boundary action S_{tot} (Eq.,(7.13))). The closure problem is systematically described within the framework of non-local memory kernels (Eqs.,(6.12)–(6.13)) and higher-order EFT terms.
4. **Structural understanding of differences** (U4) is achieved. The Chern number wall is reinterpreted as a superselection rule (C10) and is positioned not as a failure of unification but as a structural consequence of the unified framework.

4.3 Self-Assessment of Maturity

Level	Definition	Attainment
0	Discovery of formal analogies	**Complete**
1	Establishment of a common language (dictionary)	**Complete** (Part I)
2	Formulation of a variational principle	**Complete** (Part II)
3	Characterization and re-derivation of known theories	**Complete** (Quantum and EM: complete re-derivation; Gravity: principled justification of all four requirements established as effective theory)
4	Quantitative predictions	**Partially achieved** (Quantitative indicators such as $\tau_2 \sim 11M$, $c(\ell = 2) \approx 0.06M$ are established. The ϱ-no-hair test is at the proposal stage)

Level	Definition	Attainment
5	Complete dynamical unification	**2/3 achieved** (W_{GR} derivation: complete; inter-sector unification via c_1 superselection rule: complete; bulk dynamics: pathway from 4D action $\to$ NP $\to$ screen projection confirmed (Part V §7, Appendix C.3); Closure A justified as the leading term of the screen EFT)

Current level of attainment: Level 3–4 (Characterization as effective theory complete + justification of Closure A via screen projection + partial quantification + structural unification via superselection rule)

4.4 Honest Statement of Limitations

This paper collection does **not** claim:

- That the Einstein equations have been derived from HVP
- That unification of interactions (in the GUT / TOE sense) has been achieved
- That ρ is a new fundamental physical quantity (ρ is a re-description of existing structures)
- Rigorous derivation of bulk dynamics (Part V §7 is at the stage of a structural proposal)

What this paper collection claims:

- That the connection structures of the three domains can be described in a common geometric language (**proved**)
- That a variational principle can be formulated on that language (**proposed**)
- That the variational principle is consistent with known physics (**verified within the scope of effective theory**)
- That the differences are structurally understood as topological superselection rules, and inter-sector unification via c_1 is achieved (**established**)
- That the closure problem is resolved as an EFT feature indicating the scope of applicability of the effective theory and the need for higher-order corrections, rather than a fundamental inconsistency (**established**: Eqs.,(6.8)–(6.10), memory kernels (6.12)–(6.13))
- That independent diagnostic tools are obtained from the framework (**proposed**)

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§5 Progress and Outlook

5.1 Resolved Issues

The following issues have been resolved through the analysis of Part III.

1. **Principled justification for all four requirements of W_{GR}.** Requirements 1–3 follow from the Dirichlet energy principle, and Requirement 4 (cross-focusing) follows from the $\delta\lambda$ Euler–Lagrange equation of the extended boundary action S_{tot} (Eq.,(7.13)). Conditions (A)

(incorporation of ρ_R into the variational system), (B) (presymplectic consistency), and (C) (gauge invariance) have been verified (Part III §7.8).

2. **Scaling factor for flux coefficients.** $C_\alpha \approx 0.045\,\alpha/M$ ($\beta_N = 1$ convention, Eq.,(7.6a)). Only the ratios $C_\alpha : C_\beta : C_\gamma$ are physical (Eq.,(7.6b)); once the normalization of $\rho_{\mathbb{C}}$ is fixed, the remaining coefficients follow from flux-balance matching.
3. **Systematic description of the closure problem.** The effective closure A is formulated in EFT terms (Eq.,(6.8)) and quantified through the QNM consistency condition (Eq.,(6.10)). The physical origin of the discrepancy between D1 gradient flow and QNMs lies in light-ring-dominated wave scattering, and a telegrapher-type local PDE yields a negative result (Eq.,(6.11)). As an alternative, non-local memory kernels (Eqs.,(6.12)–(6.13)) are constructed, and the higher-order EFT term c_2 is identified as the horizon projection of the second-order Teukolsky source, with the procedure specified. The EFT coefficient series is positioned as a local derivative expansion (low-energy approximation) of the memory kernel.

5.2 Remaining Tasks (Numerical Implementation)

The following numerical computations remain to advance the structural unification to Level 5.

1. **Interpolation parameters for the memory kernel.** The parameters a_1, a_2 in Eq.,(6.13) are to be fitted and fixed against QNM data for $\ell = 2, 3, 4$.
2. **Explicit projection calculation for c_2.** The projection of the second-order Teukolsky source onto the horizon EFT is to be carried out, and the cubic self-interaction coefficient is to be determined numerically.
3. **Teukolsky calibration of $\beta_N^{(\ell)}(\omega)$.** The scaling factor ratios $C_\beta/C_\alpha, C_\gamma/C_\alpha$ are to be numerically determined through explicit computation of the transfer operator $\mathcal{T}_\ell$.

5.3 Mid-Term Tasks

4. **Rigorization of bulk dynamics.** The bulk dynamics from the constrained unified action proposed in Part V §7 is at the stage of a structural proposal. Rigorously deriving the spacetime evolution equation for ϱ as an Euler–Lagrange equation and demonstrating consistency with the Einstein equations is the main task for completing Level 5.
5. **Establishment of quantitative predictions.** Physical time-scale calibration of multipole relaxation ($c(\ell = 2) \approx 0.06M$ gives the estimate $\ell = 2$ decay time $\tau_2 \sim 1/\omega_I \approx 11M$, partial progress); construction of sensitivity curves for the ϱ-no-hair test.
6. **Derivation of tensor product structure.** Description of non-locality (entanglement) within the framework.
7. **Geometric origin of SU(3).** Inclusion of the strong interaction via extension from $\mathbb{CP}^1$ to $\mathbb{CP}^2$.

5.4 Long-Term Outlook

8. **Verification of the complete 4D unified action.** Self-consistency and novel predictions of the constrained unified action proposed in Part V.
9. **Connection with quantum gravity.** The relationship between area quantization and the discrete spectrum of ϱ.

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§6 Structure of the Paper Collection

Part	Title	Level	Main Results	Contribution to Challenges
I	Common Language	0–1	C1–C4, EM Hello World, three-sector W comparison	Foundation building
II	Unified Principle (HVP)	2	C5–C6	Foundation building
III	GR Consistency Verification	3	C7–C9	Foundation building
IV	Predictions and Diagnostics	4	C10–C12	Resolution of Challenges A and B
V	Extended Proposals	3–4 (bulk dynamics at structural proposal stage)	C13–C15	Resolution of Challenges C and D

Recommended reading order: - Theoretical physicists: Main → Part I → Part II → Part III → Part IV - Observational astronomers: Main → Part IV (§9 predictions) → Part I (§3 $c_1 = 0$) - Mathematical physicists: Part I → Part III → Part IV (§8 Chern number wall) - Overview only: This paper (Main Paper)

References

See the references in each Part.

Structural Unification of Gravity, Electromagnetism, and Quantum Mechanics via the Curvature Density ρ

— Variational Framework Based on U(1) Connections on $\mathbb{CP}^1$ and Characterization of Boundary Equations —

Part I: Common Language — Level 0–1

This paper series consists of five parts:

Part	Title	Level	Content
I	**Common Language**	**0–1**	**Dictionary, curvature density space, topological theorems, geometric quantization**
II	Unified Principle (HVP)	1–2	Formulation of the Holonomy Variational Principle and stationarity conditions
III	GR Consistency Verification	2–3	Characterization of Einstein boundary constraints, Ψ_2 boundary theory
IV	Observational Predictions and Open Problems	3	Horizon spectrum classification, Chern number wall, current achievements
V	Extensions	3–4	Electromagnetic field, Dirac structure, weak force, 4D unified action

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§1 Introduction: From a Common Language to Structural Unification

1.1 Results of the Dictionary

In the companion paper, it was established that both the Berry curvature on the quantum state space ($\mathbb{CP}^1$) and the rotational curvature on cross-sections of a Kerr black hole horizon (NEH) can be expressed as U(1) curvatures of the same type:

$$F = \rho(u)\,\omega_{FS} \tag{1.1}$$

Here ω_{FS} denotes the Fubini–Study 2-form (normalized as $\int_{S^2}\omega_{FS} = 2\pi$), and the scalar function ρ encodes the differences between the theories. The distinction between the two reduces solely to

the choice of curvature density ρ: in the quantum case $\rho = 1$ (constant curvature), while in the gravitational case $\rho = \kappa(\zeta)$ (variable density, $\zeta = \cos\theta$).

This "dictionary" rendered the structures "connection → curvature → holonomy" appearing in gravity (null Kerr projection), electromagnetism (U(1) gauge), and quantum mechanics (Berry) mutually comparable within a single grammar. Concretely, all three are organized as U(1) principal bundle connections on $\mathbb{CP}^1 \simeq S^2$, the curvature 2-form is expressed as a scalar density ρ with respect to the basis ω_{FS}, and the holonomy $\mathrm{Hol}(C) = \exp(i \oint_C A)$ serves as the unit of comparison.

Furthermore, within this framework, the agreement of local normal forms and the differences in global topology (Chern number c_1) were cleanly separated. The Berry curvature has $c_1 = 1$ (Hopf bundle), whereas the rotational curvature of the Kerr NEH has $c_1 = 0$ (trivial bundle); ordinary electromagnetic fields (without magnetic monopoles) likewise have $c_1 = 0$. The spherical harmonic expansion of the curvature density ρ quantifies the differences as "which multipoles are excited": Berry is a pure monopole ($\ell = 0$ only), while the Kerr solution (by axial symmetry and equatorial reflection symmetry) excites only odd multipoles ($\ell = 1, 3, 5, ...$). This odd-parity selection rule is a characteristic property of Kerr; for general NEH, only $c_1 = 0$ (§3) holds universally. For perturbed Kerr, the appearance of even-ℓ components ($\ell \geq 2$) is permitted.

1.1a Core Claims of the Present Theory

A. The U(1) connections appearing in all three domains—gravity, electromagnetism, and quantum mechanics—share the same curvature density representation $F = \rho\,\omega_{FS}$ on $\mathbb{CP}^1 \simeq S^2$, and their differences reduce solely to the choice of scalar density ρ and topological sector c_1. The core claim is not that the U(1) groups themselves are identical, but that all three can be described in the common language of curvature density.

B. For general NEH cross-sections, $c_1[\Omega^{(\mathrm{rot})}] = 0$ holds (§3; proved using Stokes' theorem alone).

C. On the horizon, ρ is linked to the Einstein equations through the physical identification $\rho = -2\,\Psi_2\big|_{\mathcal{H}}$ with the Weyl scalar, and this is numerically confirmed via Schwarzschild normalization (Part III §4).

Claim/Status List.

Claim	Status	Section
Common curvature density dictionary $F = \rho\,\omega_{FS}$	Established within the present framework	Part I §2–§4
$c_1 = 0$ for general NEH	**Proved**	Part I §3
Universality of local connection forms (isomorphism of Berry, gravitational, and electromagnetic connections)	**Proved**	Part I §2.6
Reproduction of Berry sector from HVP	Re-derivation / sanity check	Part I §4.6
Characterization of Einstein boundary constraints as HVP stationary classes	Characterization (not derivation)	Part III §3.5, §8
4D constrained unified action	Extension proposal	Part V §7

The formulation of the unified principle (HVP) is given in Part II.

1.2a Positioning of the Theory

What this paper presents is not a unification of interactions, but a unified geometric description of connection structures appearing in boundary data (structural unification). On the gravity side, the Einstein equations are not derived; rather, it is shown that the boundary equations on NEH/IH are consistently embedded (characterized) within our framework. This differs essentially from the "interaction unification" pursued by Grand Unified Theories (GUT) or a Theory of Everything (TOE); what the present theory pursues is a unified geometric description of connection structures appearing in boundary data. *The present work should therefore be viewed not as a unification of interactions but as a structure unification: a unified description of connection structures appearing in boundary data. On the gravity side, we do not derive the Einstein equations but show that the NEH/IH boundary equations are consistently characterized within our framework.*

> *"The complex curvature density $\rho = K + iB$ provides a unified geometric description linking quantum polarization space $\mathbb{CP}^1$, horizon geometry S^2, and the Weyl curvature scalar Ψ_2."*

The core of this unified picture lies in the fact that three domains of physics share the same $S^2 \simeq \mathbb{CP}^1$ geometry:

Space	Physical origin	Isomorphism with S^2
Quantum state space	Bloch sphere of spin-1/2	$\mathbb{CP}^1 \simeq S^2$
Electromagnetic polarization space	Poincaré sphere of polarization	S^2
Horizon cross-section	2-dimensional cross-section of the BH horizon	$S \simeq S^2$

All three share the U(1) grammar "connection $\rightarrow$ curvature $\rightarrow$ holonomy," and the differences reduce to only two parameters: the Chern number c_1 (topological invariant) and the multipole structure (dynamical content). The present theory is positioned as an effective boundary theory on the celestial sphere, providing a description obtained by projecting the bulk dynamics of the Einstein equations onto S^2.

The achievements and remaining challenges as a unified theory are discussed in Part IV.

1.2c Why S^2 Appears

The space $S^2 \simeq \mathbb{CP}^1$ playing the central role in this paper is not a mere geometric coincidence. The reasons why S^2 appears in the three physical systems (quantum, electromagnetic, gravitational) that are the objects of unification originate in the following fundamental symmetry structures, and the twistor-theoretic viewpoint provides a fourth, structural rationale:

1. **Quantum systems.** The spin-1/2 state space is isomorphic to the projective Hilbert space $\mathbb{CP}^1$ and is realized as the Bloch sphere. More generally, the orbit of highest-weight coherent states (Perelomov type) of a spin-j system forms $\mathbb{CP}^1$, which is a consequence of the homogeneous space structure $\mathrm{SU}(2)/\mathrm{U}(1) \simeq S^2$.

2. **Electromagnetic polarization.** By the transversality condition of Maxwell's equations, the space of completely polarized states becomes the Poincaré sphere S^2 described by Stokes parameters. This is the geometric manifestation of the fact that the helicity degrees of freedom of the photon are classified by $\mathrm{SO}(2) \simeq \mathrm{U}(1)$; identifying the overall phase of Jones vectors yields $\mathbb{CP}^1 \simeq S^2$.

3. **Black hole horizons.** By the topological censorship theorem (Friedman–Schleich–Witt [1993]) and its consequences (Jacobson–Venkataramani [1995]), the event horizon cross-section of an asymptotically flat stationary black hole is restricted to S^2. In the definition of NEH as well, the topology of the cross-section is required to be S^2 as a topological condition.

4. **Celestial sphere of null directions (twistor-theoretic viewpoint).** In Penrose's twistor correspondence, the set of null directions through a point x in 4-dimensional spacetime forms a celestial sphere $S^2 \simeq \mathbb{CP}^1$, corresponding to a twistor line L_x in twistor space $\mathbb{PT}$ (Penrose [1967], Penrose–Rindler [1986]). The $\mathbb{CP}^1$ of the present framework is positioned as the horizon version (celestial version) of this twistor line (detailed in Part V §6).

Therefore, the S^2 appearing in this work emerges naturally at the intersection of **four independent rationales**: three physical systems—quantum state space, polarization degrees of freedom, and gravitational boundary geometry—and the celestial sphere of null directions (twistor-theoretic viewpoint) providing an independent geometric rationale. In this sense, the unified structure $F = \rho\,\omega_{FS}$ does not depend on a specific model, but is a geometric consequence that necessarily arises from the structure $\mathrm{U}(1)$ principal bundle $\to S^2$ shared by the fundamental symmetries of the three domains; the twistor correspondence provides independent corroboration of this shared structure.

> *The appearance of $\mathbb{CP}^1$ in the present formulation is not coincidental but arises at the intersection of four independent structures: three physical systems — the quantum state space (Bloch sphere), the polarization manifold (Poincaré sphere), and the horizon cross-section (topological censorship) — together with the celestial sphere of null directions (twistor correspondence), which provides independent geometric corroboration. Although the geometric interpretations differ across these domains, all four constructions associate a $\mathrm{U}(1)$ principal bundle over S^2 with physically meaningful holonomies, providing the structural foundation for the unified description $F = \rho\,\omega_{FS}$. The unification itself concerns the first three systems; the twistor viewpoint furnishes an additional structural rationale for the ubiquity of S^2.*

1.3 Organization of Part I

This Part constructs the common language that forms the foundation of the unified description. In §2, the mathematical structure of the curvature density space $\mathcal{M}_\rho$ on the null Kerr screen $S^2 \simeq \mathbb{CP}^1$ is formulated, and a unified positioning of Berry, Kerr, and Dirac monopole configurations is given. In §2.6, the universality of local connection forms for Berry and gravitational rotation connections is proved. In §3, the universality of $c_1 = 0$ for general NEH is demonstrated from Stokes' theorem. In §4, the logical chain from geometric quantization to the uncertainty principle is organized; in §4.6, the re-derivation of Berry curvature and the quantization condition from HVP (Part II) is carried out (Hello World I); and in §4.7, the same argument is applied to the electromagnetic sector and the W potentials of the three sectors are compared.

The formulation and physical origin of the unified principle (HVP) are given in Part II; GR consistency verification in Part III; observational predictions and the Chern number wall in Part IV; and extension proposals for the electromagnetic field, Dirac structure, weak force, etc., in Part V.

Terminological convention (null Kerr screen). Throughout this paper series, the $S^2 \simeq \mathbb{CP}^1$ that appears as the base space of U(1) connections in the unified framework is called the **null Kerr screen**. On the quantum side it is physically realized as the Bloch sphere, on the electromagnetic side as the Poincaré sphere, and on the gravity side as the Kerr horizon cross-section. The name "null Kerr" reflects the fact that this sphere appears as the celestial sphere of null directions and is deformed by the gravitational field to carry the geometry of the Kerr horizon. The usage of null vectors ℓ, n and null hypersurfaces in standard terminology is unchanged.

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§2 Mathematical Structure of the Curvature Density Space $\mathcal{M}_\rho$

Note on notational conflicts. In this paper, ρ denotes the curvature density ($F = \rho\,\omega_{FS}$). The reader should be aware of potential confusion with the following similar symbols: - $\tilde{\rho} = |\psi|^2$: probability density (used in §4.1) - ρ_{NP}: Newman–Penrose spin coefficient (convergence; used in this paper only in the context of the NEH condition $\rho_{\mathrm{NP}} = 0$) - $\widehat{\rho}$: quantum density matrix (Appendix A.5)

These are distinguished by subscripts or superscripts.

2.1 Basic Definitions

The curvature 2-form of a U(1) connection on $S^2 \simeq \mathbb{CP}^1$ can be written in the standard form

$$F = \rho\,\omega_{FS}$$

using the Fubini–Study 2-form $\omega_{FS} = \frac{1}{2}\sin\theta\, d\theta \wedge d\phi$ ($\int_{S^2} \omega_{FS} = 2\pi$). The normalized area measure is defined as $d\mu_{FS} := \frac{1}{2\pi}\omega_{FS}$ ($\int_{S^2} d\mu_{FS} = 1$).

Definition 2.1 (Curvature density space). The curvature density space $\mathcal{M}_\rho$ is defined as

$$\mathcal{M}_\rho := \left\{ \rho \in C^\infty(S^2, \mathbb{R}) \;\middle|\; F_\rho := \rho\,\omega_{FS} \text{ is realizable as the curvature of a U(1) connection on } S^2 \right\}$$

By Chern–Weil theory, the necessary and sufficient condition for $\rho \in \mathcal{M}_\rho$ is

$$\boxed{\langle \rho \rangle := \int_{S^2} \rho\, d\mu_{FS} = c_1 \in \mathbb{Z}} \tag{2.1}$$

where c_1 is the first Chern number.

Remark. This condition imposes no constraints on the range or sign of ρ. In particular, ρ may be locally negative (the Kerr function $\kappa(\zeta)$ does in fact change sign). The only constraint is the integrality of the mean value.

Remark (real vs. complex). In §2–§4 below, ρ is treated as a real-valued function. This suffices for the dictionary-level comparison (Berry's $\rho = 1$, Kerr's $\rho = \kappa(\zeta)$). From Part III §4 onward, for the purpose of GR consistency verification, ρ is extended to a complex curvature density

$\rho = K + iB = -2\Psi_2$. In this case, the real ρ of §2–§4 corresponds to the scale transformation of the rotational curvature density $\mathrm{Im}(\rho^{\mathbb{C}})$: $\rho_{\mathrm{rot}} = 2\,\mathrm{Im}(\rho^{\mathbb{C}}) = 2B = -4\,\mathrm{Im}\,\Psi_2$.

Notational preview (Part II onward). From Part II onward, the following symbols are introduced to resolve the polysemy of ρ: ϱ (abstract curvature density variable of HVP), $\rho_{\mathrm{rot}} = -4\,\mathrm{Im}\,\Psi_2$ (real rotational curvature density), $\rho_{\mathbb{C}} = K + iB = -2\Psi_2$ (complex curvature density). In Part I, ρ is used as a real-valued curvature density to the extent sufficient for dictionary-level comparison.

2.2 Stratification by Chern Number

For each integer $n \in \mathbb{Z}$, the Chern stratum is defined as

$$\mathcal{M}_\rho^{(n)} := \left\{ \rho \in \mathcal{M}_\rho \;\middle|\; \langle \rho \rangle = n \right\}$$

The total space decomposes into a disjoint union

$$\mathcal{M}_\rho = \bigsqcup_{n \in \mathbb{Z}} \mathcal{M}_\rho^{(n)}$$

Each stratum is in one-to-one correspondence with the topological type of the U(1) bundle.

Proposition 2.2 (Affine structure). Each stratum $\mathcal{M}_\rho^{(n)}$ is an affine space whose model vector space is the space of smooth functions with zero mean, $\mathcal{M}_\rho^{(0)}$. In particular, $\rho \mapsto \rho + 1$ gives an isomorphism $\mathcal{M}_\rho^{(n)} \xrightarrow{\sim} \mathcal{M}_\rho^{(n+1)}$.

2.3 Function Space Structure

The tangent space of each stratum ($= \mathcal{M}_\rho^{(0)}$ itself) is equipped with the L^2 inner product

$$(\delta\rho_1, \delta\rho_2)_{L^2} := \int_{S^2} \delta\rho_1\, \delta\rho_2\, d\sigma$$

($d\sigma = \sin\theta\, d\theta\, d\phi$: solid angle element). The spherical harmonics $\{Y_{\ell m}\}$ form a complete orthonormal system of $L^2(S^2, d\sigma)$, and the expansion of ρ

$$\rho(\theta, \phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \rho_{\ell m}\, Y_{\ell m}(\theta, \phi)$$

provides a coordinate system on $\mathcal{M}_\rho$. The Chern number is directly encoded in the $\ell = 0$ component: $\rho_{00} = \sqrt{4\pi}\, c_1$. Within each stratum, ρ_{00} is fixed, and the free coordinates are the countably infinite set $\{\rho_{\ell m}\}_{\ell \geq 1}$.

As an extension incorporating singular curvature distributions, the completion using Sobolev spaces $H^s(S^2)$ $(s \geq 0)$

$$\overline{\mathcal{M}}_\rho^{(n),s} := \left\{ \rho \in H^s(S^2) \;\middle|\; \langle \rho \rangle = n \right\}$$

is also defined.

Kähler potential viewpoint. The Fubini–Study form is derived from the Kähler potential $K_0 = \log(1+|z|^2)$ as

$$\omega_{FS} = i\,\partial\bar{\partial}\log(1+|z|^2)$$

($z = e^{i\phi}\tan(\theta/2)$: stereographic coordinate). By the $\partial\bar{\partial}$-lemma (Hodge theory), any curvature $F = \rho\,\omega_{FS}$ on $\mathbb{CP}^1$ admits a Kähler potential K such that $F = i\,\partial\bar{\partial}K$. The potentials for the three sectors are:

Sector	Kähler potential K	Curvature density ρ
Berry (quantum)	$K = \log(1+\lvert z\rvert^2)$	$\rho = 1$ (constant curvature)
Gravity (Kerr)	$K = K_{\mathrm{GR}}(\lvert z\rvert^2)$	$\rho = \kappa(\zeta)$ (variable density)
Electromagnetic (Dirac monopole)	$K = n\log(1+\lvert z\rvert^2)$	$\rho = n$ (constant curvature)

K_{GR} is determined by the Kerr geometry and is essentially the same object as the height function Ψ (vorticity potential) of Part III §12 ($\Delta\Psi \propto \rho_{\mathrm{rot}}$). The curvature density ρ is recovered as the $\partial\bar{\partial}$-derivative of the potential K, and K encodes ρ "one level down."

Remark (distinction from the spherical harmonic expansion). The spherical harmonic expansion $\rho = \sum \rho_{\ell m} Y_{\ell m}$ in §2.3 above is a computational tool (coordinate system) on $\mathcal{M}_\rho$ and is not the geometric degrees of freedom of the theory themselves. The geometric degrees of freedom of the present theory are structured as the pair

$$\boxed{(\text{topological invariant},\ \text{geometric shape}) = (c_1,\ \rho)}$$

The Chern number c_1 is encoded in the $\ell = 0$ component ($\rho_{00} = \sqrt{4\pi}\,c_1$), and the shape of the curvature density is encoded in the $\ell \geq 1$ components. The spherical harmonics $Y_{\ell m}$ are merely an orthonormal basis for expanding this structure.

2.4 Locations and Distances of Concrete Examples

Theory	ρ	$c_1 = \langle\rho\rangle$	Multipole structure	Stratum
Berry (spin-1/2)	$\rho = 1$ (constant)	1	Pure monopole ($\ell = 0$)	$\mathcal{M}_\rho^{(1)}$
Kerr NEH	$\rho = \kappa(\zeta)$ (odd function)	0	Odd multipoles only ($\ell = 1, 3, 5, \ldots$)	$\mathcal{M}_\rho^{(0)}$
EM (no monopole)	Depends on field configuration	0	General	$\mathcal{M}_\rho^{(0)}$
Dirac monopole ($g = n/2$)	$\rho = n$ (constant)	n	Pure monopole	$\mathcal{M}_\rho^{(n)}$

The Berry curvature $\rho = 1$ is the unique SO(3)-invariant point (maximally symmetric configuration) in $\mathcal{M}_\rho^{(1)}$ and corresponds to the canonical connection of the Hopf bundle. The Kerr curvature $\rho = \kappa(\zeta)$ lies on a 1-parameter curve in $\mathcal{M}_\rho^{(0)}$ parametrized by $a/M \in [0, 1]$. At $a = 0$ (Schwarzschild) it is at the origin $\rho = 0$; as $a \to M$ (extremal Kerr), higher odd multipoles become prominent.

The standard Dirac monopole ($g = 1/2$) has $\rho = 1$ and occupies the same point as the Berry curvature in the curvature density space. This is the $\mathcal{M}_\rho$-representation of the three-way identity: Hopf bundle = minimal Dirac monopole = spin-1/2 Berry connection.

Distances within $\mathcal{M}_\rho^{(0)}$ are measured by the ℓ^2 norm of spherical harmonic coefficients, quantifying the difference in multipole distributions between Kerr and electromagnetic field configurations (e.g., a uniform magnetic field $\rho_{\text{uniform}} = 2B_{\text{mag}}R^2 \cos\theta$).

Topological consequence. Along a continuous path $\rho(t)$, $\langle\rho(t)\rangle$ is continuous and takes only integer values, hence is constant. Therefore c_1 is invariant under continuous deformations, and no continuous path within $\mathcal{M}_\rho$ connects Berry ($c_1 = 1$) and Kerr ($c_1 = 0$). This Chern number wall is a constraint of the dictionary and is discussed in detail in Part IV §2.

2.5 Complex Extension and Topological Sectors (n_R, n_I)

The curvature density ρ is extended to complex values:

$$\rho = p + iq$$

Here p is the real part corresponding to mass density, and q is the imaginary part corresponding to NUT parameter density. This complex extension is naturally motivated by the fact that Ψ_2 is generally complex in the correspondence $\rho = -2\Psi_2$ with the Weyl scalar.

Two-component U(1) structure. The term "complex curvature density" does not refer to a $\mathrm{GL}(1, \mathbb{C})$ connection, but rather to a pair of two independent U(1) connections (A_R, A_I) (a two-component U(1) structure). $A_I = \Omega^{(\text{rot})}$ (rotation 1-form) has been physically identified (Part III §4). As for A_R, the GR consistency verification from Part III §4 onward indirectly specifies it as the connection generating $K = -2\,\mathrm{Re}\,\Psi_2$ as curvature, but the independent geometric construction of A_R itself remains a problem for future work. Each component of (A_R, A_I) gives rise to an independent curvature 2-form:

$$F_R = p\,\omega_{FS}, \qquad F_I = q\,\omega_{FS} \tag{2.2}$$

The Chern–Weil conditions for each curvature are independent, and the topological sector is labeled by the pair of Chern numbers $(n_R, n_I) := (\langle p\rangle, \langle q\rangle) \in \mathbb{Z}^2$. Thus the extended curvature density space is classified as

$$\mathcal{M}_\rho^{\mathbb{C}} = \bigsqcup_{(n_R, n_I) \in \mathbb{Z}^2} \mathcal{M}_\rho^{(n_R, n_I)}$$

Identification of physical sectors.

Sector (n_R, n_I)	Physical content	Remarks
$(n_R, n_I) = (0, 0)$	Kerr sector	Dynamically selected by "roughness decay"
$(n_R, 0)$, $n_R \neq 0$	Mass-type topological sector	Related to magnetic mass
$(0, n_I)$, $n_I \neq 0$	NUT-type topological sector	Related to gravitomagnetic charge
(n_R, n_I), $n_R, n_I \neq 0$	Mixed sector	Dyonic structure

The Kerr sector $(n_R, n_I) = (0, 0)$ is dynamically selected by the exponential decay of $\ell \geq 2$ components ("roughness decay") under the D1 gradient flow (Part III §6.1). That is, starting from any initial condition, after sufficient time the $\ell \geq 2$ components vanish, and only $\ell = 0$ (mass) and $\ell = 1$ (angular momentum) survive. The physical meaning of the $n_I \neq 0$ sector is suggested to be related to magnetic mass or gravitomagnetic charge, but its complete elucidation remains a problem for future work.

2.6 Local Connection-Form Universality

In this section, it is rigorously shown that the Berry connection A_{Berry} and the horizon rotation connection $A \equiv \Omega^{(\text{rot})}$ share the same transformation rules and holonomy definition as principal bundle connections of the same kind. This provides the mathematical foundation for the unified principle (Part II §3).

Theorem 2.6 (Local connection-form universality). The following three connections are all **one-dimensional abelian principal bundle connections** on $S^2 \simeq \mathbb{CP}^1$, and share identical gauge transformation rules and holonomy formulas via the Lie algebra isomorphism $\mathfrak{u}(1) \cong \mathfrak{so}(1,1) \cong \mathbb{R}$.

Connection	Principal bundle $P \to S^2$	Structure group	Gauge transformation	Holonomy
A_{Berry}	Hopf bundle ($c_1 = 1$)	U(1)	$A \to A + d\chi$	$\exp(i \oint_C A)$
$\Omega^{(\text{rot})}$	NEH normal bundle ($c_1 = 0$)	$\mathbb{R}^+$ (boost)	$\Omega \to \Omega + d(\ln f)$	$\exp(\oint_C \Omega) \in \mathbb{R}^+$
A_{EM}	Electromagnetic bundle ($c_1 = n$)	U(1)	$A \to A + d\Lambda$	$\exp(ie \oint_C A)$

> **Convention (Wick-type identification).** The structure group on the gravity side is strictly $\mathbb{R}^+ \cong \mathrm{SO}(1,1)_0$ (boost group), and the holonomy takes real values as $\exp(\oint \Omega) \in \mathbb{R}^+$. In this paper, the $\mathbb{R}^+$ connection is mapped to the U(1) connection form via the correspondence $\omega \leftrightarrow iA$ ($\chi = \ln f$) for comparison. This identification is based on the Lie algebra isomorphism $\mathfrak{so}(1,1) \cong \mathfrak{u}(1) \cong \mathbb{R}$ and is a **convention**; the physical content (the structure of the curvature 2-form $F = \rho\, \omega_{FS}$) is preserved under this identification.

Proof. The isomorphism of the three as one-dimensional abelian principal bundle connections is demonstrated in the following four steps.

(i) Correspondence of structure groups. The structure group on the Berry side is U(1) (phase freedom of the state $|\psi\rangle \sim e^{i\chi}|\psi\rangle$), and on the electromagnetic side it is also U(1) (phase freedom

of the charged particle). On the gravity side, it arises from the boost freedom of the null tetrad $\ell^\mu \to f\ell^\mu$ ($f > 0$), with structure group $\mathbb{R}^+$. The Lie algebras of all three are one-dimensional ($\mathfrak{u}(1) \cong \mathfrak{so}(1,1) \cong \mathbb{R}$), and the local theory of connections (curvature, gauge transformations) takes an identical form through this Lie algebra isomorphism.

(ii) Isomorphism of gauge transformation rules. As connections on the Lie algebra $\mathfrak{u}(1) \cong i\mathbb{R}$, the gauge transformation rules of all three take the form

$$A \longrightarrow A + d\chi, \qquad \chi \in C^\infty(S^2, \mathbb{R}) \tag{2.3}$$

On the Berry side, χ corresponds to the phase choice of the state; on the gravity side, it corresponds to $\chi = \ln f$ (boost parameter) (see Part III §4.2, Eq.,(4.4)). The curvature 2-form $F = dA$ is gauge-invariant in all cases: $F \to F$ ($d(d\chi) = 0$).

(iii) Universal holonomy formula. Under the Wick-type identification (convention above), the holonomy along a closed loop $C \subset S^2$ is unified for all three as

$$\mathrm{Hol}_C(A) = \exp\left(i \oint_C A\right) = \exp\left(i \int_\Sigma \rho\, \omega_{FS}\right) \tag{2.4}$$

(Σ is a region on S^2 with $C = \partial\Sigma$). The second equality follows from Stokes' theorem and $F = \rho\, \omega_{FS}$. Note that the natural holonomy on the gravity side is $\exp(\oint \Omega) \in \mathbb{R}^+$, which is mapped to the above formula via the identification $\omega \leftrightarrow iA$. The holonomy is a gauge invariant, protected under $A \to A + d\chi$ by $\oint_C d\chi = 0$ (since C is a closed loop).

(iv) Determination stage of topological conditions. The fixing of the Chern number c_1 occurs at the following distinct stages:

Stage	Physical origin	Mechanism determining c_1
Berry connection	Structure of Hilbert space (choice of spin-j representation)	$c_1 = 2j$ (Chern class of prequantum bundle, §4.2)
NEH rotation connection	NEH conditions ($\theta_{(k)} = 0$, $\sigma^{(k)}_{ab} = 0$)	$c_1 = 0$ (Stokes' theorem, §3.2)
Electromagnetic connection	Presence or absence of magnetic charge	$c_1 = 2eg$ (Dirac quantization condition, Part V §2.2)

The topological condition is a "global selection" of the bundle and is determined independently of the local variational laws of the connection. In the HVP framework (Part II §3), the topological condition is fixed first as Condition 1, after which Condition 2 (stability) and Condition 3 (flux rule) are applied. □

Corollary 2.6a (Unified holonomy comparison formula). For any closed loop $C \subset S^2$, the difference between the Berry holonomy and the gravitational holonomy is given by

$$\mathrm{Hol}_C(A_{\mathrm{Berry}}) \cdot \mathrm{Hol}_C(\Omega^{(\mathrm{rot})})^{-1} = \exp\left(i \int_\Sigma (1 - \kappa(\zeta))\, \omega_{FS}\right) \tag{2.5}$$

The integrand $1-\kappa(\zeta)$ measures the "curvature deviation from the quantum case." When C encloses the entire S^2 (global holonomy), $\int_{S^2}(1-\kappa)\,\omega_{FS} = 2\pi(1-0) = 2\pi$, and the topological difference $\Delta c_1 = 1$ manifests as a shift of 2π in the Berry phase.

Remark (Lie algebra isomorphism vs. Lie group non-isomorphism). The identification in Theorem 2.6 holds at the level of the **Lie algebra** $\mathfrak{u}(1) \cong \mathfrak{so}(1,1) \cong \mathbb{R}$. On the other hand, as **Lie groups**, $\mathrm{U}(1) \cong S^1$ (compact) and $\mathbb{R}^+$ (non-compact) are not isomorphic. As a physical consequence, the Berry holonomy $e^{i\alpha} \in \mathrm{U}(1)$ is periodic as a phase angle ($\alpha \sim \alpha + 2\pi$), while the gravitational holonomy $e^{\beta} \in \mathbb{R}^+$ is non-periodic as a boost parameter ($\beta \in \mathbb{R}$). The identification $\omega \leftrightarrow iA$ in this paper uses only the Lie algebra isomorphism and does not implicitly ignore this difference in group structure. Since the local theory of connections (curvature $F = dA$, gauge transformations, Chern–Weil theory) is completed at the Lie algebra level, the unified description $F = \rho\,\omega_{FS}$ is mathematically rigorous. The group non-isomorphism manifests as a difference in the range of values of global holonomies and constitutes one aspect of the Chern number wall (Part IV §2).

Remark (limitations of the isomorphism). The three connections are mathematically isomorphic as U(1) principal bundle connections, but the topological types (c_1) of the principal bundles to which they belong generally differ. The local structure of connections (transformation rules, holonomy formulas) is universal, but the global structure (Chern numbers) differs from theory to theory. This "local isomorphism with global difference" is precisely the mathematical content of the Chern number wall discussed in Part IV §2.

Observational meaning of gravitational holonomy. The Wick-type identification ($\omega \leftrightarrow iA$) described above is a computational dictionary, and the physical content resides on the $\mathbb{R}^+$ holonomy side. The natural holonomy $\exp(\oint_C \Omega^{(\mathrm{rot})}) \in \mathbb{R}^+$ on the gravity side is a boost parameter, measuring the cumulative transformation of the null tetrad rescaling $\ell^\mu \to f\ell^\mu$ ($f > 0$) along a closed loop. This $\mathbb{R}^+$ holonomy directly corresponds to the following physical observables:

(a) **Non-uniformity of surface gravity (redshift variation).** On the NEH, the surface gravity κ_s is related to the temporal component of $\Omega^{(\mathrm{rot})}$, and when κ_s is non-uniform over the cross-section ($d\kappa_s \neq 0$), the $\mathbb{R}^+$ holonomy along a closed loop appears as a quantity integrating the spatial variation of κ_s. This is in principle observable as the angular dependence of the redshift on the horizon, and for Kerr black holes it is quantified as the difference in κ_s between the equatorial plane and the poles.

(b) **Gravitational wave memory effect.** The permanent update of $\Omega^{(\mathrm{rot})}$ accompanying the passage of gravitational waves is expressed as a change $\Delta\big(\exp(\oint_C \Omega)\big) \neq 1$ in the $\mathbb{R}^+$ holonomy. This corresponds precisely to the holonomy memory effect of Part IV §5.5 and is a physical quantity detectable by next-generation gravitational wave detectors (LISA, Einstein Telescope). The holonomy update is computed as the time integral $\int \partial_u \rho\, du$ of ρ, and the two components (mass-type and rotation-type) of the memory effect are separated into changes of the real and imaginary parts of ρ.

From the above, the identification of the $\mathbb{R}^+$ holonomy with the U(1) holonomy is a computational convenience, and the physically meaningful quantity is the $\mathbb{R}^+$ holonomy (cumulative boost) itself. Mapping to the U(1) form is merely a convention to enable dictionary-level comparison with Berry and electromagnetic connections.

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§3 Topological Constraint: Universality of $c_1 = 0$

3.1 Formulation of the Question

For the Kerr solution, $c_1 = 0$ follows immediately from the odd-function property of the curvature density $\kappa(\zeta)$. However, this derivation depends on the axial symmetry of Kerr. The question of this section is:

> Does $c_1 = 0$ always hold for general NEH (including non-Kerr, non-axisymmetric cases)?

3.2 Core of the Proof

Under the NEH conditions ($\theta_{(k)} = 0$, $\sigma^{(k)}_{ab} = 0$), the Hájíček 1-form on the cross-section $S \simeq S^2$ is defined as

$$\omega_a := -\ell_\mu \nabla_a k^\mu \qquad (a : \text{direction tangent to } S)$$

Under the identification $\Omega^{(\mathrm{rot})}_a := \omega_a$ with the rotation 1-form, $\Omega^{(\mathrm{rot})}$ functions as a U(1) connection (for the gauge transformation property $\Omega_a \to \Omega_a + D_a(\ln f)$, see Part III §4.2, Eq.,(4.4)). The structure equation gives

$$d\Omega^{(\mathrm{rot})} = -2\,(\mathrm{Im}\,\Psi_2)\,\epsilon_S \tag{3.1}$$

where Ψ_2 is the Weyl scalar and $\epsilon_S = 2\,\omega_{FS}$ is the area 2-form of the cross-section ($\int_{S^2} \epsilon_S = 4\pi$).

Proof of $c_1 = 0$ (core in three lines):

1. Fixing (k, ℓ) globally on the NEH, the Hájíček 1-form $\omega_a = -\ell_\mu \nabla_a k^\mu$ is constructed as a **globally smooth real 1-form** on S. (Note: this global smoothness follows from the physical construction via the Levi-Civita connection and does not presuppose triviality of the U(1) bundle. See the discussion below.)
2. Since $S \simeq S^2$ is a closed surface ($\partial S = \emptyset$), Stokes' theorem gives $\int_S d\omega = 0$.
3. The curvature 2-form $F := d\Omega^{(\mathrm{rot})} = d\omega$ is gauge-invariant, so $c_1 = \frac{1}{2\pi}\int_S F = 0$. □

Discussion. The following points are important in this proof.

- **Absence of circularity.** At first glance, asserting in Step 1 that "ω is a global 1-form" may appear to presuppose that "the bundle is trivial." However, $\omega_a = -\ell_\mu \nabla_a k^\mu$ is **not constructed as a section of a U(1) bundle but is directly constructed from the spacetime Levi-Civita connection and the null tetrad** (k, ℓ). The pair (k, ℓ) consists of null vector fields on the spacetime, which can be globally and smoothly fixed from the NEH structure (null generator k and auxiliary null vector ℓ). Therefore, the global smoothness of ω_a is a **consequence of the physical construction**, not an assumption about bundle topology. The result $c_1 = 0$ (triviality of the bundle) is derived as a conclusion, not assumed as a premise.
- The proof **does not assume axial symmetry**. It is a universal result independent of the odd-function property of Kerr.
- The NEH conditions ($\theta_{(k)} = 0$, $\sigma^{(k)}_{ab} = 0$) enter essentially in ensuring the smoothness of ω_a and the well-definedness of the cross-section S. Without these conditions, ω_a could have singularities, so the NEH conditions are essential implicit premises of the proof.

Global smoothness condition. The null generator k exists globally by the definition of NEH. The choice of the auxiliary null vector ℓ can be made locally at each point (since the normal bundle of the NEH is 2-dimensional), but a globally smooth choice requires triviality of the normal bundle. The crucial point here is that the curvature $F = d\Omega^{(\mathrm{rot})}$ of $\Omega^{(\mathrm{rot})}$ is itself a gauge-invariant quantity (independent of the choice of ℓ), so $c_1 = \frac{1}{2\pi}\int_S F$ does not depend on whether ℓ can be globally smoothly fixed. Even if ℓ cannot be globally smoothly fixed, one constructs $\omega_a^{(\alpha)}$ on local patches $\{U_\alpha\}$ whose curvatures are consistent, and summing $\int_{U_\alpha} d\omega^{(\alpha)} = \int_{U_\alpha} F$ yields $\int_S F = \int_S d(\text{local form}) = 0$ (boundary contributions from each patch cancel by the consistency conditions). That is, $c_1 = 0$ holds even without assuming global fixing of (k, ℓ).

3.3 Application to General NEH, Kerr–Newman, and Perturbed Kerr

Kerr–Newman (charged rotating black hole). The rotation 1-form $\omega(\zeta) = 2Mar_+/(r_+^2 + a^2\zeta^2)$ depends on the charge Q through $r_+ = M + \sqrt{M^2 - a^2 - Q^2}$, but $\kappa(\zeta) = 2\omega'(\zeta)$ remains an odd function, and $c_1 = 0$ also follows from axial symmetry. However, the important point is that $c_1 = 0$ is obtained directly from Stokes' theorem without relying on axial symmetry. The charge modifies the multipole distribution but does not affect the total flux.

Perturbed Kerr. As long as the NEH conditions are maintained, even with breaking of axial symmetry (appearance of $m \neq 0$ components), $\int_S d(\delta\omega) = 0$ follows immediately from Stokes' theorem.

Numerical relativity (post-binary-merger). In the asymptotic approach to a stationary state, if the NEH conditions hold approximately, then $c_1 = 0$ is established. During the dynamical horizon stage, the definition of $\Omega^{(\mathrm{rot})}$ itself requires modification, but due to the discreteness of $c_1 \in \mathbb{Z}$, the persistence of $c_1 = 0$ is expected even during the transition process.

3.4 Physical Meaning: "Gravitational Rotation Always Cancels"

$c_1 = 0$ can be interpreted from three perspectives.

(i) Global cancellation. The rotational curvature density $\mathrm{Im}\,\Psi_2$ on the NEH cross-section can be locally nonzero, but its integral over the entire sphere is always zero. There is no "net flux" of rotation; only a multipolar distribution exists.

(ii) Triviality of the bundle. $c_1 = 0$ means that the U(1) bundle to which $\Omega^{(\mathrm{rot})}$ belongs is topologically isomorphic to the trivial bundle $S^2 \times \mathrm{U}(1)$. There is no Dirac-string-type topological obstruction.

(iii) Contrast with Berry curvature ($c_1 = 1$). The Berry curvature of the quantum state space has $c_1 = 1$ (Hopf bundle) and carries a global topological obstruction. The difference from $c_1 = 0$ on the gravity side is not a peculiarity of Kerr but a universal property of NEH.

3.5 Compatibility with the Euler Number

The tangent bundle of S^2 is topologically nontrivial (Euler number $\chi = 2$, hairy ball theorem), and the frame bundle $\mathrm{Fr}(S^2)$ of S^2 is a $\mathrm{SO}(2) \cong \mathrm{U}(1)$ principal bundle with $c_1^{\mathrm{Fr}} = \chi(S^2) = 2$ by the Gauss–Bonnet theorem. On the other hand, the rotation 1-form of NEH has $c_1 = 0$. This compatibility is due to the fact that $\Omega^{(\mathrm{rot})}$ is not the Levi-Civita connection of the tangent bundle of S^2 but a different U(1) connection related to the normal bundle of the NEH. Since the two belong to different U(1) principal bundles, it is mathematically natural for their Chern numbers to differ.

§4 Geometric Quantization and the Uncertainty Principle

The purpose of this section is not a novel derivation of quantum mechanics, but a confirmation (sanity check) that HVP (formulated in Part II) reproduces known results of geometric quantization without breaking them.

4.1 Symplectic Structure on Null Surfaces

At each point x on a null initial surface Σ, expressing the state of a two-level system in polar coordinates, the squared amplitude (probability density) $\tilde{\rho}(x) = |\psi(x)|^2$ and the phase $\theta(x)$ are identified as canonically conjugate variables. The symplectic form is defined as

$$\omega = \int_{\Sigma} \sqrt{q(x)}\, d\tilde{\rho}(x) \wedge d\theta(x)\, d^2x \tag{4.1}$$

($\sqrt{q(x)}$ is the area element of the cross-section metric). The Poisson bracket gives

$$\{\tilde{\rho}(x),\, \theta(y)\} = \frac{1}{\sqrt{q(x)}}\, \delta^{(2)}(x-y)$$

Notational remark. Here $\tilde{\rho}(x) = |\psi(x)|^2$ is the probability density and is a different physical quantity from the curvature density ρ ($F = \rho\, \omega_{FS}$). The curvature density ρ belongs to the geometric structure of the connection (background structure), while the probability density $\tilde{\rho}$ belongs to the dynamical degrees of freedom of the quantum state. On the quantum side, $\rho = 1$ (constant curvature) means that the Berry curvature is uniform; it does not mean that the probability density $\tilde{\rho}$ is constant.

4.2 Procedure of Geometric Quantization

By the Kostant–Souriau geometric quantization, a prequantum line bundle $L \to S^2$ is constructed on the symplectic manifold (S^2, ω_{FS}).

Integrality condition. The necessary and sufficient condition for the prequantum bundle to exist is

$$\frac{1}{2\pi\hbar} \int_{S^2} \omega = n \in \mathbb{Z}$$

Under the normalization $\int_{S^2} \omega_{FS} = 2\pi$ of ω_{FS}, defining the symplectic form for a spin-j representation as $\omega_j = 2j\hbar\, \omega_{FS}$ gives

$$\frac{1}{2\pi\hbar} \int_{S^2} \omega_j = \frac{2j\hbar \cdot 2\pi}{2\pi\hbar} = 2j$$

and the integrality $n = 2j$ guarantees the half-integer quantization of j. The case $j = 1/2$ corresponds to the minimal nontrivial quantization, and the Hilbert space is constructed as

$\mathcal{H}_j = H^0(S^2, L^{\otimes 2j})$ (the space of holomorphic sections of $L^{\otimes 2j}$); the Riemann–Roch theorem gives $\dim \mathcal{H}_j = 2j + 1$.

Canonical quantization rule. By the geometric quantization prescription $\hat{f} = -i\hbar\,\nabla^L_{X_f} + f$, the commutation relation

$$\boxed{[\hat{\tilde{\rho}}(x),\, \hat{\theta}(y)] = i\hbar\,\frac{\delta^{(2)}(x-y)}{\sqrt{q(x)}}} \tag{4.2}$$

is derived. This is a consequence of the symplectic structure on the null surface, not an externally imposed assumption.

4.3 Logical Chain: Curvature → Noncommutativity → Uncertainty

The logical chain leading to the uncertainty principle is as follows:

$$\boxed{F = \rho\,\omega_{FS} \neq 0 \text{ (curvature of the prequantum bundle)} \implies \text{nontrivial holonomy} \xrightarrow{\text{quantization}} [\hat{J}_i, \hat{J}_j] \neq 0 \implies \text{unce}}$$

First step: Curvature of the prequantum bundle ⇒ nontrivial holonomy. The connection curvature $F = \rho\,\omega_{FS} \neq 0$ ($\rho = 1$ for spin-1/2) of the prequantum bundle $L \to S^2$ guarantees, by the Ambrose–Singer theorem, that the holonomy along closed loops is nontrivial. The "curvature" entering essentially here is not the Gauss curvature of S^2 itself but the curvature $F \propto \omega_{FS}$ of the connection on the prequantum bundle.

Second step: Quantization ⇒ commutation relations. By the geometric quantization prescription (§4.2), the Poisson bracket $\{J_x, J_y\} = J_z$ of angular momentum functions corresponding to the SO(3) generators on S^2 is quantized to the commutation relation $[\hat{J}_x, \hat{J}_y] = i\hbar\,\hat{J}_z$.

Third step: Commutation relations ⇒ uncertainty. Substituting $\hat{A} = \hat{J}_x$, $\hat{B} = \hat{J}_y$ into the Robertson inequality

$$\Delta A \cdot \Delta B \geq \frac{1}{2}\left|\langle [\hat{A}, \hat{B}] \rangle\right| \tag{4.3}$$

yields $\Delta J_x \cdot \Delta J_y \geq \frac{\hbar}{2}|\langle \hat{J}_z \rangle|$.

4.4 Geometric Meaning of $\hbar$

Within the present framework, $\hbar$ admits three geometric interpretations:

1. **Phase-space area quantum**: As the geometric content of the Bohr–Sommerfeld condition $\oint A = 2\pi\hbar\, n$, the phase-space area is quantized in integer multiples of $2\pi\hbar$.
2. **Chern class of the prequantum bundle**: $c_1 = \frac{1}{2\pi\hbar}\int \omega \in \mathbb{Z}$ shows that $\hbar$ sets the scale of the integrality condition.
3. **Effective curvature radius of S^2**: Through the scaling $\omega = \hbar\,\omega_{FS}$, $\hbar$ is the conversion factor linking the area of the state space to the physical phase-space area.

4.5 Three-Layer Structure of Uncertainty

Layer	Mathematical structure	Form of uncertainty	Relation to S^2
Geometric	SO(3) Lie algebra	$\Delta J_x \cdot \Delta J_y \geq \frac{\hbar}{2}\lvert\langle J_z\rangle\rvert$	Curvature $\to$ noncommutativity
Wave-mechanical	Fourier transform	$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$	Flat limit of S^2 $(R \to \infty)$
Information-theoretic	Fisher information	$\delta\xi \geq \frac{1}{\sqrt{NF_Q}}$	FS metric $= \frac{1}{4}g^{(F)}$

All three are different manifestations of the same geometric origin.

4.6 Re-derivation: Berry Curvature and Quantization Condition (Hello World I)

In this section, the Berry curvature $\rho = 1$ for spin-1/2 and the angular momentum quantization condition are re-derived from the unified principle HVP (formulated in Part II). This is the quantum-side "Hello World" of the present framework.

Starting point. In the quantum sector of HVP, one chooses $A = A_{\text{Berry}}$, $c_1 = 1$ (Hopf bundle). The energy functional of the stability condition (HVP Condition 2) is

$$\mathcal{E}_{\text{QM}}[\rho] = \int_{S^2} \left[\frac{1}{2}|d\rho|^2 + \frac{\lambda}{2}(\rho - 1)^2\right] \omega_{FS} \tag{4.4}$$

The first term penalizes non-uniformity of the curvature, and the second term is a stabilizing potential near the Chern condition $\langle\rho\rangle = 1$.

Step 1: Stationarity condition and derivation of Berry curvature $\rho = 1$.

Taking the first variation of $\mathcal{E}_{\text{QM}}$ with respect to ρ gives

$$\delta\mathcal{E}_{\text{QM}} = \int_{S^2} \left[-\Delta_{S^2}\rho + \lambda(\rho - 1)\right] \delta\rho \, \omega_{FS} = 0 \tag{4.5}$$

for arbitrary $\delta\rho$ (subject to $\langle\delta\rho\rangle = 0$). The Euler–Lagrange equation is

$$-\Delta_{S^2}\rho + \lambda(\rho - 1) = \mu \tag{4.6}$$

(μ is a Lagrange multiplier for the constraint $c_1 = 1$). The constant solution $\rho = 1$ trivially satisfies this equation with $\mu = 0$. Furthermore, since the eigenvalues of Δ_{S^2} are $-\ell(\ell + 1)$ $(\ell = 0, 1, 2, ...)$, the second variation for a perturbation $\delta\rho = \sum_{\ell \geq 1} \delta\rho_{\ell m} Y_{\ell m}$ is

$$\delta^2\mathcal{E}_{\text{QM}} = \sum_{\ell \geq 1} \sum_m \left[\ell(\ell + 1) + \lambda\right] |\delta\rho_{\ell m}|^2 > 0 \qquad (\lambda > 0) \tag{4.7}$$

and $\rho = 1$ is the **global minimum** of $\mathcal{E}_{\text{QM}}$. Thus, the spin-1/2 Berry curvature $\rho = 1$ (constant curvature, canonical connection of the Hopf bundle) is uniquely re-derived from the stability condition of HVP. □

Step 2: Re-derivation of angular momentum quantization.

The flux rule (Condition 3) of HVP requires

$$\int_{S^2} \rho\, \omega_{FS} = 2\pi c_1 = 2\pi \cdot 1 = 2\pi \tag{4.8}$$

Under $\rho = 1$, $\int_{S^2} \omega_{FS} = 2\pi$ holds immediately. For the extension to general spin-j, $c_1 = 2j$, and the existence condition for the prequantum bundle (§4.2)

$$\frac{1}{2\pi\hbar} \int_{S^2} \omega_j = 2j \in \mathbb{Z}$$

is recovered. The half-integer quantization of j is a direct consequence of the flux rule $c_1 \in \mathbb{Z}$.

Step 3: Re-derivation of commutation relations and the uncertainty principle.

By the local connection-form universality (§2.6), the Berry connection A_{Berry} obeys the general holonomy formula (Eq.,(2.4)) as a U(1) principal bundle connection. Since $\rho = 1 \neq 0$, the curvature is nondegenerate, and by the Ambrose–Singer theorem, the holonomy group coincides with all of U(1). The logical chain of §4.3

$$F = \rho\, \omega_{FS} \neq 0 \implies \text{nontrivial holonomy} \xrightarrow{\text{quantization}} [\hat{J}_i, \hat{J}_j] \neq 0 \implies \text{uncertainty principle}$$

is reproduced within the HVP framework. The Robertson inequality gives $\Delta J_x \cdot \Delta J_y \geq \frac{\hbar}{2} |\langle \hat{J}_z \rangle|$.

Summary. The core results on the quantum side have been re-derived from HVP (formulated in Part II) as follows:

Result	Basis in HVP	Status
Berry curvature $\rho = 1$	Global minimum of $\mathcal{E}_{\text{QM}}[\rho]$	Re-derivation (reproduction within HVP)
Half-integer quantization of j	Flux rule $c_1 = 2j \in \mathbb{Z}$	Re-derivation (reproduction within HVP)
Uncertainty principle	$\rho \neq 0 \Rightarrow$ nontrivial holonomy $\Rightarrow$ noncommutativity	Re-derivation (reproduction within HVP, same as §4.3)

These are in complete agreement with standard geometric quantization results, but by passing through the HVP framework, the structure is made explicit that they are derived from the **same type of stationarity condition** (Eq.,(4.6)) as the GR side (Part III §8).

4.7 Electromagnetic Sector Hello World and Three-Sector Comparison

The same logic as §4.6 is applied to the electromagnetic sector, and the potential W structures of the three sectors are compared.

Construction of W_{EM}. For the Chern number $c_1 = n$ of the electromagnetic U(1) bundle (Dirac monopole, $n \in \mathbb{Z}$), the same stability condition as §4.6 is imposed. The energy functional is

$$\mathcal{E}_{\mathrm{EM}}[\rho] = \int_{S^2} \left[\frac{1}{2}|d\rho|^2 + \frac{\alpha_E}{2}(\rho - n)^2\right] \omega_{FS} \tag{4.9}$$

The stationarity condition is

$$-\Delta_{S^2}\rho + \alpha_E(\rho - n) = \mu \tag{4.10}$$

and the constant solution $\rho = n$ satisfies this with $\mu = 0$. The second variation for $\ell \geq 1$ perturbations is $[\ell(\ell+1) + \alpha_E]|\delta\rho_{\ell m}|^2 > 0$ ($\alpha_E > 0$), and $\rho = n$ is the **global minimum** of $\mathcal{E}_{\mathrm{EM}}$ (the same stability argument as §4.6).

Finding: $W_{\mathrm{EM}}(n = 1) = W_{\mathrm{QM}}$. For the minimal Dirac monopole ($n = 1$), $W_{\mathrm{EM}} = \alpha_E(\rho - 1)^2/2$ and $W_{\mathrm{QM}} = \lambda(\rho - 1)^2/2$ become the **same potential** (under the identification $\alpha_E = \lambda$). This is a **reflection at the level of the variational principle** of the three-way identity in the $\mathcal{M}_\rho^{(1)}$ stratum noted in §2.4: Hopf bundle = minimal Dirac monopole = Berry connection. The minimal Dirac monopole and the Berry connection are stabilized by the same potential.

Inevitability of the agreement. This agreement of the quadratic forms is **nearly inevitable**. In the $c_1 = 1$ stratum, all three belong to the geometry of the same U(1) line bundle, so a local, even, smooth potential stabilizing $\rho = 1$ must have the form $W(\rho) = \kappa(\rho-1)^2/2 + O((\rho-1)^3)$ at leading order in its Taylor expansion. The Dirichlet energy minimality reinforces this structure: the energy of small deformations around the reference connection begins with the second variation, making the quadratic potential the most natural choice near a stable vacuum. However, the exact agreement of the coefficient κ ($\alpha_E = \lambda$) presupposes the same Hermitian metric, fiber radius convention, and normalization of the energy functional.

Three-sector W comparison table.

Sector	$W[\rho]$	Critical point	c_1	Closure
Quantum (Berry)	$\lambda(\rho-1)^2/2$	$\rho = 1$ (constant)	1	Automatic
EM ($n = 0$)	$\alpha_E\rho^2/2$	$\rho = 0$	0	Automatic
EM (monopole n)	$\alpha_E(\rho-n)^2/2$	$\rho = n$ (constant)	n	Automatic
Gravity	$\alpha\|\Psi_2\|^2 + \beta(\mathrm{Re}\,\Psi_2)^2 + V_{\mathrm{cf}}$	Kerr (variable)	0	Problematic

Pattern: The closure problem and the complexity of W share the same root. In the quantum and electromagnetic sectors, the critical points are **constant configurations** ($\rho = \mathrm{const}$), and a quadratic W suffices. Only in the gravitational sector is the critical point a **variable configuration** (Kerr: $\Psi_2 = -M/(r_+ + ia\cos\theta)^3$), requiring multiple terms in W ($|\Psi_2|^2$, $(\mathrm{Re}\,\Psi_2)^2$, V_{cf}).

This difference originates from spin. For spin-1/2 (quantum) and spin-1 (electromagnetic), the variables on S^2 automatically constitute a closed variational problem. For spin-2 (gravity), the NP variables ($\Psi_2, \Psi_3, \Psi_4, \sigma, ...$) form a coupled system, and a closed equation for Ψ_2 alone is not obtained (Part III §2.2, §6.3). That is, **the closure problem and the complexity of W share the same root**, both originating from the self-coupling structure of the spin-2 field.

Quantification via Teukolsky variables. More precisely, the essence of the obstruction lies not in the value of spin per se but in **whether the physical degrees of freedom close in terms of Teukolsky variables alone**. In the Teukolsky formalism, the separated variable for each spin-s is expanded in spin-weighted spherical harmonics ${}_sY_{\ell m}$, with the lowest ℓ being $\ell \geq |s|$.

- $s = 0$ **(scalar)**: The field itself closes as a scalar.
- $s = 1$ **(electromagnetic)**: The extremal Maxwell components $(\alpha, \underline{\alpha})$ carry the gauge-invariant degrees of freedom and can be transformed to the Fackerell–Ipser equation. Treating the $\ell = 0$ charge mode separately, the system nearly closes.
- $s = 2$ **(gravitational)**: Ψ_0, Ψ_4 capture only the radiative Weyl components and do not uniquely determine the full metric perturbation. The $\ell = 0$ (mass change) and $\ell = 1$ (angular momentum change) modes are not contained in ${}_2Y_{\ell m}$, and stationary perturbation and gauge completion are required separately (Wald [1973]; Green–Hollands–Zimmerman [2020]).

The fact that $s = 2$ lacks the low-multipole modes $\ell = 0, 1$ precisely means that these are entangled with mass/angular momentum changes and gauge freedom, which is the essential reason why the Teukolsky radiative variables alone "do not close." This is the Teukolsky-formalism counterpart of the closure problem in Part III §6.3.

Structural Unification of Gravity, Electromagnetism, and Quantum Mechanics via the Curvature Density ρ

— Variational Framework Based on a U(1) Connection on $\mathbb{CP}^1$ and Characterization of Boundary Equations —

Part II: Unified Principle (Holonomy Variational Principle) — Level 2

This paper series consists of five parts:

Part	Title	Level	Content
I	Common Language	0–1	Dictionary, curvature density space, topological theorems, geometric quantization
II	**Unified Principle (HVP)**	**2**	**Formulation of the Holonomy Variational Principle, physical origins, clarification of variational targets**
III	GR Consistency Verification	2–3	Characterization of Einstein boundary constraints, Ψ_2 boundary theory
IV	Observational Predictions and Open Problems	3	Horizon spectrum classification, Chern number wall, current status
V	Extended Proposals	3–4	Electromagnetic field, Dirac structure, weak force, 4D unified action

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§1 Introduction: From Dictionary to Principle

In Part I, it was established that the U(1) connections appearing in the three domains of gravity, electromagnetism, and quantum mechanics share a common curvature density representation on the null Kerr screen $S^2 \simeq \mathbb{CP}^1$:

$$F = \rho\, \omega_{FS} \tag{1.1}$$

and that the differences reduce solely to the choice of scalar density ρ and the topological sector c_1 (Part I §1.1, §2.6). By the universality of local connection forms (Part I Theorem 2.6), the Berry

connection, gravitational rotation connection, and electromagnetic connection share identical gauge transformation rules and holonomy formulas through the Lie algebra isomorphism $\mathfrak{u}(1) \cong \mathfrak{so}(1,1) \cong \mathbb{R}$.

However, this "dictionary" remains a static comparative framework. The dynamical equations relating different ρ have not been unified. In each theory, the principle determining ρ has a distinct origin—the Einstein equations, Maxwell equations, and symmetry of the state space, respectively. The purpose of this part is to **propose (postulate)** a unified variational principle (Holonomy Variational Principle; HVP) that describes the physics of all three domains in a common variational formalism, and to clarify its physical motivation and conditions of applicability.

1.1 Structure of Part II

§2 formally introduces a three-symbol system to resolve the polysemy of the ρ notation. §3 formulates the HVP as an axiomatic proposal and demonstrates its concrete application to all three domains. §4 supplements the physical origins of the HVP from the perspectives of gauge theory, the Aharonov–Bohm effect, Wilson loops, and lattice gauge theory. §5 clarifies the variational target (A-variation vs. ϱ-variation vs. C-variation). §6 distinguishes three levels of claim strength for the HVP (three-tier declaration), and §7 specifies the positioning as an effective theory and its conditions of applicability.

Claim/Status summary (Part II).

Claim	Status	Section
Introduction of the ϱ symbol decomposition	Definition	§2
Formulation of HVP (C5)	Axiomatic proposal (postulate)	§3
Physical origins of HVP (C6)	Motivation	§4
Clarification of variational targets	Clarification	§5
Three-tier declaration	Positioning	§6
Conditions of applicability as an effective theory	Specification of conditions	§7

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§2 Decomposition of the ϱ Symbol: Three-Symbol System for Curvature Density

In Part I, ρ was used as a real-valued curvature density for dictionary-level comparison (Part I §2.1 Remark). From Part II onward, to resolve the polysemy whereby ρ refers to multiple physical quantities, we formally introduce the following three symbols.

Definition 2.1 (Three-symbol system for curvature density).

Symbol	Name	Definition	Range	Context of use
ϱ	Abstract curvature density	Variational variable of HVP	$\mathbb{R}$ or $\mathbb{C}$	Formulation of HVP (dictionary level)
ρ_{rot}	Real rotational curvature density	$\rho_{\mathrm{rot}} := -4\,\mathrm{Im}\,\Psi_2$	$\mathbb{R}$	Rotational dynamics of GR
$\rho_{\mathbb{C}}$	Complex curvature density	$\rho_{\mathbb{C}} := K + iB = -2\Psi_2$	$\mathbb{C}$	Ψ_2 boundary theory

Here $K = -2\,\mathrm{Re}\,\Psi_2$ is the mass density (corresponding to Gauss curvature) and $B = -2\,\mathrm{Im}\,\Psi_2$ is the rotation density (corresponding to the normal fundamental form).

Relations among symbols. $\rho_{\mathrm{rot}} = 2B = 2\,\mathrm{Im}(\rho_{\mathbb{C}})$, and the real ρ used in Part I §2–§4 corresponds to $\varrho = 1$ (as a value of the abstract variable) in the Berry sector and to a rescaling of ρ_{rot} in the GR sector.

Principle of usage. In discussions of HVP at the level of principle, ϱ is used to describe the structure of the theory. In concrete GR calculations, $\rho_{\mathbb{C}}$ or ρ_{rot} is used to make the direct correspondence with physical quantities explicit. This separation avoids confusions such as "Is the ρ of HVP the same as the ρ of Berry?" or "Which ρ is the one corresponding to Ψ_2?"

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§3 Formulation of the Holonomy Variational Principle (HVP)

3.1 Proposal of the HVP

Building on the universality of local connection forms established in Part I §2.6, we **propose (postulate)** a single generative principle common to all three domains.

> **Unified Principle (Holonomy Variational Principle; HVP — proposed variational postulate).** *For a U(1) principal bundle $P \to S^2$ over the null Kerr screen $S^2 \simeq \mathbb{CP}^1$ and the curvature density ϱ ($F_A = \varrho\,\omega_{FS}$) determined by its connection A, the physics is determined by the stationarity condition of the holonomy functional*
>
> $$\mathcal{Z}[A] := \exp\left(i \oint_C A\right) = \exp\left(i \int_\Sigma \varrho\,\omega_{FS}\right) \qquad (\partial\Sigma = C) \tag{3.1}$$
>
> *The admissible ϱ are selected by the following three conditions:*
>
> 1. **Topological condition (Chern selection)**: $\langle\varrho\rangle = c_1 \in \mathbb{Z}$ (Chern–Weil integrality, Part I §2.1)
> 2. **Stability condition**: ϱ is a critical point of an appropriate energy functional $\mathcal{E}[\varrho]$
> 3. **Flux rule**: The global flux $\int_{S^2} \varrho\,\omega_{FS} = 2\pi c_1$ is tied to conserved charges

Character of the conditions. Condition 1 (topological condition) is a mathematical consequence of Chern–Weil theory and has been **proved**. Condition 3 (flux rule) corresponds to known conservation laws in each domain, and on the GR side it is proved in Part I §3. Condition 2 (stability condition) **assumes** the existence of a potential W; the complete determination of W_{GR} is partially

achieved in Part III §3.5 but retains a part that depends on the resolution of the closure problem (Part III §6.3).

3.2 Rationale for Holonomy as the Unit of Comparison

The connection A itself changes under gauge transformations $A \to A + d\alpha$ and is therefore not a local observable. On the other hand, the line integral along a closed loop C is gauge-invariantly defined by virtue of $\oint_C d\alpha = 0$, and the minimal loop observable

$$W(C) := \exp\left(i \oint_C A \right)$$

coincides with the Aharonov–Bohm phase (in the U(1) case). Requiring $W(C) = W(C_1)W(C_2)$ for the composition of loops $C = C_1 \circ C_2$, continuity and the group homomorphism property naturally select the exponential form.

3.3 From Holonomy Variation to Local Equations

The answer to "why vary the holonomy?" is to extract local equations, as in the action principle. By Stokes' theorem, for an infinitesimal surface Σ (with boundary $\partial\Sigma = C$),

$$\oint_C A = \int_\Sigma F, \qquad W(C) = \exp\left(i \int_\Sigma F \right) \simeq 1 + i \int_\Sigma F + \cdots,$$

and infinitesimal deformations of the loop are governed by the local curvature $F = dA$. Therefore, imposing the stationarity condition on the holonomy is a standard procedure for projecting from loop quantities (gauge invariants) to local equations; in this paper, it is implemented as a variational problem for ϱ.

Writing $F = \varrho\, \omega_{FS}$ on S^2, a quadratic action (e.g., a functional obtained by adding a potential term to $\int_{S^2} \frac{1}{2} F \wedge *F$) induces an effective functional $\mathcal{E}[\varrho]$ for ϱ, whose stationarity condition yields the common form (3.5).

3.4 Application to the Gravity Sector

On the horizon cross-section $S \simeq S^2$, one chooses $A = \Omega^{(\mathrm{rot})}$ (the Hájíček 1-form). $\Omega^{(\mathrm{rot})}$ is a 1-form connection describing boost transformations in the **normal bundle** of the cross-section (associated with the choice of (k, ℓ)), with structure group $\mathbb{R}^+ \simeq \mathrm{SO}(1,1) \cong \mathbb{R}$ (in Part I §2.6 it is compared in U(1) form via a Wick-type identification). Its holonomy is interpreted as the total transformation of the null frame along a closed loop (a loop quantity corresponding to a phase), and is formally compared with the Berry phase as "holonomy of a connection."

The stability condition of HVP reproduces the isolated horizon condition as a stationary solution ($\partial_u \Psi_2 = 0$) of the cross-section projection of the Bianchi identity

$$\partial_u \Psi_2 = -\eth \Psi_3 + \cdots \tag{3.2}$$

Under the definition $\rho_{\mathbb{C}} = -2\Psi_2$ (Part III §4), the Kerr solution is positioned as a critical point of

$$\delta\mathcal{E}_{\mathrm{GR}}[\varrho]\big|_{\varrho=\kappa_{\mathrm{Kerr}}}=0,\qquad \mathcal{E}_{\mathrm{GR}}[\varrho]:=\int_{S^2}\left[\frac{1}{2}|d\varrho|^2+W_{\mathrm{GR}}(\varrho,\bar{\varrho})\right]\omega_{FS} \tag{3.3}$$

(the explicit form of W_{GR} is constructed in Part III §3.5). The flux rule yields $c_1=0$ (the universal result of Part I §3) and the Bondi mass-loss formula $dM/du=-\mathcal{F}_E$.

3.5 Application to the Quantum Sector

On the state space $\mathbb{CP}^1$, one chooses $A=A_{\mathrm{Berry}}$ (Berry connection). The stability condition of HVP, by the SO(3) symmetry of the Fubini–Study metric, selects the minimum of

$$\delta\mathcal{E}_{\mathrm{QM}}[\varrho]\big|_{\varrho=1}=0,\qquad \mathcal{E}_{\mathrm{QM}}[\varrho]:=\int_{S^2}\left[\frac{1}{2}|d\varrho|^2+\frac{\lambda}{2}(\varrho-c_1)^2\right]\omega_{FS} \tag{3.4}$$

at $\varrho=c_1=1$ (constant curvature, the canonical connection of the Hopf bundle). This corresponds uniquely to the spin-1/2 Berry connection (re-derivation is completed in Part I §4.6). The flux rule yields $c_1=1$ (topological type of the Hopf bundle) and the Bohr–Sommerfeld quantization condition.

3.6 Common Stationarity Condition

The stationarity conditions of both sectors share the common structure

$$\boxed{\Delta_{S^2}\varrho=\frac{\partial V}{\partial\varrho}} \tag{3.5}$$

Here V is a potential (on the quantum side $V=\frac{\lambda}{2}(\varrho-1)^2$; on the gravity side $V=W_{\mathrm{GR}}(\Psi_2,\bar{\Psi}_2)$), and in the gravity sector where ϱ is complex ($\rho_{\mathbb{C}}=-2\Psi_2$), $\partial V/\partial\varrho$ is to be read as $\partial W/\partial\bar{\Psi}_2$. The differences are entirely concentrated in the choice of potential V and topological sector c_1.

	Gravity	Quantum	Electromagnetism
Connection A	$\Omega^{(\mathrm{rot})}$	A_{Berry}	EM gauge field A_{EM}
c_1	0	1	0 or n
Potential V	$W_{\mathrm{GR}}(\Psi_2,\bar{\Psi}_2)$	$\frac{\lambda}{2}(\varrho-1)^2$	Derived from the Maxwell action
Stationary solution	Kerr	Hopf bundle ($\varrho=1$)	Solutions of Maxwell's equations
Type of stationarity condition	$\Delta_{S^2}\varrho=\partial V/\partial\varrho$	$\Delta_{S^2}\varrho=\partial V/\partial\varrho$	$\Delta_{S^2}\varrho=\partial V/\partial\varrho$

3.7 Theoretical Status of the HVP

HVP is not a "unification of interactions" in the sense of a Grand Unified Theory (GUT). What HVP asserts is a "structural unification": that the physics of all three domains can be described in a common geometric grammar (variation of a U(1) connection on the null Kerr screen S^2). The

specific dynamics of each domain (Einstein equations, Schrödinger equation, Maxwell equations) are recovered as choices of the HVP potential W and topological condition c_1.

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§4 Physical Origins of the HVP

4.1 Three-Step Logic from Holonomy Stability to Local Action

We supplement the physical motivation for the HVP proposed in §3. The purpose of this section is not to **derive** the HVP from other principles, but to argue from multiple independent perspectives "why it is natural to take holonomy stability as the starting point."

The local observable is the curvature F, and the global observable measurable along a closed curve γ is the holonomy phase

$$Z[A;\gamma] \equiv \exp\Big(i \oint_{\gamma} A\Big) \tag{4.1}$$

Requiring holonomy stability (the neighborhood of $\delta \log Z = 0$), via Stokes' theorem $\oint_{\gamma} A = \int_{\Sigma} F$, is translated into a stationarity condition on the area integral of the curvature. Thus "holonomy stability" naturally leads in the direction of minimizing the energy of the curvature on S^2:

$$S_{\text{curv}}[A] = -\frac{1}{4} \int F \wedge *F \tag{4.2}$$

This is nothing other than a Maxwell/Yang–Mills type action. That is, the HVP connects to a local field action principle through the following three-step logic:

1. **Holonomy stability**: Stationarity condition on the phase $\oint_C A$ along a closed curve
2. **Lifting to a surface via Stokes**: Translation to local curvature via $\oint_C A = \int_{\Sigma} F$
3. **Minimization of the quadratic form of curvature**: Minimization of $\int F \wedge *F \to$ Maxwell/Yang–Mills type action

4.2 Motivation from the Aharonov–Bohm Effect

The physical rationale for adopting holonomy (Wilson loop) as the fundamental quantity in the HVP is clearly manifest in the Aharonov–Bohm effect. When a charged particle traverses a loop around a solenoid, a phase shift

$$\Delta\phi = \frac{e}{\hbar} \oint_C A_{\text{EM}} \cdot dl \tag{4.3}$$

appears in the interference pattern, even though the particle passes only through a region where no magnetic field exists. This phase shift is determined not by the local field strength F but by the global information of the connection A (holonomy). In other words, **the minimal physically observable unit is the holonomy, neither the connection nor the curvature.** The HVP takes this fact as its starting point and constructs dynamics from the stationarity condition of the holonomy functional.

4.3 Wilson Loops and Gauge Theory

In lattice gauge theory (Wilson 1974), the continuum theory action is understood as a reconstruction from the minimal plaquettes of Wilson loops. For the plaquette variable on the lattice

$$U_P = \exp\left(ig \oint_{\partial P} A \right)$$

Wilson's lattice action is defined as

$$S_W = \beta \sum_P \left(1 - \frac{1}{N} \operatorname{Re} \operatorname{Tr} U_P \right) \tag{4.4}$$

and reproduces the Yang–Mills action $S_{\mathrm{YM}} = -\frac{1}{4} \int \operatorname{Tr}(F \wedge *F)$ in the continuum limit $a \to 0$.

In this construction, **the starting point is the holonomy (plaquette variable U_P), and the curvature F and the action S_{YM} appear secondarily as the small-loop limit of the holonomy**. The HVP is structurally isomorphic to Wilson's construction: holonomy stability → quadratic form of curvature → local action.

Structural correspondence between HVP and lattice gauge theory.

	Lattice gauge theory	HVP
Fundamental variable	Plaquette U_P	$\mathcal{Z}[A] = \exp(i \oint A)$
Gauge invariance	Automatic (loop quantity)	Automatic (loop quantity)
Emergence of local action	S_{YM} as $a \to 0$	$\mathcal{E}[\varrho]$ for small loops
Variational target	Lattice link variable $U_\mu(x)$	Curvature density ϱ

4.4 "Motivation," Not "Derivation," of the HVP

What was presented in §4.1–4.3 is not a **proof** that the HVP is a fundamental principle. What was shown is the following:

- In gauge theory, the minimal physical observable is the holonomy (Aharonov–Bohm)
- Holonomy stability naturally induces a Maxwell/Yang–Mills type action (three-step logic)
- This construction is structurally isomorphic to Wilson's lattice construction

The HVP is a reformulation of these known structures as a ϱ-variation on S^2, **modeling** local dynamics as the lowest-order gauge-invariant functional induced from small-loop holonomies. It does not possess a principled answer to "why holonomy stability?", but the consistency with the structure of observables in gauge theory provides the physical motivation.

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§5 Clarification of the Variational Target

In the formulation of the HVP, it is necessary to clarify "what is being varied." The following three variational targets are distinguished.

5.1 A-Variation (Variation of the Connection)

When the connection A itself is taken as the variational target, the treatment of gauge degrees of freedom becomes necessary. In Yang–Mills theory, $\delta S_{\mathrm{YM}}/\delta A = 0$ yields the field equations, but this depends on the choice of gauge-fixing condition. On S^2, since A is varied along orbits of gauge transformations $A \to A + d\alpha$, the effective degrees of freedom reduce to the information contained in the curvature $F = dA$, namely ϱ (the curvature density).

5.2 ϱ-Variation (Variation of the Curvature Density)

Under $F = \varrho\,\omega_{FS}$, when ϱ is taken directly as the variational target, gauge degrees of freedom are automatically eliminated (ϱ is a gauge invariant). The common stationarity condition (3.5) of §3.6 is the result of ϱ-variation in this sense.

Advantages. Since a gauge invariant is handled directly, there is no ambiguity of gauge fixing. Direct comparison across the three domains is possible.

Constraints. ϱ-variation is applicable only within the range where $F = \varrho\,\omega_{FS}$ holds (restricted to U(1) connections on S^2). For extension to 4D spacetime, this constraint itself must be imposed via Lagrange multipliers (see Part V §7).

5.3 C-Variation (Variation of the Closed Loop)

When the loop C of the Wilson loop is itself taken as the variational target, this falls within the category of loop equations (Makeenko–Migdal equations). The HVP does not perform C-variation in this sense; it fixes C and varies the curvature distribution ϱ within its interior.

5.4 Choice in the HVP

The HVP adopts **ϱ-variation**. The reasons for this choice are:

1. **Automatic guarantee of gauge invariance.** Since ϱ is a gauge invariant, the gauge-fixing procedure is unnecessary.
2. **Direct comparison across three domains.** By taking ϱ as the common variable, differences are concentrated solely in the potential V and c_1.
3. **Physical transparency.** ϱ carries a direct physical meaning in each domain (Berry curvature, rotational curvature, electromagnetic field strength).

The transition from A-variation to ϱ-variation is based on the fact that the space of gauge equivalence classes $\mathcal{A}/\mathcal{G}$ on S^2 is isomorphic to $\mathcal{M}_\varrho$ (the curvature density space, Part I §2.1).

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§6 Three-Tier Declaration: Claim Levels of the HVP

The strength of the claim in each sector of the HVP differs essentially and is distinguished into the following three tiers.

6.1 (i) HVP itself is a "unified postulate" (variational ansatz)

$F = \varrho\,\omega_{FS}$ and the Holonomy Variational Principle are **proposed axioms** serving as the starting point of the theory and are not derived from other principles. The physical motivation of §4 argues

for the naturalness of the HVP but does not constitute a proof. The HVP is a variational ansatz that is justified a posteriori by the physical consistency of its results.

6.2 (ii) The GR-side results constitute a "consistent embedding (characterization)"

The Einstein equations have not been **derived** from the HVP; rather, it is shown that the NEH/IH Einstein boundary constraints are consistently **characterized** as stationarity conditions of the HVP (Part III §8). This distinction is essential:

- **Derivation**: The Einstein equations follow from HVP → HVP contains GR → **not claimed** in this paper
- **Characterization**: The boundary projection of the Einstein equations is consistent with the stationarity conditions of HVP → HVP can describe the boundary structure of GR in a unified manner → **claimed** in this paper

6.3 (iii) The quantum-side results constitute a "re-derivation of known theory (Hello World)"

It is confirmed that Berry curvature $\varrho = 1$ and the quantization condition are reproduced as stationarity conditions of the HVP quantum sector (Part I §4.6). This serves as a sanity check for the internal consistency of the framework; it is not that "quantum mechanics was derived from HVP" but that "it was confirmed that HVP does not contradict quantum mechanics."

> *HVP is a unifying variational ansatz; we prove compatibility on horizons and reproduce Berry curvature as a sanity check.*

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§7 Positioning as an Effective Theory

7.1 Effective-Theory Character

The stability condition (Condition 2) of the HVP is, at the present stage, positioned as an effective-theory assumption. The gravity-sector potential W_{GR} (Part III §3.5) is constructed from the boundary projection of the Einstein equations, but its derivation lacks a principled justification from known NEH/IH/DH actions (Ashtekar–Krishnan boundary terms, Carlip–Teitelboim horizon action). That is, Condition 2 does not possess a fundamental answer to "why this potential?" and relies on an empirical construction via effective closure of the Bianchi identities (Part III §6.3).

7.2 Conditions of Applicability

Accompanying this effective-theory character, the domain of applicability of the HVP gravity sector is restricted to the following:

1. **Near-Kerr regime**: Background Kerr spacetime + small perturbations ($\delta\Psi_2/\Psi_2^{\mathrm{Kerr}} \ll 1$)
2. **Small shear**: $|\sigma| \ll 1$ (regime where the NEH condition holds approximately)
3. **Quasi-stationary evolution**: $|\partial_u \Psi_2| \ll |\Delta_{S^2}\Psi_2|$ (temporal variation slower than spatial variation)

Outside these conditions (dynamical horizons immediately after binary mergers, highly non-axisymmetric configurations), the quantitative predictive power of HVP is lost together with the breakdown of the effective closure A (Part III §6.3).

7.3 Conditions for Promotion to a Principle

The complete principled foundation of Condition 2 (promotion to a "principle") requires deriving W_{GR} variationally from known NEH/IH/DH actions. Specifically, the most promising route is to derive the form of W_{GR} from the Ashtekar–Krishnan boundary symplectic structure

$$\Omega_{\mathrm{IH}} = \frac{1}{8\pi G}\int_S \delta\Omega^{(\mathrm{rot})} \wedge \delta\epsilon_S \tag{7.1}$$

This is positioned as Open Problem 2 of the Main Paper §5.1.

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§8 Summary

The following was achieved in this part.

1. **Introduction of the ϱ symbol decomposition** (§2): A three-symbol system (ϱ, ρ_{rot}, $\rho_{\mathbb{C}}$) resolving the polysemy of the curvature density was formally defined.
2. **Formulation of the HVP (C5)** (§3): A variational principle unifying the physics of all three domains was formulated as an axiomatic proposal with three conditions (topological condition, stability condition, flux rule). The common stationarity condition $\Delta_{S^2}\varrho = \partial V/\partial\varrho$ is shared across all three domains, and the differences reduce solely to the choice of potential V and c_1.
3. **Physical origins of the HVP (C6)** (§4): The three-step logic—holonomy stability → lifting to a surface via Stokes → minimization of the quadratic form of curvature—was presented. Structural correspondences with the Aharonov–Bohm effect and Wilson's lattice construction were discussed, supplementing the physical motivation for the HVP.
4. **Clarification of variational targets** (§5): A-variation, ϱ-variation, and C-variation were distinguished, and the reasons for the HVP's adoption of ϱ-variation (automatic guarantee of gauge invariance, direct comparison across three domains, physical transparency) were made explicit.
5. **Three-tier declaration** (§6): The claim levels of the HVP were made explicit as (i) variational ansatz, (ii) characterization (GR side), and (iii) sanity check (quantum side).
6. **Positioning as an effective theory** (§7): The conditions of applicability (near-Kerr regime, small shear, quasi-stationary evolution) were made explicit, and the conditions necessary for promotion to a principle were identified.

What Part II has established. A variational principle unifying the physics of all three domains has been formulated under explicit claim levels. The common stationarity condition $\Delta_{S^2}\varrho = \partial V/\partial\varrho$ is shared across all three domains, and the differences reduce solely to the choice of potential V and c_1. The HVP is a postulate, and its justification comes from the physical consistency of its results (the GR characterization in Part III, the Berry re-derivation in Part I §4.6).

Structural Unification of Gravity, Electromagnetism, and Quantum Mechanics via the Curvature Density ρ

— Variational Framework Based on a U(1) Connection on $\mathbb{CP}^1$ and Characterization of Boundary Equations —

Part III: GR Compatibility Verification (Horizon Compatibility) — Level 2–3

This collection of papers consists of five parts:

Part	Title	Level	Content
I	Common Language	0–1	Dictionary, curvature density space, topological theorems, geometric quantization
II	Unifying Principle (HVP)	2	Formulation of the Holonomy Variational Principle, physical origin, clarification of the variational target
III	**GR Compatibility Verification**	**2–3**	**Characterization of Einstein boundary constraints, Ψ_2 boundary theory**
IV	Observational Predictions and Open Problems	3	Horizon spectral classification, the Chern number wall, state of the art
V	Extension Proposals	3–4	Electromagnetic field, Dirac structure, weak force, 4D unified action

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§1 Introduction: Verification of the Gravitational Sector of the HVP

We verify whether the common language ($F = \varrho\,\omega_{FS}$) established in Part I and the unified variational principle (HVP) proposed in Part II are **compatible with the boundary constraints of the Einstein equations in the gravitational sector**. This is the purpose of Part III.

The core results of this part are as follows:

1. **Discovery of a fundamental obstacle** showing that ϱ alone does not yield a closed variational problem, and its resolution via a boundary theory with Ψ_2 as the fundamental variable (§2–§3)
2. **Construction of the Ψ_2 boundary theory**: promoting the Weyl scalar Ψ_2 to a fundamental variable and confirming that the Einstein boundary constraints are **characterized** within the stationarity conditions of the HVP (§3, §8)

3. **Identification of the closure problem** and explicit specification of the applicability conditions as an effective theory (§6–§7)

Declaration of claim level. This part does **not derive** the Einstein equations from the HVP. What is shown is that the Einstein boundary constraints on NEH/IH are **consistently characterized** (= embedded) as stationarity conditions of the HVP. This distinction is essential for defining the scope of the claims of the entire collection of papers (see Part II §6.2).

1.1 Structure of Part III

§2 organizes the current status of the dynamics in the three domains and analyzes the fundamental obstacle to a variational principle for ϱ alone. §3 constructs a boundary theory with Ψ_2 as the fundamental variable. §4 establishes the physical identification $\rho_{\mathbb{C}} = -2\Psi_2$, and §5 discusses the connection to the Raychaudhuri equation. §6 treats gradient flow dynamics and effective evolution equations, and §7 addresses the closure problem and the determination of flux coefficients. §8 performs the HVP characterization of the Einstein boundary constraints (Hello World II), and §9 condenses the results as the Minimal GR Compatibility Set. §10 discusses the connection with the Ashtekar × Damour hybrid framework, and §11 provides a concrete illustration via a toy model. §12 introduces the unified variable Ψ as a future prospect.

Claim/Status list (Part III).

Claim	Status	Section
Fundamental obstacle to a variational principle for ϱ alone	Established	§2
Construction of the Ψ_2 boundary theory (C7)	Characterization (not derivation)	§3, §8
Physical identification $\rho_{\mathbb{C}} = -2\Psi_2$	Established	§4
Effective Closure A and identification of the closure problem (C8)	Effective theory (conditional)	§6
Conditional determination of flux coefficients α, β, γ (C9)	Conditionally determined	§7
Minimal GR Compatibility Set	Established	§9

Notation conventions. Following the three-symbol system introduced in Part II §2, we distinguish ϱ (abstract variable), $\rho_{\mathrm{rot}} = -4\,\mathrm{Im}\,\Psi_2$ (real rotational curvature density), and $\rho_{\mathbb{C}} = K + iB = -2\Psi_2$ (complex curvature density). In §2–§5, we explicitly distinguish ϱ (general theory) from $\rho_{\mathbb{C}}$ (GR calculations) according to context.

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§2 Fundamental Obstacle: Limitations of the Variational Problem for ϱ Alone

2.1 Current Status of the Dynamics in the Three Domains

Within the framework $F = \varrho\,\omega_{FS}$, the dynamics determining ϱ differ in each theory:

	Gravity	Quantum	Electromagnetism
ϱ	$\kappa(\zeta)$	1	$\rho_{\mathrm{EM}}(\theta,\phi)$
Determining principle	Cross-section projection of the Einstein equations	Symmetry of the state space	Maxwell equations
c_1	0	1 (spin-1/2)	0 or $2g$
Multipoles	Odd ℓ only (Kerr)	$\ell = 0$ only	Determined by the Maxwell equations

2.2 Candidate for the Unified Action and the Obstacle

As a candidate for an action that governs ϱ in a unified manner, one may consider

$$S[\varrho] = \int_{S^2} \left[\frac{1}{2}G(\varrho)\,(D_a\varrho)(D^a\varrho) + V_{\mathrm{theory}}(\varrho)\right]\omega_{FS} \;+\; \mu\left(\int_{S^2}\varrho\,\omega_{FS} - 2\pi\,c_1\right) \tag{2.1}$$

However, analysis of GR compatibility reveals that a local variational principle for ϱ alone faces a **fundamental obstacle**.

Relation between ϱ and Ψ_2. The correspondence $\rho_{\mathrm{rot}} = -4\,\mathrm{Im}\,\Psi_2$ established from the dictionary means that the rotational curvature density ρ_{rot} is only the imaginary part of the Weyl scalar Ψ_2. Reducing the Bianchi identities in the Newman–Penrose formalism under the NEH conditions ($\rho_{\mathrm{NP}} = 0$, $\sigma = 0$, $\kappa_{\mathrm{NP}} = 0$), one obtains for $\Psi_2 = P + iQ$ ($P = \mathrm{Re}\,\Psi_2$, $Q = \mathrm{Im}\,\Psi_2$, $\rho_{\mathrm{rot}} = -4Q$):

$$(\delta - \bar{\delta})P = -2(\mathrm{Im}\,\mu)\,Q, \qquad (\delta - \bar{\delta})Q = 2(\mathrm{Im}\,\mu)\,P \tag{2.2}$$

This possesses a **Cauchy–Riemann-type structure**[1]:

$$(\delta - \bar{\delta})\Psi_2 = 2i(\mathrm{Im}\,\mu)\,\Psi_2 \tag{2.3}$$

The governing equation for $\rho_{\mathrm{rot}} = -4Q$ is

$$(\delta - \bar{\delta})\rho_{\mathrm{rot}} = -8\,(\mathrm{Im}\,\mu)\,\mathrm{Re}\,\Psi_2 \tag{2.4}$$

where $\mathrm{Re}\,\Psi_2$ appears on the right-hand side. That is, ρ_{rot} **is coupled to** $\mathrm{Re}\,\Psi_2$**, and a closed equation for** ρ_{rot} **alone cannot be obtained**.

Furthermore, $\mathrm{Re}\,\Psi_2$ is determined by the cross-focusing equation

$$\mathrm{Re}\,\Psi_2 = -\frac{1}{4}\,{}^{(2)}R + \frac{1}{2}|\Omega^{(\mathrm{rot})}|^2 \tag{2.5}$$

but $|\Omega^{(\mathrm{rot})}|^2 = 4\pi^2\,|d(\Delta_{S^2}^{-1}\rho_{\mathrm{rot}})|^2$ is a **nonlocal** functional of ρ_{rot}.

[1]This CR-type structure holds when the null tetrad $(k^\mu, \ell^\mu, m^\mu, \bar{m}^\mu)$ is fixed in the Kinnersley gauge ($\kappa_{\mathrm{NP}} = 0$, $\epsilon_{\mathrm{NP}} = 0$). Under a different gauge choice, the concrete form of the $\delta - \bar{\delta}$ operator changes, so the coefficients of (2.3) are gauge-dependent. However, the coupling structure between P and Q (the fact that it is CR-type) is gauge-invariant.

Consequence. It is in general impossible to reproduce the Einstein equations for ϱ on the NEH as the Euler–Lagrange equations of a local variational problem for a single scalar field ϱ on S^2. The reasons are:

1. **Nonlocality**: $|\Omega^{(\mathrm{rot})}|^2$ is a nonlocal functional of ρ_{rot} and cannot be expressed as a local potential $V(\varrho)$.
2. **Coupled system structure**: ρ_{rot} is coupled to $\mathrm{Re}\,\Psi_2$ through the Bianchi identities and does not form a closed equation for ρ_{rot} alone.
3. **Zero-point problem**: At the zeros of $\kappa(\zeta)$, the Fisher-type kinetic term $G(\varrho) = 1/\varrho$ diverges.

2.3 Local Potential near the Equator

When adopting $G(\varrho) = 1$ (standard kinetic term), analysis near the equator of the Kerr solution ($\kappa \approx c\zeta$) yields

$$V_{\mathrm{GR}}^{(\mathrm{local})}(\varrho) \approx -\varrho^2 + O(\varrho^4) \tag{2.6}$$

The negative sign of the ϱ^2 term suggests that $\varrho = 0$ (non-rotating, Schwarzschild) is an **unstable equilibrium**. This is in contrast to the quantum side, where $V_{\mathrm{QM}} = \lambda(\varrho - 1)^2$ selects $\varrho = 1$ as a stable minimum.

---

§3 Construction of the Ψ_2 Boundary Theory

3.1 Comparison of Amendments

In light of the obstacle identified in §2, three amendments are examined.

Amendment I: Nonlocal action.

$$S_{\mathrm{I}}[\varrho] = \int_{S^2} \left[\frac{1}{2} (d\varrho)^2 - \varrho^2 + \alpha \, |d(\Delta_{S^2}^{-1} \varrho)|^2 \right] \omega_{FS} \tag{3.1}$$

Amendment II: Complex action with Ψ_2 as the fundamental variable.

$$S_{\mathrm{II}}[\Psi_2] = \int_{S^2} \left[\frac{1}{2} \, |d\Psi_2|^2 + W(\Psi_2, \bar{\Psi}_2) \right] \omega_{FS} \tag{3.2}$$

Amendment III: Action with constrained connection variables.

$$S_{\mathrm{III}}[\Omega, \lambda] = \int_{S^2} \left[\frac{1}{2} \, |d\Omega|^2 + \lambda \big({}^{(2)}R + 4 \, \mathrm{Re}\, \Psi_2[\Omega] - 2|\Omega|^2 \big) \right] \omega_{FS} \tag{3.3}$$

Amendment II is adopted. By promoting Ψ_2 to the fundamental variable, (1) ρ_{rot} is included as the imaginary part of Ψ_2, thereby avoiding the zero-point problem, and (2) the Cauchy–Riemann-type coupling of the Bianchi identities is naturally absorbed into the interior of Ψ_2.

3.2 Promotion of the Fundamental Variable

Definition. The fundamental variable is taken to be the Weyl scalar $\Psi_2 \in C^\infty(S^2, \mathbb{C})$, and its relation to the complex curvature density is

$$\boxed{\rho_{\mathbb{C}} := -2\Psi_2} \tag{3.4}$$

incorporated as a **definition** (promoting the physical identification of §4 to a definition). Writing the real and imaginary parts as

$$\Psi_2 = P + iQ, \qquad P := \operatorname{Re}\Psi_2, \quad Q := \operatorname{Im}\Psi_2 \tag{3.5}$$

the correspondence with $\rho_{\mathbb{C}} = K + iB$ is given by $K = -2P$, $B = -2Q$.

3.3 Evolution Equation for Ψ_2 (Bianchi Projection)

Under the NEH conditions, the cross-section projection of the Bianchi identities yields the evolution equation for Ψ_2:

$$\boxed{\partial_u \Psi_2 = -\eth \Psi_3 + \sigma \Psi_4 + (\text{matter terms})} \tag{3.6}$$

[Closure A] Rewriting the effective closure of §6 in Ψ_2 variables, the effective evolution equation near Kerr becomes

$$\boxed{\partial_u \Psi_2 = c\, \Delta_{S^2} \Psi_2 + S_{\text{flux}}(\Psi_2)} \tag{3.7}$$

The crucial point is that the equation which could not be closed for ρ_{rot} ($= -4Q$) alone in §2 **closes when Ψ_2 as a whole is taken as the variable**. The coupling between P and Q is absorbed into the interior of Ψ_2 through the Bianchi identities.

3.4 Natural Emergence of the CR Structure

The Cauchy–Riemann-type coupling equation of §2.2,

$$(\delta - \bar{\delta})\Psi_2 = 2i(\operatorname{Im}\mu)\, \Psi_2 \tag{3.8}$$

is naturally interpreted as a CR structure on S^2. For the complex coordinate $z = e^{i\phi}\tan(\theta/2)$ on S^2 (stereographic projection), one defines the CR condition for Ψ_2:

$$\bar{\partial}_b \Psi_2 = i\, \mu_{\text{Im}}\, \Psi_2 \tag{3.9}$$

This CR structure couples the real part P (mass component) and imaginary part Q (rotational component) of Ψ_2 in the following sense:

- The angular derivative of P on S^2 generates Q, and vice versa (see equation (2.2))
- The coupling constant is $\operatorname{Im}\mu$ (the imaginary part of the NP spin coefficient), which encodes the degree of rotation of the spacetime

- In the Schwarzschild limit ($\mathrm{Im}\,\mu = 0$), P and Q decouple, and $Q = 0$, $P = -M/r^3$ hold independently

3.5 Construction of the Potential $W(\Psi_2, \bar{\Psi}_2)$

Remark (explicit statement of the design character). The potential W below is **constructed (designed)** from physical requirements and is not derived from first principles. This design character is consistent with the positioning of the HVP as an effective theory (Part II §7). However, as shown in §7.8, through consistency with the stationarity condition of the D1 gradient flow, 3 out of the 4 requirements possess principled justification from the Dirichlet energy principle.

The potential W of Amendment II is constructed from the following physical requirements.

Requirement 1 (Kerr critical point). The Kerr solution $\Psi_2 = \Psi_2^{\mathrm{Kerr}}(\theta; M, a)$ is a critical point of W:

$$\left.\frac{\delta W}{\delta \bar{\Psi}_2}\right|_{\Psi_2 = \Psi_2^{\mathrm{Kerr}}} = 0 \tag{3.10}$$

Requirement 2 (CR compatibility). W is compatible with the CR structure.

Requirement 3 (Schwarzschild stability). The Schwarzschild solution ($\Psi_2 = -1/(8M^2)$ is real, $Q = 0$) is a stable critical point.

Requirement 4 (cross-focusing compatibility). At the critical point, the cross-focusing equation (equation (2.5)) is reproduced.

Construction of the potential. The simplest potential satisfying the above requirements takes the form

$$\boxed{W(\Psi_2, \bar{\Psi}_2) = \alpha\,|\Psi_2|^2 + \beta\,(\mathrm{Re}\,\Psi_2)^2 + V_{\mathrm{cf}}(\Psi_2)} \tag{3.11}$$

where

- $\alpha\,|\Psi_2|^2 = \alpha(P^2 + Q^2)$: an SO(2)-invariant mass term. Stability of Schwarzschild requires $\alpha > 0$.
- $\beta\,P^2$: an anisotropy term favoring the real part. It effectively incorporates the nonlocality of the cross-focusing equation.
- $V_{\mathrm{cf}}(\Psi_2)$: a nonlocal correction determined by the cross-focusing equation:

$$V_{\mathrm{cf}}(\Psi_2) := \gamma\left(\mathrm{Re}\,\Psi_2 + \frac{1}{4}\,{}^{(2)}R - \frac{1}{2}|\Omega[\Psi_2]|^2\right)^2 \tag{3.12}$$

This variationally imposes the cross-focusing equation. $|\Omega[\Psi_2]|^2$ is a **nonlocal functional** computed via $\mathrm{Im}\,\Psi_2$ **[Closure A dependent]**.

Verification of the Kerr critical point. For the Kerr solution, $\Psi_2^{\mathrm{Kerr}} = -M/(r_+ + ia\cos\theta)^3$ satisfies $V_{\mathrm{cf}} = 0$ (exactly satisfying the cross-focusing equation) and meets the critical point condition.

Schwarzschild verification. For Schwarzschild ($a = 0$), $\Psi_2 = -1/(8M^2)$ is a real constant, and $V_{\rm cf} = 0$. The stationarity condition when the $\ell = 0$ constraint is imposed via a Lagrange multiplier μ_0 is

$$\Delta_{S^2}\Psi_2 = \frac{\partial W}{\partial \bar{\Psi}_2} - \mu_0$$

and for the constant solution $\Psi_2 = \Psi_2^{\rm Schw}$, it is satisfied with $\mu_0 = (\alpha + \beta)\Psi_2^{\rm Schw}$. Stability against $\ell \geq 1$ perturbations is guaranteed by $\alpha + \beta > 0$. Consistency with $\rho_{\mathbb{C}} = -2\Psi_2 = 1/(4M^2)$ agrees with the Schwarzschild normalization of §4.

Remark (unresolved elements). The numerical determination of the potential coefficients α, β, γ of W depends on the resolution of the closure problem (§6.3). What is given in this section is the **structural form** of W; the unique determination of the coefficients remains an open problem. The nonlocality contained in $V_{\rm cf}$ does not completely resolve the fundamental obstacle of §2, and the variational problem for Ψ_2 strictly remains a **nonlocal variational problem**. However, compared to the case of $\rho_{\rm rot}$ alone, the structural obstacle of the coupled system is resolved by the promotion to Ψ_2.

3.6 Characterization Theorem

Theorem 3.6 (HVP Characterization of Einstein Boundary Constraints). Under the NEH/IH conditions and $c_1 = 0$, $\rho_{\mathbb{C}} = -2\Psi_2$, the boundary data of the Einstein equations close as a function of Ψ_2, and the following characterization holds:

(i) **Closure of data**: All geometric data on the NEH ($^{(2)}R$, $\Omega^{(\rm rot)}$, σ, κ_s) are reconstructed from Ψ_2 and its derivatives. $\mathrm{Re}\,\Psi_2$ encodes the cross-section Gauss curvature, and $\mathrm{Im}\,\Psi_2$ encodes the rotational vorticity.

(ii) **Permitted mode structure of ϱ**: The spherical harmonic expansion of $\rho_{\rm rot}$ for the Kerr solution contains only odd ℓ (Kerr selection rule), and the conserved quantities (M, J) correspond to the $\ell = 0$ and $\ell = 1$ components. For perturbed Kerr, even-ℓ ($\ell \geq 2$) components are permitted, and their decay rates are determined by the gradient flow of §6.1 as $-\ell(\ell + 1)$.

(iii) **Compatibility with the variational structure**: The stationarity condition of the potential W (equation (3.11)), $\Delta_{S^2}\Psi_2 = \partial W/\partial\bar{\Psi}_2$, recovers the cross-focusing equation as the variation of the $V_{\rm cf}$ term, and the curvature expression for the rotation 1-form from the definition $\rho_{\mathbb{C}} = -2\Psi_2$.

Remark (distinction between characterization and derivation). This theorem does not **derive** the Einstein equations from the HVP. The cross-focusing equation is used as input in the construction of $V_{\rm cf}$ (equation (3.12)), introducing a circular element, and the NEH/IH conditions themselves are consequences of the Einstein equations. What this theorem asserts is that the Einstein boundary constraints are **characterized** within the unified variational structure of HVP stationarity conditions, and this characterization renders the gravitational sector and the quantum sector (Part I §4.6) comparable as variational problems of the same type.

3.7 Completeness Status of the Ψ_2 Boundary Theory

Element	Status	Basis
Identification of the fundamental variable	**Complete**	Promotion to Ψ_2
Structure of the evolution equation	**Complete**	Bianchi projection (equation (3.7)) **[Closure A]**
CR coupling structure	**Complete**	Equations (3.8)–(3.9)
Structural form of the potential W	**Complete (designed)**	Equations (3.11)–(3.12)
Determination of potential coefficients	**Structurally complete; numerically undetermined**	§7: structural fixing from 3 conservation laws
Kerr critical point	**Verified**	Verification of Requirement 1
Schwarzschild consistency	**Verified**	Consistent with $\rho_{\mathbb{C}} = 1/(4M^2)$
Complete justification of Closure A	**Partial**	See §6.3

3.8 Conclusion: Ψ_2 as a Whole is the True Dynamical Variable

The above analysis demonstrates that ϱ is not a free variational variable but a dependent quantity determined by the constraint system of the Einstein equations. Completion of the unified action requires promotion from ϱ to Ψ_2 (a complex variable). The real and imaginary parts of Ψ_2 are coupled through the Bianchi identities, and there is an intrinsic limitation to treating only one of them as a variational variable. **Ψ_2 is the protagonist of the gravitational sector of the present theory**, and the abstract variable ϱ of Part II is concretized through Ψ_2.

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§4 Physical Identification: $\rho_{\mathbb{C}} = -2\Psi_2$

4.1 Identification of the Connection and Curvature Density

The GR compatibility verification has established the physical identification of the connection A and the curvature density.

Identification of the connection. The U(1) connection A on the null Kerr screen S^2 is identified with the H'aj'iček 1-form Ω_a on the horizon:

$$A \equiv \Omega_a, \qquad \Omega_a := -\ell_\mu \nabla_a k^\mu \tag{4.1}$$

where (k^μ, ℓ^μ) is a pair from the null tetrad.

Definition of the curvature density. The complex curvature density $\rho_{\mathbb{C}}$ determined from the curvature 2-form $F = dA$ is identified as

$$\boxed{\rho_{\mathbb{C}} := K + iB, \qquad K := -2\,\mathrm{Re}\,\Psi_2\big|_{\mathcal{H}}, \quad B = *d\Omega} \tag{4.2}$$

Here K is minus twice the real part of the Weyl scalar, and B is the vorticity of the rotation 1-form.

Remark. In the vacuum Schwarzschild limit ($\Omega = 0$), the cross-focusing equation gives $K = \frac{1}{2}\,{}^{(2)}R$. For a general rotating spacetime, $K = \frac{1}{2}\,{}^{(2)}R - |\Omega^{(\mathrm{rot})}|^2$.

Correspondence with the Weyl scalar. Under vacuum conditions,

$$\boxed{\rho_{\mathbb{C}} = -2\,\Psi_2\big|_{\mathcal{H}}} \tag{4.3}$$

This convention is confirmed by numerical computation in Schwarzschild spacetime:

$$\rho_{\mathbb{C}}\big|_{\text{Schw}} = K + i\cdot 0 = \frac{1}{2}\,{}^{(2)}R = \frac{1}{r_H^2} = \frac{1}{4M^2}, \qquad -2\Psi_2\big|_{\text{Schw}} = -2\cdot\left(-\frac{M}{r^3}\right)\Big|_{r=2M} = \frac{1}{4M^2}$$

The two expressions agree exactly.

4.2 Consistency of the Gauge Structure

Under the boost freedom $\ell \to f\ell$ of the null tetrad, the H'aj'iček 1-form transforms as

$$\Omega_a \to \Omega_a + D_a(\ln f) \tag{4.4}$$

This is isomorphic to the U(1) gauge transformation $A \to A + d\chi$ ($\chi = \ln f$) (the Wick-type identification of Part I Theorem 2.6), guaranteeing the consistency of the identification as a connection.

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§5 The Raychaudhuri Equation and the Curvature Density

The physical identification $\rho_{\mathbb{C}} = -2\Psi_2$ is directly linked to the dynamics of null congruences. The Raychaudhuri equation (null version) is written in general as

$$\dot{\theta} = -\frac{1}{2}\theta^2 - \sigma_{ab}\sigma^{ab} + \omega_{ab}\omega^{ab} - R_{ab}k^a k^b \tag{5.1}$$

Under the NEH conditions (twist-free: $\omega = 0$) and vacuum conditions, this is reduced to

$$\dot{\theta} = -\frac{1}{2}\theta^2 - |\sigma|^2 \tag{5.2}$$

Ψ_2 does not appear directly in this equation, but through the shear evolution equation $\dot{\sigma} = -\theta\,\sigma + \Psi_0$ and the cross-focusing equation (equation (2.5)), $\rho_{\mathbb{C}} = -2\Psi_2$ contributes indirectly to the dynamics of the null congruence.

Physical meaning. The boundary data on the null Kerr screen S^2 (the curvature density $\rho_{\mathbb{C}}$) constrain the cross-section geometry and shear evolution of the null congruence via the cross-focusing equation and Bianchi identities, and thereby determine the bulk spacetime geometry:

$$\rho_{\mathbb{C}}\ \text{(curvature density)} \xrightarrow{\rho_{\mathbb{C}}=-2\Psi_2} \Psi_2 \xrightarrow{\text{cross-focusing + Bianchi}} ({}^{(2)}R, \sigma) \xrightarrow{\text{Raychaudhuri}} \theta \xrightarrow{\text{reconstruction}} \text{spacetime geo}$$

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§6 Gradient Flow Dynamics and Effective Evolution Equations

6.1 D1 Gradient Flow and the Kerr Attractor

> **Switching the meaning of ρ.** In this section, we use $\rho_{\mathrm{rot}} = -4\,\mathrm{Im}\,\Psi_2$ (real rotational curvature density; the $\ell = 0$ component is zero by $c_1 = 0$). From §6.2 onward, we switch to $\rho_{\mathbb{C}} = K + iB = -2\Psi_2$ (complex curvature density).

[Closure A] The D1 gradient flow below is not an exact consequence of the Einstein equations but an effective dynamical model based on the effective closure of the Bianchi identities (§6.2–§6.3).

D1 gradient flow equation.

$$\boxed{\partial_u \rho_{\mathrm{rot}} = \Delta_{S^2} \rho_{\mathrm{rot}} + \lambda_1(u) \cos\theta} \tag{6.1}$$

$\lambda_1(u)$ is a Lagrange multiplier ensuring the conservation of angular momentum.

Mode decomposition. For the spherical harmonic expansion, the time evolution of each mode is

$$\rho_{\ell m}(u) = \rho_{\ell m}(0)\, e^{-\ell(\ell+1)u} \qquad (\ell \geq 2) \tag{6.2}$$

ℓ	Decay rate	Physical meaning
$\ell = 0$	0 (conserved)	Mass (Bondi mass)
$\ell = 1$	0 (conserved by constraint)	Angular momentum
$\ell = 2$	-6	Quadrupole deformation (slowest decaying mode)
$\ell \geq 2$	$-\ell(\ell+1)$	Exponential decay

Kerr dipole attractor. As $u \to \infty$, all $\ell \geq 2$ components decay exponentially, and

$$\rho_{\mathrm{rot}}(u, \theta) \xrightarrow{u \to \infty} \frac{3J}{2\pi} \cos\theta \tag{6.3}$$

The Kerr dipole is the attractor of this dynamical system. This is qualitatively similar to Price's theorem (though the functional form of the decay differs — Price gives an inverse power law $t^{-(2\ell+2)}$, while D1 gives exponential decay $e^{-\ell(\ell+1)u}$).

6.2 Effective Evolution Equation from the Bianchi Identities

Remark. From this section onward, $\rho_{\mathbb{C}} = K + iB$ (complex curvature density) is used. The $\ell = 0$ mode is nonzero (corresponding to mass).

Starting point. The projection of the Bianchi identities onto the NEH gives

$$\partial_u \Psi_2 = -\eth \Psi_3 + \sigma \Psi_4 + \Phi_2 \bar{\Phi}_2 + \cdots \tag{6.4}$$

This serves as the source of time evolution for $\rho_{\mathbb{C}}$ via $\rho_{\mathbb{C}} = -2\Psi_2$.

Effective Closure A. [Closure A] The angular derivative term $-\eth\Psi_3$ is approximated by the cross-section Laplacian:

$$-\eth\Psi_3 \approx c\,\Delta_{S^2}\rho_{\mathbb{C}} \tag{6.5}$$

Naturality from screen projection. This closure is not an arbitrary assumption but is understood as the leading term naturally arising from the geometry of the screen S^2. Adopting $\rho_{\mathbb{C}} = -2\Psi_2$, by equation (6.4) the evolution of $\rho_{\mathbb{C}}$ is governed by the angular derivative $\eth\Psi_3$ of Ψ_3. Considering an effective theory expansion in which Ψ_3 is approximated as a local functional of $\rho_{\mathbb{C}}$ on the screen, the minimal second-order differential operator compatible with the spin-weight structure and rotational symmetry is the Laplace–Beltrami operator Δ_{S^2}. Therefore, equation (6.5) is positioned as the **leading term of the screen effective theory**.

Consistency with spherical harmonic expansion. Since the screen is S^2, one can expand $\rho_{\mathbb{C}} = \sum_{\ell,m} \rho_{\ell m}(u)\, Y_{\ell m}(\theta,\phi)$, and from $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$, the linear part of Closure A has the eigenvalue structure $-\eth\Psi_{3,\ell m} \approx -c\,\ell(\ell+1)\rho_{\ell m}$ for each mode. This structure is consistent with the mode tower interpretation of the present paper: $\ell = 0$ (spherically symmetric / mass mode) has $\Delta_{S^2}\rho = 0$ and is static; $\ell = 1$ (tilt / rotational mode) is the minimal nontrivial degree of freedom; $\ell \geq 2$ (radiation modes) correspond to gravitational wave quadrupole and higher deformations.

Validity conditions of the closure. In addition to the screen EFT justification above, the approximation accuracy of Closure A is guaranteed under the following physical conditions:

- Near Kerr (the background is close to Kerr)
- Low-order modes (small ℓ)
- Small shear ($|\sigma| \ll 1$)

Effective Closure B: Flux source term.

$$S_{\text{flux}} = \alpha|\sigma|^2 + \beta\, T_{\ell\ell} + \gamma\, T_{\ell\varphi} + \nabla \cdot V \tag{6.6}$$

Effective evolution equation. **[Closure A]** Combining Closures A and B,

$$\boxed{\partial_u \rho_{\mathbb{C},\geq 2} = \Delta_{S^2}\rho_{\mathbb{C},\geq 2} + S_{\text{flux}}} \tag{6.7}$$

- $\ell = 0$: $\frac{d}{du}\langle\rho_{\mathbb{C}}\rangle \propto -\mathcal{F}_E$ (Bondi mass loss formula)
- $\ell = 1$: $\dot{J}_\rho \propto -\mathcal{F}_J$ (angular momentum flux)

6.3 EFT Resolution of the Closure Problem

Effective Closure A (equation (6.5)) is not automatically derived from the Bianchi identities. However, as shown in §6.2, its **form** is naturally determined as the leading term of an effective theory by the geometry, spin-weight structure, and rotational symmetry of the screen S^2. The remaining problems are the rigorous bulk matching of the closure coefficient c and the systematic determination of higher-order corrections, which are addressed within the EFT framework.

Clarification of the meaning of "closure." The "closure" referred to here is not a spatially local, pointwise closure, but rather a **closure in terms of variables**. In the Newman–Penrose formalism, $\Psi_2, \Psi_3, \Psi_4, \theta, \sigma, \kappa, \Omega_a$ constitute a coupled system of equations; in the present framework, the goal is to express all of these as functionals of Ψ_2 and reduce the system to equations for Ψ_2 alone.

Validity conditions of Closure A.

1. Linear response near Kerr
2. Small shear: $|\sigma| \ll 1$
3. Quasi-stationary evolution

Regimes where the closure breaks down.

- Under strong gravitational wave radiation (immediately after a binary merger)
- Highly non-axisymmetric configurations
- Dynamical horizons with large shear

Order-of-magnitude assessment for systematic improvement. Expanding in the perturbation order $\epsilon := \delta\Psi_2/\Psi_2^{\mathrm{Kerr}}$:

- **Zeroth order (Kerr background)**: The closure is exact ($\Psi_3 = 0$ is guaranteed by Petrov type D).
- **First order (linear perturbation)**: $\eth\Psi_3 = c_1\,\Delta_{S^2}\delta\Psi_2 + O(\epsilon^2)$, and this is determinable from the Teukolsky equation. Closure A is quantitatively justified.
- **Second order and higher**: Nonlinear coupling enters, requiring an extended theory.

The validity range is estimated at $|\epsilon| \lesssim 0.1$ (late ringdown phase).

Interpretation as an effective field theory. Closure A can be naturally understood from the hierarchical structure of degrees of freedom in the complete Newman–Penrose system. Viewing Ψ_2 as an infrared (IR) degree of freedom and Ψ_3, Ψ_4 as ultraviolet (UV) degrees of freedom, Closure A corresponds to an **effective action obtained by integrating out the UV degrees of freedom**:

$$e^{-S_{\mathrm{eff}}[\Psi_2]} = \int \mathcal{D}\Psi_3\,\mathcal{D}\Psi_4\; e^{-S_{\mathrm{NP}}[\Psi_2,\Psi_3,\Psi_4]} \tag{6.8}$$

The stationarity condition of this effective action reproduces the Closure A of equation (6.5). On the Petrov type D (Kerr) background, $\Psi_3 = \Psi_4 = 0$, and fluctuations of the UV degrees of freedom are suppressed as powers of ϵ, so Closure A is justified at the 1-loop (first-order perturbation) level.

Systematic improvement. This EFT perspective predicts the structure of systematic correction terms. At second order,

$$-\eth\Psi_3 = c_1\,\Delta_{S^2}\rho_{\mathbb{C}} + c_2\,|\rho_{\mathbb{C}}|^2\,\rho_{\mathbb{C}} + O(\epsilon^3) \tag{6.9}$$

The coefficient c_2 is in principle determinable from second-order perturbation theory of the Teukolsky equation, providing a quantitative extension of the validity range of Closure A.

QNM consistency condition for the closure coefficient. In the mode expansion of the effective evolution equation (equation (6.7)), the decay rate of the ℓ-th multipole perturbation is $\partial_u\rho_{\ell m} = -c\,\ell(\ell+1)/r_+^2\,\rho_{\ell m}$. In the ρ language, $\ell \geq 2$ perturbations are deviations from the spherically symmetric configuration ($\rho = \mathrm{const}$) and describe the relaxation process toward the Kerr attractor. Comparing this relaxation rate with the quasi-normal mode (QNM) decay rates $\omega_I^{(\ell)}$ predicted by GR linear perturbation theory, the ℓ-dependence of the closure coefficient c is quantified:

$$\boxed{c(\ell) = \frac{\omega_I^{(\ell)}\, r_+^2}{\ell(\ell+1)}} \tag{6.10}$$

From QNM data for the spherically symmetric background (Leaver [1985]; Berti, Cardoso & Starinets [2009]), for the fundamental overtone ($n = 0$):

ℓ	$M\omega_I$	$c(\ell)/M$	$c(\ell)/c(2)$
2	0.0890	0.0593	1.00
3	0.0927	0.0309	0.52
4	0.0942	0.0188	0.32

Interpretation of the ℓ-dependence. The fact that $c(\ell)$ is not constant provides **quantitative evidence** for the approximate nature of Closure A (the assumption of constant c). For QNM data on the spherically symmetric background, $M\omega_I^{(\ell)} \approx 0.09$ (nearly uniform in ℓ), so $c(\ell) \propto 1/\ell(\ell+1)$ decreases. That is, there is a quantitative discrepancy between the D1 gradient flow (diffusive decay rate with $\ell(\ell+1)$ dependence) and the GR QNMs (decay rate $\sim 1/(4M)$ nearly uniform in ℓ). Absorbing this discrepancy is the role of the EFT corrections (higher-order terms in equation (6.9)), which are systematically incorporated as ℓ-dependence of c_2.

Physical origin of the discrepancy: photon ring dominance vs. horizon diffusion. This discrepancy is essential and reflects the fact that the ringdown (relaxation toward the Kerr attractor) of multipole perturbations is governed not by pure diffusion on the horizon but by **wave scattering with an external potential barrier (the photon ring at $r = 3M$)**. In the eikonal limit, the QNM frequencies are determined by the orbital frequency Ω_c and Lyapunov exponent λ_L of the photon ring: $\omega_{\ell n} \approx \Omega_c(\ell+1/2) - i(n+1/2)\lambda_L$ (Cardoso et al. [2009]). The imaginary part depends primarily on the overtone number n and λ_L, and the weak ℓ-dependence is a consequence of the photon ring geometry.

Therefore, the D1 gradient flow (equation (6.7)) as a horizon effective theory is merely a **pure diffusion approximation**, and the discrepancy with the ℓ-uniform decay rate exhibited by QNMs is structural.

Limitations of local PDE extensions (lesson from the telegrapher type). As a natural extension, consider the telegrapher-type equation $\tau\,\partial_u^2\rho + \partial_u\rho = v^2\Delta_{S^2}\rho$. For mode solutions $\rho_{\ell m} \propto e^{-s_\ell u}$, in the underdamped regime $4\tau v^2\ell(\ell+1) > 1$,

$$\mathrm{Re}\, s_\ell = \frac{1}{2\tau} \tag{6.11}$$

is constant independent of ℓ, structurally matching the ℓ-uniformity of QNMs (from averaging over $\ell = 2, 3, 4$, $\tau \approx 5.4M$). However, this "success" is a **spurious agreement**: in the underdamped regime, v^2 can take **any value** as long as the lower bound condition $v^2 > 1/(4\tau\cdot 6) \approx 7.7\times 10^{-3}\, M^{-2}$ is met. That is, the spatial coupling term $v^2\Delta_{S^2}\rho$ does not contribute to the decay rate, and the ℓ-uniformity is equivalent to pure relaxation $\partial_u\rho = -(1/2\tau)\rho$ without spatial structure. The fact that v^2 does not affect observables means that this local PDE contains **spurious spatial degrees of freedom**.

This is an important negative result: local PDE extensions on S^2 cannot correctly capture the nonlocal scattering resonance mediated by the photon ring.

Construction of the nonlocal memory kernel. As an alternative to local PDEs, a memory-kernel formulation is adopted that encapsulates the scattering resonance via the photon ring into a nonlocal temporal response on the horizon:

$$\partial_u \rho_{\ell m}(u) + \int_0^u K_\ell(u-u')\,\rho_{\ell m}(u')\,du' = 0$$

The minimal practical ansatz for each ℓ is a single-exponential kernel, using the dominant pole γ_ℓ (QNM decay rate) and an auxiliary pole Λ_ℓ:

$$\boxed{K_\ell(\Delta u) = \gamma_\ell\,\Lambda_\ell\,e^{-(\gamma_\ell+\Lambda_\ell)\Delta u}\,\Theta(\Delta u)} \tag{6.12}$$

By Laplace transform, the pole equation $p + \widetilde{K}_\ell(p) = 0$ becomes $p^2 + (\gamma_\ell + \Lambda_\ell)p + \gamma_\ell\Lambda_\ell = 0$, which exactly reproduces the dominant pole $p = -\gamma_\ell$. For γ_ℓ, an interpolation function connecting low-ℓ QNM data with the eikonal asymptotics is used:

$$\gamma_\ell = \frac{\lambda_L}{2} + \frac{a_1}{\ell+\frac{1}{2}} + \frac{a_2}{(\ell+\frac{1}{2})^2} \tag{6.13}$$

($\lambda_L = 1/(3\sqrt{3}\,M)$ is the Lyapunov exponent of the photon sphere). Fitting to QNM data for $\ell = 2, 3, 4$ determines a_1, a_2, and $\gamma_\ell \to \lambda_L/2$ ($\ell \to \infty$) holds automatically.

Physical interpretation. K_ℓ represents the fact that the local degrees of freedom $\rho_{\ell m}$ on the horizon do not relax immediately but undergo a **delayed response** — passing through the exterior scattering region, being trapped near the photon sphere, and then returning to the horizon. The unstable null orbits near the photon ring ($r = 3M$) project the bulk scattering dynamics onto the horizon as memory.

Relation to EFT higher-order terms. The nonlocal memory kernel and the EFT corrections (the $c_2, c_3, \ldots$ of equation (6.9)) are different descriptions of the same bulk scattering physics. In the low-frequency, short-memory limit, moment-expanding the kernel gives $\widetilde{K}_\ell(p) = k_{\ell,0} - k_{\ell,1}\,p + k_{\ell,2}\,p^2 - \cdots$, and re-expanding the ℓ-dependence as polynomials of Δ_{S^2} recovers the local effective equations of equation (6.9). Therefore, the EFT coefficient sequence is the local derivative expansion (low-energy approximation) of the memory kernel, and the complete kernel description is its "parent theory."

[Closure A] $c(\ell = 2) \approx 0.06M$ is positioned as the benchmark value (reference) for Closure A. Closure A gives the best approximation for $\ell = 2$ (the lowest dynamical mode), and the contribution of EFT corrections increases as ℓ grows.

Determination of the EFT second-order correction term c_2. The c_2 of equation (6.9) is defined as the cubic self-interaction coefficient that appears when projecting the source term of the second-order Teukolsky equation onto the horizon/screen and then onto the (ℓ, m) modes of $\rho_{\mathbb{C}}$. The concrete procedure is: (i) solve the first-order master mode $X^{(1)}_{\ell m}$, (ii) construct the second-order source $S^{(2)}[X^{(1)}, X^{(1)}]$, (iii) project onto spin-weighted spherical harmonics to obtain the effective

source $S^{(2)}_{\ell m}$, (iv) identify the component corresponding to $|\rho_{\mathbb{C}}|^2\rho_{\mathbb{C}}$ as c_2 (Campanelli & Lousto [1999]; Loutrel et al. [2023]).

ℓ-dependence of c_2. While c_1 has the dependence $c(\ell) \propto 1/\ell(\ell+1)$, c_2 is determined through the mode coupling rule $(\ell_1, m_1) \otimes (\ell_2, m_2) \to (\ell, m)$ of the second-order source, so from the behavior of the 3-j symbols, $c_2(\ell) \sim \chi_2/[\ell(\ell+1)]^p$ $(1 < p < 2)$ and decays. That is, c_2 carries higher-order angular structure than c_1 and is suppressed at high ℓ.

Extension of the validity range. The main effect of c_2 is a self-interaction correction in the intermediate-amplitude regime. Substituting the first-order solution $\rho^{(1)}_{\mathbb{C}} \sim \varepsilon\, e^{-\omega_I u}$, the c_2 term enters as $\sim c_2\varepsilon^3 e^{-3\omega_I u}$, so the relative correction is $O(c_2\varepsilon^2)$. In the late ringdown ($\varepsilon^2 \ll c_1/c_2$), c_1 dominates, and the improvement from Closure A is confined primarily to the early ringdown regime of relatively large amplitude.

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§7 Determination of the Flux Coefficients

7.1 Three Conservation Laws → Three Coefficients

Remark (preconditions). The following discussion is entirely predicated on **effective Closure A [Closure A]** and applies only within its domain of validity.

Three independent global conservation laws provide constraint conditions for the three undetermined coefficients α, β, γ of the flux source term (equation (6.6)):

$$\begin{cases} \text{(A)} & \dfrac{dM_B}{du} = -\dfrac{1}{4\pi}\displaystyle\int_{S^2} |N_{ab}|^2\, d\sigma \quad \text{(Bondi mass loss formula)} \\ \text{(B)} & \dfrac{dA}{dv} \geq 0 \quad \text{(area increase theorem)} \\ \text{(C)} & \dfrac{dJ}{du} = -\dfrac{1}{4\pi}\displaystyle\int_{S^2} \mathrm{Im}(\bar{N}\eth N)\, d\sigma \quad \text{(angular momentum flux)} \end{cases} \tag{7.1}$$

7.2 Translation into $\rho_{\mathbb{C}}$ Language

Relation between the news function and $\dot{\rho}_{\mathbb{C}}$. [Closure A] Within the domain of validity of Closure A, the following approximate correspondence holds:

$$\boxed{\dot{\rho}_{\mathbb{C}} \approx \mathcal{N}_\rho, \qquad \mathcal{N}_\rho := 2\eth\Psi_3\big|_{\text{Closure A}}} \tag{7.2}$$

$\mathcal{N}_\rho$ is the "effective news function" in $\rho_{\mathbb{C}}$ language, and it has a proportionality relation to the exact news function N only in the near-Petrov-type-D, small-shear regime.

7.3 Determination of Each Coefficient

Determination of α (from the Bondi mass loss):

$$\boxed{\alpha = \frac{1}{4\pi} \cdot C_\alpha} \tag{7.3}$$

C_α is fixed by standard normalization from consistency in the Schwarzschild limit as $C_\alpha = 1$.

Determination of β (from the area increase theorem): From the Ashtekar–Krishnan DH flux formula,

$$\boxed{\beta > 0} \tag{7.4}$$

is obtained as a necessary condition, and $\beta = C_\beta/(8\pi)$ is fixed.

Determination of γ (from the angular momentum flux):

$$\boxed{\gamma = \frac{1}{4\pi} \cdot C_\gamma} \tag{7.5}$$

7.4 Summary of the Simultaneous System

$$\boxed{\begin{cases} \dfrac{dM}{du} = -\alpha \displaystyle\int_{S^2} |\mathcal{N}_\rho|^2 \, d\sigma & \Longrightarrow \quad \alpha = \frac{1}{4\pi} C_\alpha \\ \dfrac{dA}{dv} = \beta \displaystyle\int_{S^2} |\nabla_{S^2} \rho_{\mathbb{C}}|^2 \, d\sigma + (\text{matter terms}) & \Longrightarrow \quad \beta = \frac{1}{8\pi} C_\beta \\ \dfrac{dJ}{du} = \gamma \displaystyle\int_{S^2} \rho_{\mathbb{C}} \, \eth \rho_{\mathbb{C}} \, d\sigma & \Longrightarrow \quad \gamma = \frac{1}{4\pi} C_\gamma \end{cases}} \tag{7.6}$$

Chain linking the scaling factors and closure coefficient. Once $c(\ell = 2)$ is quantified from QNM data (equation (6.10)), numerical estimates of $C_\alpha, C_\beta, C_\gamma$ become possible through the correspondence with the news function (equation (7.2)).

Concrete estimate of C_α. Writing the effective news as $\mathcal{N}_\rho = \beta_N \, \partial_u \rho_{\mathbb{C}}$ (β_N is a proportionality constant absorbing the normalization difference between the horizon variable $\rho_{\mathbb{C}}$ and the Bondi news N at null infinity), and substituting the $\ell = 2$ ringdown $\rho_{2m}(u) = \rho_{2m}(0)\, e^{-\omega_I u}$ and comparing with the first equation of (7.6),

$$C_\alpha = \frac{\alpha \, \beta_N^2 \, \omega_I}{2} \tag{7.6a}$$

is obtained. Adopting $\beta_N = 1$ (the convention $\mathcal{N}_\rho := \partial_u \rho_{\mathbb{C}}$) gives $C_\alpha \approx 0.045 \, \alpha/M$.

Remark (structure of the matching). The proportionality constant β_N is not universal. On the Schwarzschild background, $\delta\Psi_2|_{\mathcal{H}}$ and $\Psi_4^0|_{\mathcal{J}^+}$ are both obtained as different projections of the same Regge–Wheeler / Zerilli master variable $\Phi_{\ell m}$, but since the differential operators acting on each differ, their relation is not a simple constant proportionality but is given by an ℓ-dependent, frequency-dependent **transfer operator** $\mathcal{T}_\ell$:

$$\Psi^0_{4,\ell m} = \mathcal{T}_\ell \, \delta\Psi_{2,\ell m}$$

Therefore, β_N should strictly be written as $\beta_N^{(\ell)}(\omega)$, and $\beta_N = 1$ is merely a **convention choice** — "normalizing $\rho_{\mathbb{C}}$ in the same convention as the (ℓ, m) components of the shear C_{AB}." In this paper, we adopt $\beta_N = 1$.

Determination of C_β. The area growth rate $dA/dv = 2\beta\, E[\rho_{\mathbb{C}}]$ (equation (7.9)) is compared with the horizon version of the Raychaudhuri equation. Normalizing $E[\rho_{\mathbb{C}}]$ in the same convention as the horizon shear flux $\int_{S_v} |\sigma^{(1)}|^2\, dA$, C_β is absorbed to 1 under this convention. That is, C_β, like C_α, is primarily a normalization issue.

Determination of C_γ. The C_γ in the angular momentum flux $dJ/du = \gamma \int_{S^2} \rho_{\mathbb{C}}\, \eth \rho_{\mathbb{C}}\, d\sigma$ is fixed by matching the edth operation of $\rho_{\mathbb{C}}$ to the standard normalization of the horizon angular momentum flux.

Normalization structure. Under the redefinition $\rho_{\mathbb{C}} \to a\, \rho_{\mathbb{C}}$, all quadratic flux quantities scale as a^2, so $C_\alpha, C_\beta, C_\gamma$ are not independent physical constants but consist of one normalization freedom and two dimensionless ratios:

$$\boxed{C_\alpha : C_\beta : C_\gamma \quad \text{(only the ratios are physical)}} \tag{7.6b}$$

Under the convention $\beta_N = 1$, once $C_\alpha \approx 0.045\, \alpha/M$ is determined, C_β/C_α and C_γ/C_α are determined subordinately from the matching of each flux. This completes the **chain of determination**: closure coefficient (§6.3) $\to$ flux coefficients (§7.3) $\to$ scaling factors (§7.4).

Re-expression of the area increase theorem in $\rho_{\mathbb{C}}$ language. Under $\beta > 0$, the area increase theorem is automatically satisfied, with equality holding only when $\nabla_{S^2} \rho_{\mathbb{C}} = 0$ (spherically symmetric configuration = Schwarzschild limit).

7.5 Schwarzschild Verification

Schwarzschild limit ($a = 0$): $\Psi_2 = -1/(8M^2)$ (real constant), $\rho_{\mathbb{C}} = 1/(4M^2)$, $\Omega = 0$, $\sigma = 0$. All three conservation laws are automatically satisfied, and no constraints contradicting α, β, γ arise.

7.6 Verification near Kerr

Kerr solution: $\Psi_2^{\text{Kerr}} = -M/(r_+ + ia\cos\theta)^3$. In the small-spin limit ($a \ll M$),

$$\rho_{\mathbb{C}} \approx \frac{1}{4M^2} - \frac{3ia\cos\theta}{8M^3} + O(a^2)$$

The $\ell = 0$ component is the mass (Schwarzschild value), and the $\ell = 1$ component is purely imaginary, encoding the angular momentum $J = Ma$.

7.7 Mode Decomposition

Energy flux: $\int_{S^2} |\mathcal{N}_\rho|^2\, d\sigma = \sum_{\ell,m} |\dot{\rho}_{\ell m}|^2 + O(\text{closure corrections})$ (Parseval's identity)

Area flux: $\int_{S^2} |\nabla_{S^2} \rho_{\mathbb{C}}|^2\, d\sigma = \sum_{\ell,m} \ell(\ell+1)\, |\rho_{\ell m}|^2$

Angular momentum flux: Since the edth operator does not change ℓ, the coupling between different m describes the transfer of angular momentum.

7.8 Connection between the Gradient Flow and the Potential

The D1 gradient flow of §6 and the W_{GR} potential of §3.5 are nontrivially connected through the stationarity condition. In this section, this connection is made explicit, and the design character of W_{GR} is partially reassessed.

Agreement of the stationarity conditions. **[Closure A]** The stationarity condition $\partial_u \rho_{\mathbb{C}} = 0$ of the D1 gradient flow (equation (6.7)) gives

$$\Delta_{S^2} \rho_{\mathbb{C},\geq 2} = -S_{\mathrm{flux}} \tag{7.7}$$

This has the same structure as the common stationarity condition of the HVP (Part II equation (3.5)):

$$\Delta_{S^2} \varrho = \frac{\partial V}{\partial \varrho}$$

Rewriting in the Ψ_2 variable under $\rho_{\mathbb{C}} = -2\Psi_2$, the stationarity condition is $\Delta_{S^2}\Psi_2 = \partial W/\partial \bar{\Psi}_2$, and comparing with (7.7),

$$\boxed{\left.\frac{\partial W}{\partial \bar{\Psi}_2}\right|_{\mathrm{stationary}} = -S_{\mathrm{flux}}(\Psi_2)} \tag{7.8}$$

is obtained. That is, **the gradient of W equals the flux source term at stationarity**.

Partial determination of W_{GR}. The structure of the flux source term $S_{\mathrm{flux}} = \alpha|\sigma|^2 + \beta\, T_{\ell\ell} + \gamma\, T_{\ell\varphi} + \nabla \cdot V$ (equation (6.6)) and the coefficients α, β, γ determined from conservation laws in §7.3–§7.4 partially fix the gradient of W_{GR} through (7.8). Complete reconstruction of W requires a closed expression for the Ψ_2 dependence of S_{flux}, which depends on Closure A; however, the stationarity condition (7.8) provides a nontrivial constraint on the structure of W.

Reassessment of the principled basis for the 4 requirements. Through the agreement of the stationarity conditions, the basis for the 4 requirements of §3.5 is reassessed as follows:

Requirement	Content	Basis	Character
Requirement 1 (Kerr critical point)	Kerr is a critical point of W	Constrained minimum of the Dirichlet energy $E[\rho] = \frac{1}{2}\int \|\nabla\rho\|^2 \to$ only $\ell = 0, 1$ (smoothest nontrivial configuration)	**Principled**
Requirement 2 (CR compatibility)	W is compatible with the CR structure	Structurally automatic from the complex structure of Ψ_2	**Structural**
Requirement 3 (Schwarzschild stability)	Schwarzschild is a stable critical point	Absolute minimum of the Dirichlet energy ($\ell = 0$, constant configuration)	**Principled**

Requirement	Content	Basis	Character
Requirement 4 (cross-focusing compatibility)	Reproduction of the cross-focusing equation	$\delta\lambda$ Euler–Lagrange equation of the extended boundary action S_{bdy} (equations (7.11)–(7.12))	**Proposed** (conditional)

[Closure A] All 4 requirements of W_{GR} possess principled justification. Requirements 1–3 follow from the Dirichlet energy principle and the complex structure of Ψ_2; Requirement 4 (cross-focusing) follows from the $\delta\lambda$ Euler–Lagrange equation of the extended boundary action S_{tot} (equation (7.13)). Conditions (A)–(C) are verified (below).

Variational derivation of Requirement 4: Proposal of the extended boundary action. In the standard IH covariant phase space, the boundary symplectic structure is described in Chern–Simons / BF type (Ashtekar & Krishnan [2004]), and the cross-focusing is treated as a constraint of Einstein equation origin (a geometric relation of Gauss–Codazzi type) — it does not ordinarily appear as an Euler–Lagrange equation of the presymplectic structure. To go beyond this limitation, an **extended boundary action** is proposed with the cross-section metric q_{AB}, rotation 1-form $\Omega^{(\mathrm{rot})}$, and Lagrange multiplier λ as independent boundary variables:

$$S_{\mathrm{bdy}}[q,\Omega,\lambda] = \int_{\mathcal{H}} \left[\Omega^{(\mathrm{rot})} \wedge d\epsilon[q] \;+\; \lambda\left({}^{(2)}R[q] - 2|\Omega^{(\mathrm{rot})}|_q^2 + 4\rho_R\right)\epsilon[q]\right] \tag{7.11}$$

where $\epsilon[q]$ and ${}^{(2)}R[q]$ are geometrically determined from q_{AB}, and $\rho_R = \mathrm{Re}\,\Psi_2$ is the horizon curvature variable. The first term is a 1-form that provides an action-level origin for the IH presymplectic structure $\Omega_{\mathrm{IH}} \sim \int_H \delta\Omega \wedge \delta\epsilon$; the second term embeds the cross-focusing constraint via a Lagrange multiplier.

The $\delta\lambda$ variation directly reproduces the cross-focusing equation:

$$\delta_\lambda S_{\mathrm{bdy}} = 0 \quad\Longrightarrow\quad {}^{(2)}R[q] - 2|\Omega^{(\mathrm{rot})}|_q^2 + 4\,\mathrm{Re}\,\Psi_2 = 0 \tag{7.12}$$

This upgrades Requirement 4 from "external input" to "constraint of the boundary action principle." The following three consistency conditions are verified.

(A) Incorporation of ρ_R into the variational system. As long as ρ_R is treated as an externally fixed function, (7.12) remains merely "having imposed the constraint." To incorporate it as a dynamical variable, one augments the Ψ_2 boundary theory (§3) and extends the total action to

$$S_{\mathrm{tot}} = S_{\mathrm{bdy}}[q,\Omega,\lambda;\rho_R] + S_\Psi[\Psi_2,q], \qquad S_\Psi = \int_{\mathcal{H}} \left(\kappa_\Psi\,|D\Psi_2|_q^2 + W(\Psi_2,\bar\Psi_2)\right)\epsilon[q] \tag{7.13}$$

(with $\rho_R = \frac{1}{2}(\Psi_2+\bar\Psi_2)$). The stationarity condition for Ψ_2 becomes $-\kappa_\Psi\,\Delta_{S^2}\Psi_2 + \partial W/\partial\bar\Psi_2 + 2\lambda = 0$, and λ serves as a conjugate variable linking the Ψ_2 sector and the cross-focusing sector. Identifying W with W_{GR} of §3.5, ρ_R becomes a consistent dynamical variable appearing in both S_{bdy} and S_Ψ. □

(B) Consistency with the presymplectic structure. The first term of (7.11), $\int_{\mathcal{H}} \Omega^{(\mathrm{rot})} \wedge d\epsilon[q]$, generates the pair $\delta\Omega$ and $\delta\epsilon$ as fundamental conjugate variables, and the principal component of the presymplectic 2-form coincides with the standard IH $\delta\Omega \wedge \delta\epsilon$ (Ashtekar et al. [1999]). The second term (constraint term) contains no time derivatives and serves to cut out the constraint surface rather than changing the "shape" of the presymplectic structure. As long as dynamical terms involving u-derivatives are not introduced for λ, no additional conjugate pairs arise, maintaining compatibility with the minimal structure of the standard IH. □

(C) Gauge invariance. Under diffeomorphisms of the cross-section S^2, q_{AB}, $\epsilon[q]$, and ${}^{(2)}R[q]$ are metric-derived geometric quantities, and each term of (7.11) is integrated as a cross-section scalar density, hence well-defined under reparametrization. Under U(1) gauge transformations (rescaling of the null normal $\Omega \to \Omega + d\chi$), $|\Omega|_q^2$ is not gauge-invariant, so the necessary and sufficient condition is to use $\Omega^{(\mathrm{rot})}$ as the coexact component of the Hodge decomposition (the gauge-invariant rotational part). λ is treated as a cross-section scalar, U(1)-neutral multiplier field, since the constraint is a cross-section scalar. □

Declaration of claim level. Through the verification of conditions (A)–(C), the status of W_{GR} is upgraded from "complete design" to "**all 4 requirements possess principled justification**." Requirements 1–3 follow from the Dirichlet energy principle, and Requirement 4 follows from the $\delta\lambda$ Euler–Lagrange equation of the extended boundary action S_{tot} (equation (7.13)). However, this reassessment is predicated on effective Closure A, and does not hold outside the validity range of Closure A (§6.3).

Connection between the Dirichlet energy and area increase. Rewriting the area flux formula (equation (7.6), second equation) of §7.4 in terms of the Dirichlet energy $E[\rho] := \frac{1}{2}\int_{S^2} |\nabla_{S^2}\rho_{\mathbb{C}}|^2\, d\sigma$,

$$\boxed{\frac{dA}{dv} = 2\beta\, E[\rho_{\mathbb{C}}] + (\text{matter terms})} \tag{7.9}$$

is obtained. That is, **the rate of area increase is proportional to the Dirichlet energy (the magnitude of inhomogeneity of the curvature density)**. Meanwhile, under the D1 gradient flow, the Dirichlet energy **[Closure A]**

$$\frac{dE}{du} = -\int_{S^2} |\Delta_{S^2}\rho_{\mathbb{C}}|^2\, d\sigma \le 0 \tag{7.10}$$

monotonically decreases (Lyapunov function). Combining (7.9) and (7.10) yields the following picture:

1. The larger $E[\rho]$ (more inhomogeneous), the faster the area increases
2. Since $E[\rho]$ always decreases, the rate of area increase progressively slows
3. $E[\rho] = 0$ ($\nabla\rho = 0$, constant configuration = Schwarzschild) is when area increase stops
4. For Kerr ($\ell = 0, 1$ only), $E_{\mathrm{Kerr}} = \sum_m 2\,|\rho_{1m}|^2 \propto J^2$, and the rate of area increase is proportional to the square of the angular momentum

[Closure A] This connection shows that the decrease of Dirichlet energy (equation (7.10), a structural consequence of the gradient flow) and area increase (equation (7.9), a consequence of the Einstein equations) are **quantitatively linked** in $\rho_{\mathbb{C}}$ language. The irreversibility of both has the same geometric origin — the vanishing of inhomogeneity in the curvature density. However, since the area increase theorem itself presupposes the Einstein equations (§7.1 equation (7.1)(B)), this

connection is not a "derivation" of area increase from the Dirichlet principle, but a confirmation of the **quantitative correspondence** between the two in ρ language.

7.9 Completeness and Remaining Tasks

Element	Status	Remaining tasks
Identification of the 3 conservation laws	**Complete**	—
Structural determination of α	**Complete**	$C_\alpha \approx 0.045\,\alpha/M$ ($\beta_N = 1$ convention)
Proof of $\beta > 0$	**Complete**	—
Structural determination of β, γ	**Complete**	C_β, C_γ are physical as ratios $C_\beta/C_\alpha, C_\gamma/C_\alpha$ (equation (7.6b))
Schwarzschild consistency	**Verified**	—
Near-Kerr consistency	**Partial**	Quantitative comparison with Teukolsky
News–$\dot{\rho}_{\mathbb{C}}$ correspondence **[Closure A]**	**Conditional**	Extension to non-perturbative regime
Identification of $dA/dv = 2\beta\, E[\rho]$ **[Closure A]**	**Verified**	Quantification of matter term contributions
Screen projection justification of Closure A (§6.2)	**Established**	Form is natural from S^2 symmetry; coefficient from bulk matching
EFT interpretation of Closure A (equation (6.8))	**Formulated**	Determination procedure for c_2 identified (§6.3)
QNM quantification of the closure coefficient c (equation (6.10))	**Formulated**	Unification across all ℓ via the nonlocal memory kernel (equations (6.12)–(6.13))
Normalization structure of $C_\alpha, C_\beta, C_\gamma$	**Established**	Only ratios are physical (equation (7.6b))
Limitation of the telegrapher type (equation (6.11))	**Negative result**	Superseded by the nonlocal memory kernel (equation (6.12))
Nonlocal memory kernel (equations (6.12)–(6.13))	**Constructed**	Numerical fitting of a_1, a_2
Second-order Teukolsky projection of c_2	**Procedure identified**	Execution of the explicit projection calculation
Transfer operator structure of β_N	**Identified**	Concrete computation from the RW/Z master variable
Variational derivation of Requirement 4 (cross-focusing)	**Verified** (equations (7.11)–(7.13))	—

§8 Characterization: Positioning the Einstein Boundary Constraints within the HVP Framework (Hello World II)

In this section, we show that the fundamental relations of GR fall within a natural class of the HVP and Ψ_2 boundary theory. The purpose here is not to "derive" the Einstein equations, but to confirm that the Einstein boundary constraints are **characterized** as stationarity conditions of the HVP.

8.1 Characterization of the Cross-Focusing Equation

In the Ψ_2 boundary theory (§3), the stationarity condition of V_{cf} (equation (3.12)) is

$$\frac{\partial V_{\mathrm{cf}}}{\partial P} = 2\gamma \left(P + \frac{1}{4}\,{}^{(2)}R - \frac{1}{2}|\Omega|^2 \right) = 0 \tag{8.1}$$

Under $\gamma \neq 0$, the cross-focusing equation

$$\mathrm{Re}\,\Psi_2 = -\frac{1}{4}\,{}^{(2)}R + \frac{1}{2}|\Omega^{(\mathrm{rot})}|^2 \tag{8.2}$$

is reproduced. □

Remark (honest recognition of circularity). This characterization contains a circular element: the cross-focusing equation is used as input in the construction of V_{cf} (equation (3.12)), and the "output" from the stationarity condition is merely a faithful recovery of the input. What is achieved here is the confirmation that the Einstein boundary constraints are **consistently embedded** within the variational structure of the HVP.

Schwarzschild verification. With $|\Omega| = 0$ and ${}^{(2)}R = 1/(2M^2)$, we obtain $\mathrm{Re}\,\Psi_2 = -1/(8M^2)$. On the other hand, $\Psi_2\big|_{\mathrm{Schw}} = -1/(8M^2)$, which agrees exactly.

8.2 Characterization of the Curvature Expression for the Rotation 1-Form

From the universality of local connection forms (Part I Theorem 2.6) and the physical identification of §4,

$$d\Omega^{(\mathrm{rot})} = -2\,(\mathrm{Im}\,\Psi_2)\,\epsilon_S \tag{8.3}$$

is obtained, where $\epsilon_S = 2\,\omega_{FS}$ is the area 2-form of the cross-section. This derivation follows directly from the combination of $\rho_{\mathbb{C}} = -2\Psi_2$ and $F = \varrho\,\omega_{FS}$, and holds **without circularity**.

8.3 Summary of the Gravitational-Side Characterization

Result	Basis within HVP	Status
Cross-focusing equation	Stationarity condition of W (V_{cf} term)	Characterization (circular element present)
$d\Omega^{(\mathrm{rot})} = -2\,(\mathrm{Im}\,\Psi_2)\,\epsilon_S$	Universality of local connection forms + $\rho_{\mathbb{C}} = -2\Psi_2$	Derivation (no circularity)
Schwarzschild verification	$\rho_{\mathbb{C}} = 1/(4M^2)$	Verified

Result	Basis within HVP	Status
Kerr attractor property	D1 gradient flow = effective closure of Ψ_2 evolution **[Closure A]**	Established
Structure of W_{GR} (all 4 requirements)	Dirichlet principle + $\delta\lambda$ EL of S_{tot} (equation (7.13)) **[Closure A]**	All requirements with principled basis (§7.8)
$dA/dv \propto E[\rho]$	Area flux = Dirichlet energy **[Closure A]**	Quantitative correspondence (§7.8)
Screen projection justification of Closure A	Minimal local effective term from S^2 geometry, spin-weight, and rotational symmetry	Established (§6.2)
EFT interpretation of Closure A	Integration out of $\Psi_3, \Psi_4 \to$ effective action (equation (6.8))	Formulated (§6.3)
QNM consistency of the closure coefficient	$c(\ell) = \omega_I r_+^2/\ell(\ell+1)$ (equation (6.10))	Quantified (§6.3)
Nonlocal memory kernel	Horizon projection of photon ring scattering (equations (6.12)–(6.13))	Constructed (§6.3)
Second-order Teukolsky projection of c_2	Horizon EFT projection of cubic self-interaction	Procedure identified (§6.3)
Normalization structure of $C_\alpha : C_\beta : C_\gamma$	Only ratios are physical (equation (7.6b))	Established (§7.4)
Variational derivation of cross-focusing	$\delta\lambda$ EL of S_{tot} (equations (7.11)–(7.13))	Verified (§7.8)

Structural correspondence with the quantum side. The characterization of the cross-focusing equation arises as part of the **same type** of stationarity condition $\Delta_{S^2}\Psi_2 = \partial W/\partial\bar{\Psi}_2$ as the Berry curvature condition $\varrho = 1$ on the quantum side (the type of Part II equation (3.5)).

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§9 Minimal GR Compatibility Set

The results of §2–§8 are condensed, and the minimal complete set ensuring compatibility of the $\rho_{\mathbb{C}} = -2\Psi_2$ framework with GR is summarized below.

Minimal GR Compatibility Set.

(a) Gauge transformation and curvature invariance. Under the boost freedom $\ell \to f\ell$ of the null tetrad, the H'aj'iček 1-form transforms as $\Omega \to \Omega + D(\ln f)$, and the curvature 2-form $F = d\Omega$ is invariant (§4.2).

(b) Structure equation.

$$d\Omega^{(\mathrm{rot})} = -2\,\mathrm{Im}\,\Psi_2\,\epsilon_S \tag{9.1}$$

(c) Cross-focusing equation.

$$\mathrm{Re}\,\Psi_2 = -\frac{1}{4}\,{}^{(2)}R + \frac{1}{2}|\Omega^{(\mathrm{rot})}|^2 \tag{9.2}$$

(d) Schwarzschild limit verification.

$$\rho_{\mathbb{C}}\big|_{\mathrm{Schw}} = -2\Psi_2\big|_{\mathrm{Schw}} = \frac{1}{4M^2}$$

Completeness claim. The above 4 conditions constitute the **minimal complete set** ensuring that $\rho_{\mathbb{C}} = -2\Psi_2$ in the present framework is compatible with the Einstein equations on the NEH/IH. Condition (a) guarantees the well-definedness of the connection definition; conditions (b)–(c) guarantee the two independent components of the Einstein boundary constraints; and condition (d) guarantees quantitative agreement on a nontrivial solution.

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§10 The Ashtekar × Damour Hybrid Framework

The $\rho_{\mathbb{C}}$ equations bridge two major existing frameworks for the dynamics of black hole horizons.

Ashtekar side (Dynamical Horizon). The Ashtekar–Krishnan DH theory takes the geometric data $(q_{ab}, \Omega_a, \sigma_{ab})$ as fundamental variables and rigorously derives the area increase theorem and flux laws. $\rho_{\mathbb{C}} = K + iB$ is directly constructed from these geometric data.

Damour side (Membrane Paradigm). The Damour membrane paradigm treats the horizon as a viscous fluid membrane and provides Navier–Stokes-type equations. The diffusion equation for $\rho_{\mathbb{C}}$ is structurally analogous to this fluid-dynamical description.

Bridging role of the $\rho_{\mathbb{C}}$ equations. The $\rho_{\mathbb{C}}$ equations "encapsulate" Ashtekar's rigorous geometric data into the form of Damour's fluid-dynamical evolution equations. The complex curvature density $\rho_{\mathbb{C}} = K + iB$ structurally matches the "curvature + vorticity" package in the membrane paradigm of Damour [1979]. The diffusion equation for $\rho_{\mathbb{C}}$ (the D1 gradient flow) structurally matches the $\ell \geq 2$ sector of Damour's viscous membrane equations.

10.1 Equation Correspondence Table in ϱ Language

Equation	Correspondence in ϱ language
Einstein equations (NEH projection)	$\rho_{\mathrm{rot}} \propto \mathrm{Im}\,\Psi_2$; the Bianchi identities provide the evolution equation
Maxwell equations ($dF = 0$)	$c_1 = \langle\varrho\rangle$ is spacetime-invariant
Maxwell equations ($d{*}F = {*}J$)	PDE determining the spacetime dependence of each multipole mode $\rho_{\ell m}$
Schr"odinger equation	Fubini–Study transport of states on the background $\varrho = 1$

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§11 Minimal Toy Model

To illustrate the concrete behavior of the theory, a minimal truncation with axial symmetry is constructed.

Definition of the model. ρ_{rot} is truncated to $\ell = 0, 1, 2$ modes:

$$\rho_{\mathrm{rot}}(u, \theta) = a_0(u) + a_1(u)\cos\theta + a_2(u)\, P_2(\cos\theta) \tag{11.1}$$

[Closure A] Under the D1 gradient flow,

$$\boxed{\dot{a}_0 = s_0(u), \qquad \dot{a}_1 = s_1(u), \qquad \dot{a}_2 = -6\, a_2 + s_2(u)} \tag{11.2}$$

Isolated system case ($s_\ell = 0$). The quadrupole deformation a_2 relaxes exponentially, and only the Kerr dipole ($a_0 + a_1\cos\theta$) remains. This demonstrates the fundamental behavior of the theory — decay of higher multipoles and convergence to the Kerr attractor — in the minimal setting.

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§12 The Unified Variable Ψ: Prospects

Remark (positioning). The unified variable Ψ of this section is a **prospective construction** that geometrically clarifies the foundation of the Ψ_2 boundary theory (§3); it is not a prerequisite for the main results (C7–C9) of Part III. Ψ_2 is the protagonist of this part, and Ψ is positioned as the underlying potential variable for future developments.

12.1 Definition of Ψ

Definition (Unified variable Ψ). $\Psi(u, t, \theta, \phi)$ is defined as a scalar field on the null Kerr screen $S \simeq S^2$, serving as the null hypersurface height function. Its relation to the H'aj'iček 1-form is

$$\Omega_a = \Omega_a^{(0)} + \lambda\, \varepsilon_a{}^b D_b \Psi \tag{12.1}$$

Ψ is uniquely fixed as the "vorticity potential" of the H'aj'iček 1-form.

12.2 Chain of Definitions

From Ψ alone, all null-geometric quantities are generated hierarchically.

Level	Geometric quantity	Definition from Ψ	Physical meaning
0	Height Ψ	(fundamental variable)	Displacement of the null surface
1	Tilt $D_a\Psi$	Spherical gradient of Ψ	Inclination of the null congruence
2	Rotation $\varepsilon_a{}^b D_b\Psi$	Vorticity of Ψ	H'aj'iček 1-form
3	Distortion $D^{\mathrm{TF}}_{ab}\Psi$	Trace-free Hessian	Shear tensor σ_{ab}
4	Expansion $\Delta\Psi$	Spherical Laplacian	Relation to expansion θ

Level	Geometric quantity	Definition from Ψ	Physical meaning
5	Acceleration $\ell^\mu \nabla_\mu \Psi$	Null directional derivative	Surface gravity κ

Generation of curvature. From the chain of definitions,

$$F_{ab} = 2D_{[a}\Omega_{b]} = \lambda(\Delta\Psi)\,\varepsilon_{ab} \tag{12.2}$$

The curvatures in all three domains are expressed in a unified manner as the spherical Laplacian of a scalar potential on S^2.

Remark (Ψ as a K"ahler potential). Equation (12.2) can be rewritten in the language of K"ahler geometry on $\mathbb{CP}^1$ as $F = i\,\partial\bar{\partial}K$ (see Part I §2.3). That is, Ψ is none other than the **K"ahler potential** K_{GR} of the gravitational sector. In the Berry sector, $K = \log(1+|z|^2)$ gives the constant curvature $\rho = 1$, while in the gravitational sector, $\Psi = K_{\mathrm{GR}}$ gives the variable curvature $\rho = \kappa(\zeta)$. The chain of definitions in the table above is interpreted as the standard differential hierarchy from the K"ahler potential on $\mathbb{CP}^1$ (gradient → vorticity → Hessian → Laplacian).

Relation between Ψ and Ψ_2. From Ψ, the 1-form Ω_a is generated, and $\lambda\,\Delta\Psi = -4\,\mathrm{Im}\,\Psi_2$ is obtained. The imaginary part of Ψ_2 is encoded by the spherical Laplacian of Ψ, and the real part is determined through the cross-focusing equation. Ψ is a potential variable "one level before" Ψ_2.

12.3 Automatic Emergence of the Quadrupole

> **Proposition 12.3.** The trace-free Hessian $\sigma_{ab} = D^{\mathrm{TF}}_{ab}\Psi$ of Ψ contains only modes with $\ell \geq 2$. The $\ell = 0, 1$ components are identically zero.

Proof. $\ell = 0$: $D_a D_b Y_{00} = 0$. $\ell = 1$: $D^{\mathrm{TF}}_{ab} Y_{1m} = 0$ (by the symmetry of Killing vectors). For $\ell \geq 2$, nontrivial STF tensors are generated. □

The fundamental property of GR that gravitational waves are radiated only in modes with $\ell \geq 2$ (quadrupole and higher) is directly derived from the differential-geometric structure of Ψ.

12.4 Connection with BMS Symmetry

The Bondi news tensor $N_{ab} = D^{\mathrm{TF}}_{ab}\dot{\Psi}$ has a natural correspondence with the supertranslation parameter.

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§13 Summary

The following has been accomplished in this part.

1. **Fundamental obstacle to a variational problem for ϱ alone** (§2): Three obstacles were identified — nonlocality, Cauchy–Riemann coupling, and the zero-point problem.
2. **Construction of the Ψ_2 boundary theory (C7)** (§3): Ψ_2 was promoted to the fundamental variable, and the potential W was **constructed** (designed) from physical requirements. However, as shown in §7.8, through consistency with the stationarity condition of the D1 gradient flow, 3 out of 4 requirements possess principled justification from the Dirichlet energy

principle, and W_{GR} is upgraded from "complete design" to "**partial principled basis via the Dirichlet principle + completion via Einstein input**" [**Closure A**]. It was confirmed that the Einstein boundary constraints are **characterized** as stationarity conditions of the HVP (§8). The characterization of the cross-focusing equation contains a circular element, but the curvature expression for the rotation 1-form is derived without circularity.

3. **Effective closure and identification of the closure problem (C8)** (§6): The validity conditions of effective Closure A ($-\eth\Psi_3 \approx c\,\Delta_{S^2}\rho_{\mathbb{C}}$) — near Kerr, small shear, quasi-stationary evolution — were made explicit, and the closure problem was positioned as the largest open problem of the present theory. Furthermore, Closure A was formulated as an effective action obtained by integrating out Ψ_3, Ψ_4 (equation (6.8)), and a path toward systematic higher-order corrections (equation (6.9)) was indicated.

4. **Conditional determination of flux coefficients (C9)** (§7): The structural determination of α, β, γ was carried out from three physical conservation laws, and $\beta > 0$ was proved from the area increase theorem. The numerical determination of the scaling factors $C_\alpha, C_\beta, C_\gamma$ remains an open task.

5. **Minimal GR Compatibility Set** (§9): The 4 conditions necessary and sufficient for GR compatibility of the present framework — gauge consistency, structure equation, cross-focusing, and Schwarzschild verification — were condensed.

6. **Prospects for the unified variable** Ψ (§12): As a potential variable one level before Ψ_2, the chain of definitions and the automatic emergence of the quadrupole were demonstrated.

What Part III has established. The gravitational sector of the HVP is compatible with the Einstein boundary constraints as an **effective theory near Kerr**. However, complete dynamical unification (resolution of the closure problem, principled derivation of the remaining 1 requirement of W_{GR}) has not been achieved, and this will be organized as open problems in Part IV. Ψ_2 is the protagonist of the gravitational sector, and the abstract variable ϱ of Part II is concretized through Ψ_2.

Structural Unification of Gravity, Electromagnetism, and Quantum Mechanics via the Curvature Density ρ

— Characterization of Variational Framework and Boundary Equations Based on a U(1) Connection on $\mathbb{CP}^1$ —

Part IV: Predictions and Diagnostics — Level 3

This paper collection is organized in five parts:

Part	Title	Level	Content
I	Common Language	0–1	Dictionary, curvature density space, topological theorems, geometric quantization
II	Unified Principle (HVP)	2	Formulation of the Holonomy Variational Principle, physical origin, clarification of variational targets
III	GR Consistency Verification	2–3	Characterization of Einstein boundary constraints, Ψ_2 boundary theory
IV	**Predictions and Diagnostics**	**3**	**Horizon spectral classification, Chern number wall, current status**
V	Extension Proposals	3–4	Electromagnetic field, Dirac structure, weak force, 4D unified action

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§1 Introduction

From the framework of structural unification constructed in Parts I–III, we organize what is predicted and what remains unresolved. The purposes of this part are threefold:

1. **Interpretation of the Chern number wall**: Reinterpret the discrepancy between $c_1 = 0$ (gravity) and $c_1 = 1$ (Berry) not as a failure of unification but as a **superselection rule** (C10)
2. **Original predictions from the framework**: Present new diagnostic tools and predictions generated by the ϱ language (C11, C12)
3. **Current status and open problems**: Self-assess the maturity of the theory and organize remaining challenges

Claim/Status list (Part IV).

Claim	Status	Section
Chern number wall = superselection rule (C10)	Structural interpretation	§2
Horizon spectral classification and ϱ-no-hair test (C11)	New proposal	§3
Five theory-specific predictions (C12)	Qualitative predictions	§5

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§2 The Chern Number Wall and the Superselection Rule

2.1 Precise statement of the wall

The discreteness of $c_1 \in \mathbb{Z}$ is a consequence of three independent mathematical facts: (1) the discrete structure of cohomology ($H^2(S^2, \mathbb{Z}) \cong \mathbb{Z}$), (2) the discreteness of homotopy classes of transition functions ($\pi_1(\mathrm{U}(1)) \cong \mathbb{Z}$), and (3) the compactness of U(1); it cannot be changed by continuous deformation of the connection.

Proposition 2.1.

$$\mathcal{M}_\varrho^{(0)} \cap \mathcal{M}_\varrho^{(1)} = \emptyset \quad \text{and} \quad \pi_0(\mathcal{M}_\varrho) \cong \mathbb{Z} \tag{2.1}$$

2.2 Theorems within the framework (proved)

Theorem 2.2a (Impossibility via DH). On a DH, the Hájíček 1-form ω_a is also globally smoothly defined, and Stokes' theorem $\int_S d\omega = 0$ holds without using the NEH condition. Even via a DH, one cannot escape $c_1 = 0$ within the gravitational sector.

Theorem 2.2b (Nonexistence of instantons). Since $M^3 = S^2 \times [0, 1]$ is contractible to S^2, we have $H^2(M^3, \mathbb{Z}) \cong \mathbb{Z}$, and no instanton on a cobordism connecting $c_1 = 0$ and $c_1 = 1$ exists.

2.3 Considerations beyond the framework (unproved)

Mechanism	Status	Remarks
Topological phase transition	Open	Identification of the "gap" in gravity is required
Quantum gravity effects	In principle possible but suppressed	$P \sim e^{-A_H/\ell_P^2}$ at the Planck scale
Extension of the base space	$c_1 \in \mathbb{Z}$ is maintained	Integrality is preserved even upon extension to $\mathbb{CP}^n$

2.4 c_1 as a superselection rule (C10)

The very fact that crossing the wall is difficult carries an important physical message.

The stratification $\mathcal{M}_\varrho = \bigsqcup_n \mathcal{M}_\varrho^{(n)}$ of $\mathcal{M}_\varrho$ possesses the structure of a **superselection rule** in quantum mechanics. c_1 functions as a "superselection quantum number," prohibiting continuous transitions between different sectors.

Coexistence as "different topological sectors within the same framework":

- Gravity ($c_1 = 0$) and quantum ($c_1 = 1$) share the same local dynamical law for ϱ (Part II Eq. (3.5))
- The differences are concentrated in the global boundary conditions (the value of $\langle\varrho\rangle$)
- "Unification" refers to the commonality of local dynamical laws and does not require identity of global topological types
- This is entirely analogous to different plateaux of the quantum Hall effect ($c_1 = 0, 1, 2, ...$) being different topological sectors of the same theory

2.5 Reassessment of the dictionary

The c_1 wall can be interpreted not as "a limitation of the dictionary" but as "a physical difference correctly detected by the dictionary." What the ϱ dictionary has achieved is:

1. **Agreement of local structure**: Sharing of the local normal form $F = \varrho\,\omega_{FS}$
2. **Quantification of differences via multipoles**: Description of differences as selection rules in the spherical harmonic expansion
3. **Comparison of holonomies**: Comparable within the same framework
4. **Deformation theory**: Definition of "distance" and "direction" via the affine structure of $\mathcal{M}_\varrho$

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§3 Horizon Spectral Classification (C11)

As a "return on unification" provided by the unified framework of the ϱ language, we propose a systematic classification scheme for NEH.

3.1 Definition of the horizon spectrum

Definition. The spherical harmonic expansion of the rotational curvature density $\rho_{\mathrm{rot}} := -4\,\mathrm{Im}\,\Psi_2$ (notation system of Part II §2) on the NEH cross-section $S \simeq S^2$

$$\rho_{\mathrm{rot}}(\theta,\phi) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \rho_{\ell m}\, Y_{\ell m}(\theta,\phi)$$

is called the **horizon spectrum** of the NEH. By $c_1[\Omega^{(\mathrm{rot})}] = 0$ (Part I §3), $\rho_{00} = 0$ holds universally.

Rotationally invariant power spectrum. As an invariant independent of the choice of coordinates,

$$C_\ell := \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} |\rho_{\ell m}|^2 \tag{3.1}$$

3.2 Cross-theory comparison

C_ℓ serves as a unified comparison index across theories:

Theory	C_0	C_1	$C_{\ell\geq 2}$	Spectral characteristics
Berry (spin-1/2)	4π ($c_1 = 1$)	0	0	Pure monopole
Dirac monopole ($g = n/2$)	$4\pi n^2$ ($c_1 = n$)	0	0	Pure monopole
Kerr NEH ($c_1 = 0$, $a \neq 0$)	0	> 0	Odd ℓ only	Odd multipole sequence
General NEH ($c_1 = 0$)	0	Arbitrary	Arbitrary	Unconstrained

Remark (cross-Chern-sector comparison). The table above juxtaposes theories belonging to different Chern sectors. Since C_ℓ is a rotationally invariant quantity defined independently on each stratum, comparison across sectors is well-defined as a comparison of the functional form of C_ℓ.

3.3 NEH classification by horizon spectrum

Class	Support of the spectrum	Physical identification
Kerr type	Odd ℓ only ($m = 0$, axisymmetric)	Kerr BH
Perturbed Kerr type	Odd ℓ + even ℓ ($\ell \geq 2$)	Post-merger ringdown BH
Non-Kerr type	Arbitrary spectrum	Hairy BH, modified gravity

Origin of the Kerr selection rule. The fact that ρ_{rot} of the Kerr solution contains only odd ℓ is derived from the double symmetry of (1) axial symmetry ($m = 0$ only) and (2) equatorial reflection symmetry (vanishing of even ℓ).

3.4 The ϱ-no-hair test

ϱ-no-hair test. A statistically significant detection of $\rho_{\ell m} \neq 0$ for even $\ell \geq 2$ constitutes direct evidence of deviation from the Kerr hypothesis.

- **Null hypothesis** H_0 (Kerr): $\rho_{\ell m} = 0$ for all even ℓ
- **Alternative hypothesis** H_1 (non-Kerr): $\exists\, \ell$ (even), m such that $\rho_{\ell m} \neq 0$

3.5 Inverse transform protocol from the QNM spectrum

1. **Forward direction**: $\rho_{\ell m} \to$ boundary conditions for the Teukolsky equation $\to$ QNM frequencies
2. **Inverse direction**: Observed QNM $\to$ extraction of $(M, a) \to$ reconstruction of $\rho_{\ell m}^{\mathrm{Kerr}} \to$ attribution of residuals $\Delta\omega$ to even-ℓ components

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§4 ℓ-Spectrum Heuristic Correspondence Table

Note. The following correspondence table is a **heuristic** organization of the hierarchical structure formed by the ℓ spectrum across the three domains. It is not a correspondence based on rigorous derivation but is positioned as an identification of structural similarities.

ℓ	Null geometry	Gravity	Electromagnetism	Quantum
0	Chern number (c_1)	Mass (M)	Charge (Q)	Berry phase ($\oint A$)
1	Connection (Ω_a)	Rotation ($J = Ma$)	Magnetic dipole	Spin (SU(2) representation)
2	Waves ($\sigma_{ab} = D^{\mathrm{TF}}_{ab}\Psi$)	Gravitational waves (quadrupole radiation)	Electromagnetic waves (quadrupole radiation)	Transitions (interference between states)
≥ 3	Higher multipoles	Higher deformations	Higher radiation modes	—

Structural meaning of the correspondence table:

1. $\ell = 0$ **(conserved quantities)**: The total integral over the sphere defines conserved quantities. The relation $c_1[\Omega^{(\mathrm{rot})}] = 0$ from Part I §3 is an expression of the special nature of $\ell = 0$ in the gravitational sector.
2. $\ell = 1$ **(connection)**: By the universality of the local connection form established in Part I Theorem 2.6, all three are organized as principal bundle connections of the same type.
3. $\ell = 2$ **(waves)**: By the proposition in Part III §12.3, the shear tensor $\sigma_{ab} = D^{\mathrm{TF}}_{ab}\Psi$ contains only modes with $\ell \geq 2$.

The flow "Connection → Curvature → Wave." The hierarchy of the ℓ spectrum reflects the fundamental flow of differential geometry — from connection ($\ell = 1$) through curvature (the topological quantity at $\ell = 0$ and local fluctuations at $\ell \geq 2$) to waves ($\ell = 2$). This flow is shared across the three domains, though the quantitative rigor of each correspondence varies by domain.

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§5 Five Theory-Specific Predictions (C12)

5.1 List of predictions

Prediction	Content	Strength of evidence	Detectability
1. $c_1 = 0$ for any stationary BH	Universal vanishing of rotational flux	Strong (general proof exists, Part I §3)	Medium-term
2. ρ_{rot} of Kerr BH contains only odd multipoles	Absence of even-ℓ components	Strong (Kerr symmetry)	Short- to medium-term
3. Dephasing Bell threshold	Bell violation when $\gamma_1\gamma_2 > 1/\sqrt{2}$	Moderate (standard result of QM)	Already achieved
4. Area quantization	$A_{\mathrm{hor}} = 2\pi\hbar \cdot N$	Conceptual	Extremely difficult

Prediction	Content	Strength of evidence	Detectability
5. ϱ-no-hair test	$\rho_{\ell m} \neq 0$ for even $\ell \geq 2$ = deviation from Kerr	Strong (§3)	Medium-term (LISA + ET)

5.2 Multipole structure of BH shadows (EHT)

The spherical harmonic expansion of the curvature density $\kappa(\zeta) = \sum_{\ell\,\mathrm{odd}} a_\ell\, P_\ell(\zeta)$ of a Kerr BH corresponds to the non-circularity of the shadow. A diagnostic criterion specific to the ϱ language:

$$\boxed{a_{\ell\,\mathrm{even}} \neq 0 \;\; (\ell \geq 2) \quad \Longrightarrow \quad \text{evidence of non-Kerr}}$$

5.3 QNM ringdown

The time evolution of $\rho_{\mathbb{C}}$ after a binary merger, $\rho_{\mathbb{C}}(t, \zeta) \to \kappa_{\mathrm{Kerr}}(\zeta; M_f, a_f)$, is described as a path within $\mathcal{M}_\varrho^{(0)}$, and $c_1 = 0$ is preserved throughout the entire process.

5.4 Gravitational Faraday effect

The photon polarization rotation angle in Kerr spacetime is $\Delta\Phi_{\mathrm{grav}} \sim 2aM/b^2$. In the ϱ language, this is described as a coupling between Berry curvature and gravitational curvature. The quantitative prediction of the cross-term depends on the completion of the unified variational principle.

5.5 Holonomy memory effect

After the passage of gravitational wave radiation, the holonomy along a closed loop γ on the null Kerr screen S^2 is permanently updated:

$$\Delta\left(\oint_\gamma A\right) = \int_0^{u_1} du \int_\Sigma \partial_u \rho_{\mathbb{C}}\, \omega_{FS} \neq 0 \tag{5.1}$$

Due to the complex nature of $\rho_{\mathbb{C}}$, the holonomy update has two components: the real part corresponds to the memory of cross-sectional curvature (mass-type), and the imaginary part corresponds to the memory of rotational vorticity (angular-momentum-type).

Geometric realization. By Stokes' theorem, $\oint_\gamma A = \int_\Sigma \rho_{\mathbb{C}}\, \omega_{FS}$. The mechanism of the update: (1) gravitational wave radiation induces $\partial_u \Psi_2 \neq 0$ through the Bianchi identity, (2) redistribution of $\rho_{\mathbb{C}}$, (3) irreversible imprint on the holonomy. This can also be interpreted as the ϱ-language expression of BMS supertranslations.

5.6 Multipole relaxation rates

[Closure A] From the D1 gradient flow of Part III §6.1:

ℓ	Decay rate $-\ell(\ell+1)$	Physical process
$\ell = 2$	-6	Relaxation of quadrupole deformation
$\ell = 3$	-12	Relaxation of octupole deformation
$\ell = 4$	-20	Relaxation of hexadecapole deformation

ℓ	Decay rate $-\ell(\ell+1)$	Physical process

5.7 Challenges toward quantification of predictions

Among the above predictions, 1–3 are qualitative predictions (structural claims) and follow directly from existing mathematical results. To reach Level 4 (quantitative predictions):

(i) Physical time scale of multipole relaxation. The relationship between the time u in the $\rho_{\mathbb{C}}$ equation and the physical Bondi time depends on calibration against the Kerr background.

(ii) Sensitivity curve for the ϱ-no-hair test. Quantification of error propagation from the measurement precision $\delta\omega_{\ell mn}$ of QNM frequencies to $\delta\rho_{\ell m}$.

(iii) Amplitude of the holonomy memory effect. Depends on the numerical determination of the flux coefficients α, β, γ (Part III §7).

5.8 Verification roadmap

Short-term (through $\sim$ 2030): EHT shadow observations, LVK ringdown, laboratory Berry phase experiments

Medium-term ($\sim$ 2030–2040): LISA, ngEHT / space VLBI, Einstein Telescope / Cosmic Explorer, precision CMB polarization

Long-term (after 2040): Space-based quantum experiments, signatures of area quantization in primordial BH evaporation

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§6 Current Status: Self-Assessment of Maturity

6.1 Maturity levels

Level	Definition	Status
0	Discovery of formal analogies	**Complete**
1	Establishment of a common language (dictionary)	**Complete** (Part I)
2	Formulation of the variational principle	**Complete** (Part II)
3	Characterization and re-derivation of known theories	**Partially complete** (Parts I, III: quantum side complete, gravity side is effective theory)
4	Quantitative predictions	**Proposal stage** (§5: qualitative predictions presented, quantification is future work)
5	Complete dynamical unification	**Not achieved**

Current status: Level 2.5.

6.2 Multi-axis evaluation

Evaluation axis	Score	Basis
Structural unification	**4/5**	Unified representation $F = \varrho\,\omega_{FS}$, ℓ-spectrum correspondence. -1 for lack of dynamical unification
Mathematical consistency	**4/5**	Proof of $c_1 = 0$, universality theorem for connections. -1 for the closure problem
Dynamical derivation	**3/5**	Berry re-derivation is complete. Gravity side remains at the level of characterization
Observational predictions	**2/5**	Formulated but quantitative numerical predictions are still lacking
Weak force	**2/5**	Only the identification $\mathbb{CP}^1 \simeq \mathrm{SU}(2)/\mathrm{U}(1)$. Chirality and Higgs not achieved
Strong force	**1/5**	$\mathbb{CP}^2$ extension identified as a candidate. Formulation not achieved

6.3 Success as a dictionary (Level 0–1)

- Berry, Kerr, and electromagnetic connections are positioned as distinct points in $\mathcal{M}_\varrho$ via $F = \varrho\,\omega_{FS}$
- Stratification of $\mathcal{M}_\varrho$ by Chern number and the spherical harmonic coordinate system have been constructed
- $c_1 = 0$ has been proved as a universal property of general NEH
- The physical identification $\rho_{\mathbb{C}} = -2\Psi_2$ has been numerically confirmed with Schwarzschild normalization

6.4 Progress as a structural unification theory (Level 1–2)

(i) HVP and effective theory. The common stationarity condition $\Delta_{S^2}\varrho = \partial W/\partial\bar{\varrho}$ has the same structure on both the quantum side and the gravity side.

(ii) Ψ_2 boundary theory. The obstruction that ϱ alone does not close is resolved by promotion to Ψ_2. The structural form of the potential W has been established.

(iii) Proof of connection universality. It has been proved that Berry connections and NEH rotational connections are U(1) principal bundle connections of the same type.

(iv) Two Hello Worlds. Both the quantum side (Berry curvature) and the gravity side (cross-focusing equation) are characterized from stationarity conditions of the same type.

(v) Horizon spectral classification. A "return on unification" — a systematic framework that would not naturally arise without the ϱ language.

§7 Structural Limitations and an Honest Assessment

(i) The closure problem has partially progressed. The structural determination of the flux coefficients α, β, γ is complete (Part III §7). $\beta > 0$ has been proved from the area increase theorem. The remaining task is the numerical determination of the scaling factors $C_\alpha, C_\beta, C_\gamma$. The validity range of effective closure A is limited to the near-Kerr regime.

(ii) The nonlocality of V_{cf} persists. Although the obstruction of the P-Q coupling has been resolved by promotion to Ψ_2, the nonlocality of the variational problem remains essential.

(iii) The c_1 wall cannot be crossed within the framework. The most consistent interpretation is the superselection rule of §2.4.

(iv) The stability condition of HVP is an effective-theory assumption. The complete determination of W_{GR} depends on the closure problem.

(v) Circularity of the cross-focusing equation. The characterization in Part III §8.1 uses the Einstein equations as input in the construction of V_{cf}. This is a characterization, not a derivation.

(vi) Structural similarity with celestial holography. The chain bulk GR $\to$ null projection $\to$ S^2 boundary $\to$ ϱ dynamics is close in spirit to celestial holography, but this is not a claim of holographic correspondence.

§8 List of Open Problems

8.1 Highest-priority problems

0. **Complete calculation of Kerr $\rho_{\mathbb{C}}$.** Direct verification that $F = \rho_{\mathrm{rot}}\, \omega_{FS}$ holds exactly on the Kerr horizon from $\Psi_2^{\mathrm{Kerr}} = -M/(r_+ + ia\cos\theta)^3$, over the entire range $a/M \in [0, 1)$.
1. **Resolution of the closure problem and determination of the coefficients of W.** Numerical determination of the scaling factors $C_\alpha, C_\beta, C_\gamma$ (from boundary conditions of the Teukolsky equation).
2. **Elevation of HVP Condition 2 to a principle.** Derivation of W_{GR} variationally from the boundary symplectic structure of Ashtekar–Krishnan. Promotion of the stability condition from an "effective-theory assumption" to a "principle."

8.2 Medium-term problems

3. **Null-geometric derivation of microcausality.** Justification of the tensor product structure as a consequence of null geometry.
4. **Precise connection to observations.** Application of the ϱ-no-hair test to LIGO/Virgo/KAGRA data. Implementation of the QNM inverse transform protocol.
5. **Refinement of the c_1 superselection rule.** Its place in quantum gravity. Physical translation rules for the inter-sector affine map $\varrho \mapsto \varrho + 1$.

8.3 Long-term outlook

6. **Verification of the complete 4D unified action.** Self-consistency of the constrained unified action proposed in Part V.
7. **Derivation of** SU(3) **from screen geometry.** Formulation of the $\mathbb{CP}^1 \to \mathbb{CP}^2$ extension.

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§9 Structural Correspondence with Six Preceding Theories

Preceding theory	Core structure	Correspondence with the present framework	Depth of correspondence
(1) Penrose Twistor	$\mathbb{CP}^1$ fiber, complex-geometric encoding of Weyl curvature	Agreement of the base space $S^2 \simeq \mathbb{CP}^1$, $\rho_{\mathbb{C}} = -2\Psi_2$	Structural interchangeability
(2) Newman–Penrose	Spin coefficients, $\eth$ operator, Weyl scalars	Variable correspondence: $\theta \leftrightarrow \mathrm{Re}(\rho_{\mathrm{NP}})$, etc. $\Delta_{S^2} \leftrightarrow \eth\bar{\eth}$	Full interchangeability
(3) BMS symmetry	Supertranslation, Bondi news	$N_{ab} = D^{\mathrm{TF}}_{ab}\dot{\Psi}$; holonomy memory (§5.5) as the ϱ representation of supertranslations	Structural consistency
(4) Teukolsky equation	Spin-weighted wave equation	Isomorphic structure of mode equations; QNM inverse transform (§3.5)	Structural similarity
(5) Isolated horizon	NEH/IH boundary conditions, Ashtekar data	$\rho_{\mathbb{C}} = K + iB$ as the "encapsulation" of Ashtekar data; IH condition $\partial_u \rho_{\mathbb{C}} = 0$	Full consistency
(6) Berry connection	A_{Berry}, Fubini–Study curvature, Hopf bundle	$\varrho = 1$ special case; HVP quantum sector; connection universality theorem	Re-derived within the theory

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§10 Summary

This part has achieved the following.

1. **Chern number wall = superselection rule (C10)** (§2): The discrepancy in c_1 is understood structurally not as a failure of unification but as different topological sectors within the same framework. Through the analogy with the quantum Hall effect, unification is defined not as identity of topological sectors but as commonality of local dynamical laws.

2. **Horizon spectral classification and ϱ-no-hair test (C11)** (§3): A systematic framework has been constructed for classifying NEH into Kerr type / perturbed Kerr type / non-Kerr type by the spherical harmonic spectrum of ρ_{rot}, and the detection of even $\ell \geq 2$ has been formulated as a hypothesis test for the Kerr hypothesis.

3. **Five theory-specific predictions (C12)** (§5): The universality of $c_1 = 0$, the Kerr odd-parity selection rule, the dephasing Bell threshold, area quantization, and the ϱ-no-hair test have been presented as qualitative predictions.

4. **Self-assessment of maturity** (§6–§7): The current status has been assessed as Level 2.5 (establishment of structural unification + partial characterization), and the structural limitations (closure problem, nonlocality, circularity) have been honestly acknowledged.

5. **Open problems and outlook** (§8): The complete calculation of Kerr $\rho_{\mathbb{C}}$, the closure problem, and the elevation of HVP to a principle have been organized as highest-priority problems.

What Part IV has established. The Chern number wall is compatible with structural unification as a superselection rule, and the ϱ language yields original diagnostic tools (horizon spectral classification, ϱ-no-hair test) and five qualitative predictions. These are "returns on unification" and not mere formal rewriting.

Structural Unification of Gravity, Electromagnetism, and Quantum Mechanics via the Curvature Density ρ

— Variational Framework Based on U(1) Connections on $\mathbb{CP}^1$ and Characterization of Boundary Equations —

Part V: Proposed Extensions (Extensions & Proposed Generalizations) — Level 3–4

This collection of papers consists of five parts:

Part	Title	Level	Content
I	Common Language	0–1	Dictionary, curvature density space, topological theorems, geometric quantization
II	Unified Principle (HVP)	2	Formulation of the Holonomy Variational Principle, physical origin, clarification of variational objects
III	GR Consistency Verification	2–3	Characterization of Einstein boundary constraints, Ψ_2 boundary theory
IV	Observational Predictions and Challenges	3	Horizon spectrum classification, Chern number wall, current status
V	**Proposed Extensions**	**3–4**	**Electromagnetic field, Dirac structure, weak force, 4D unified action**

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§1 Introduction

Parts I–IV established the structural unification of gravity and quantum (Berry) structures using the curvature density ϱ of U(1) connections on the null Kerr screen $S^2 \simeq \mathbb{CP}^1$ as a common language. In this part, we extend this framework to the electromagnetic field, spinor fields, the weak interaction, and the 4D bulk, aiming to resolve Challenge C (electromagnetic and Dirac inclusion) and Challenge D (bulk extension) identified in Main Paper §2.

Declaration on the claim level of this part. The content of this part differs in claim level from the boundary theory of Parts I–III (the established foundation of the structural unification). The electromagnetic and Dirac inclusions in §2–§3 are reorganizations of existing mathematical facts in the ϱ language and are classified as "established" or "consistency verification." The SU(2)

structure in §4 is limited to pointing out a group-theoretical analogy. The non-locality and twistor compatibility in §5–§6 are verifications of structural consistency. The unified action in §7–§10 is positioned as a **proposed extension** and is not necessarily derived from the boundary structural unification of Parts I–III.

Claim/Status list (Part V).

Claim	Status	Section
Inclusion of the electromagnetic field in $\mathcal{M}_\varrho$ (C13)	Established	§2
Consistency with Dirac structure (C14)	Consistency verified	§3
Proposal of a 4D constrained unified action (C15)	Proposed extension (speculative)	§7

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§2 Natural Inclusion of the Electromagnetic Field (C13)

2.1 Positioning of the Electromagnetic Curvature Density

In the geometric formulation of the electromagnetic field, the magnetic flux through S^2 surrounding a magnetic monopole is given by $\Phi_B = \int_{S^2} F = 4\pi g$, and the Chern number of the U(1) bundle is $c_1 = 2eg = n$. The Dirac quantization condition is equivalent to the integrality of the Chern number:

$$\boxed{\text{Dirac quantization condition} \quad eg = \frac{n}{2} \quad \Longleftrightarrow \quad c_1 = n \in \mathbb{Z}} \tag{2.1}$$

This is a direct physical consequence of the stratification of the curvature density space $\mathcal{M}_\varrho$.

Concrete examples of the electromagnetic curvature density ρ_{EM}:

Configuration	c_1	ρ_{EM}	Multipole
No magnetic field	0	$\varrho = 0$	None
Uniform magnetic field (on S^2)	0	$\varrho = 2B_{\mathrm{mag}}R^2\cos\theta$	$\ell = 1$ only
Dipole magnetic field	0	$\varrho \propto P_1, P_3, \ldots$	Odd ℓ
Dirac monopole ($g = n/2$)	n	$\varrho = n$ (constant)	$\ell = 0$ only

2.2 Interpretation of Dirac Quantization in the ϱ Language

The U(1) bundle with $c_1 = 1$ corresponding to the minimum magnetic charge $g = 1/(2e)$ is the Hopf bundle, and $\rho_{\mathrm{Dirac}} = 1 = \rho_{\mathrm{Berry}}$. That is, the Dirac minimum monopole, the Hopf bundle, and the spin-1/2 Berry connection are three physical interpretations corresponding to the same point in $\mathcal{M}_\varrho^{(1)}$.

2.3 Nonlinear Coupling in Kerr–Newman

For the Kerr–Newman black hole, an exact solution of the Einstein–Maxwell system,

$$\kappa_{\mathrm{KN}}(\zeta; M, a, Q) \neq \kappa_{\mathrm{Kerr}}(\zeta; M, a) + \rho_{\mathrm{EM}}(\zeta; Q) \tag{2.2}$$

holds. The curvature densities of gravity and electromagnetism do not separate linearly in general. However, this nonlinear coupling is not inconsistent with the affine structure of $\mathcal{M}^{(0)}_{\varrho}$: the nonlinearity is a dynamical constraint (the question of which point in $\mathcal{M}^{(0)}_{\varrho}$ the Einstein–Maxwell equations select) and is independent of the mathematical structure of the space.

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§3 Consistency with Dirac Structure (C14)

Remark. This section does not claim to "derive the Dirac equation from our theory," but rather confirms (as a consistency check) that the null Kerr screen geometry of our framework **naturally accommodates** the vierbein, spin connection, and Dirac operator.

3.1 Construction from the Vierbein and Spin Connection

Starting point. On a spacetime manifold $(M, g_{\mu\nu})$, we introduce the vierbein (orthonormal frame field) e^{μ}_{a} $(a = 0, 1, 2, 3)$ and the spin connection $\omega_{\mu}{}^{ab}$. The torsion-free condition $de^{a} + \omega^{a}{}_{b} \wedge e^{b} = 0$ (Cartan's first structure equation) uniquely determines the spin connection.

The Dirac operator is geometrically and uniquely constructed as

$$\not{D} = \gamma^{a} e^{\mu}_{a} \nabla^{(\mathrm{spin})}_{\mu} \tag{3.1}$$

where γ^{a} are the generators of the Clifford algebra $\{\gamma^{a}, \gamma^{b}\} = 2\eta^{ab}$, and $\nabla^{(\mathrm{spin})}_{\mu} = \partial_{\mu} + \frac{1}{4}\omega_{\mu}{}^{ab}\gamma_{a}\gamma_{b}$ is the spinorial covariant derivative. The Dirac equation $(\not{D} - m)\psi = 0$ is nothing but a dynamical consequence of this geometric construction.

Lemma 3.1 (Geometric uniqueness of the Dirac operator). On a Riemannian manifold (M, g), given a representation of the Clifford algebra and a torsion-free connection as input, $\not{D}$ is the unique first-order differential operator independent of coordinate choice.

Remark. The inputs required for the appearance of the Dirac equation are only (i) differentiable manifold structure, (ii) metric (vierbein), and (iii) a representation of the Clifford algebra—all of which are contained in the standard structures of Riemannian geometry.

3.2 Necessity of Spin: Double Cover Structure

Spin arises as an inevitable consequence of the double cover of the spacetime rotation group $\mathrm{SO}(1, 3)$:

$$\mathrm{Spin}(1, 3) \simeq \mathrm{SL}(2, \mathbb{C}) \tag{3.2}$$

Since $\pi_1(\mathrm{SO}^{+}(1, 3)) = \mathbb{Z}_2$, the universal covering group is $\mathrm{SL}(2, \mathbb{C})$.

Connection with the ϱ language. In the universality of local connection forms established in Part I Theorem 2.6, both the Berry connection and the gravitational rotation connection were organized as U(1) principal bundle connections on $S^2 \simeq \mathbb{CP}^1$. The spin connection is the lift of this U(1) structure to $\mathrm{SL}(2,\mathbb{C})$, and the minimal representation (fundamental representation) of $\mathrm{SL}(2,\mathbb{C})$ gives the spin-1/2 Weyl spinor.

3.3 Null Decomposition and Spin Weight

Under decomposition by the null tetrad $(\ell^\mu, k^\mu, m^\mu, \bar{m}^\mu)$, spinor fields are described as spin-weighted fields on $\mathbb{CP}^1 \simeq S^2$ (the celestial sphere). A Weyl spinor ξ^A $(A = 0, 1)$ defines a null direction $\ell^\mu \propto \xi^A \bar{\xi}^{A'}$, and the phase freedom of ξ^A gives spin weight $s = \pm 1/2$:

$$\xi^A \in \mathbb{CP}^1, \qquad s = \pm\frac{1}{2} \tag{3.3}$$

The unified variable Ψ of Part III §12 has spin weight $s = 0$, the shear σ has spin weight $s = 2$, and the Weyl scalars $\Psi_0, \ldots, \Psi_4$ have spin weights $s = 2, 1, 0, -1, -2$ respectively. Dirac fields live on $\mathbb{CP}^1$ as fields of spin weight $s = \pm 1/2$, corresponding to the half-integer ℓ strata in the ℓ-spectrum classification of Part IV §4.

Positioning within this framework. The verification of consistency with Dirac structure guarantees that this framework naturally encompasses spin-1/2 fields. As part of the answer to "why does S^2 appear?" discussed in Part I §1, it confirms the dual role of $S^2 \simeq \mathbb{CP}^1$ as both the space of null directions and the projective space of spinors.

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§4 Group-Theoretical Analogy with the Weak Interaction

Remark. This section points out a **group-theoretical analogy** between the symmetry group of the screen geometry and the gauge group of the Standard Model. It is not a derivation or motivation of the weak interaction.

4.1 Symmetry Group of $\mathbb{CP}^1$ and SU(2)

The null Kerr screen cross-section $S^2 \simeq \mathbb{CP}^1$ has SU(2) as its isometry group:

$$\mathbb{CP}^1 \simeq \mathrm{SU}(2)/\mathrm{U}(1) \tag{4.1}$$

This relation is geometrically realized as the Hopf fibration $S^3 \to S^2$ (fiber $S^1 \simeq \mathrm{U}(1)$), and is consistent with the geometric quantization of Part I §4 and the definition of the unified variable Ψ in Part III §12.

4.2 SU(2) Connection and Yang–Mills Curvature

The spin structure on $\mathbb{CP}^1$ naturally carries an SU(2) connection A^i_μ $(i = 1, 2, 3)$, whose curvature is

$$F^i = dA^i + \epsilon^{ijk}\, A^j \wedge A^k \tag{4.2}$$

which has structural similarity to the Yang–Mills SU(2) curvature. Since the weak interaction in the Standard Model is described as an $\mathrm{SU}(2)_L$ gauge theory, the rotational symmetry of the screen geometry exhibits a group-theoretical analogy with the gauge structure of the weak interaction.

4.3 Decomposition to U(1) and Positioning of the Berry Connection

Via the $\mathrm{SU}(2) \to \mathrm{U}(1)$ decomposition (Cartan decomposition), the U(1) connection A introduced in Part I is understood as the Cartan component of the SU(2) connection:

- SU(2): Full symmetry of the screen geometry (group-theoretical analogy with the weak interaction)
- U(1): Its Cartan component (correspondence with the electromagnetic interaction / Berry connection)

Therefore, the screen structure of our theory suggests a group with group-theoretical similarity to the electroweak unification group $\mathrm{SU}(2) \times \mathrm{U}(1)$ from the geometry of $\mathbb{CP}^1$.

4.4 Verification of Twistor Compatibility

In Penrose twistors $Z^\alpha = (\omega^A, \pi_{A'})$, $\pi_{A'}$ provides coordinates on $\mathbb{CP}^1$. Since the screen bundle $S^2 \simeq \mathbb{CP}^1$ of our theory coincides with the fiber of twistor space, consistency is confirmed regarding the origin of the gauge group, in addition to the twistor compatibility conditions of §6.

4.5 Outlook toward SU(3)

The realization of a complete unification ($\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$) requires a geometric origin for SU(3). A natural candidate is the extension from $\mathbb{CP}^1$ to $\mathbb{CP}^2$, where

$$\mathbb{CP}^2 \simeq \mathrm{SU}(3)/(\mathrm{SU}(2) \times \mathrm{U}(1)) \tag{4.3}$$

holds. The formulation of the $\mathbb{CP}^2$ extension and the geometric realization of confinement are among the most important future challenges.

4.6 Limitations of This Section and Geometric Consistency of the Higgs Mechanism

The structure $\mathbb{CP}^1 \simeq \mathrm{SU}(2)/\mathrm{U}(1)$ is a mathematical fact, but the following gaps remain for a complete reproduction of the weak interaction in the Standard Model:

1. The $\mathrm{SU}(2)_L$ of the Standard Model is a **chiral symmetry** acting only on left-handed fermions, but the screen geometry of our framework lacks a chirality selection mechanism.

The resolution of chirality is a necessary condition for this framework to progress from "group-theoretical analogy" to "derivation," and remains an important future challenge.

Geometric consistency of the Higgs mechanism. On the other hand, for the Higgs mechanism, a consistent geometric interpretation can be obtained from the existing structures of our framework. Mass is interpreted as the **geometric inertia** (gyroscopic resistance) of the curvature density ϱ configuration. That is, the variational stability of the ϱ configuration on S^2 (the stationarity condition of Part II Eq.(3.5)) corresponds to "resistance to change" = inertial mass.

Under this interpretation:

- **Explanation of the difficulty of observing the Higgs particle.** Directly exciting the geometric inertia itself (= producing the Higgs particle) requires globally perturbing the stable structure of the ϱ configuration, demanding extreme energy (acceleration). The high production threshold of 125 GeV is structurally understood as the excitation cost of this geometric inertia.
- **Geometric identity of the Kerr horizon and the Higgs field.** In this framework, observation of the Kerr black hole horizon amounts to directly observing the gravity sector ($c_1 = 0$) of the same geometric foundation (the ϱ configuration on S^2) as the Higgs field. This is a consequence of C10 (superselection rule of the Chern number: different topological sectors are described in the same language) and requires no additional assumptions.

Claim level. The above is not a quantitative derivation of the Higgs potential $V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4$, but a confirmation (geometric consistency) that the Higgs mechanism is **structurally consistent** with the inertial structure of the curvature density in our framework. The geometric derivation of the Higgs potential remains a future challenge.

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§5 Non-Locality and Entanglement

5.1 Setting the Question

Bell's theorem shows that local hidden-variable theories cannot reproduce the predictions of quantum mechanics (CHSH quantity $|S| = 2\sqrt{2}$). The questions are whether the internal parameter λ of the null fiber model functions as a hidden variable, and whether the tensor product structure can be derived from null geometry.

5.2 Verification Results

Conclusion C1. For the null fiber model to reproduce $|S| = 2\sqrt{2}$, it must contain the tensor product Hilbert space $H_1 \otimes H_2$ and the Born rule on it.

Conclusion C2. As long as the null internal parameter λ is treated as a local variable, Bell's theorem prevents exceeding $|S| \leq 2$.

Conclusion C3. The source of non-local correlations lies in the geometric structure of $H_1 \otimes H_2$. The fact that $\mathbb{CP}^3$ is "larger" than $\mathbb{CP}^1 \times \mathbb{CP}^1$ (the Segre image) is what permits entanglement.

Conclusion C4. The effective CHSH quantity reduces to $|\bar{S}| = \gamma_1\gamma_2 \cdot 2\sqrt{2}$. If $\gamma_1\gamma_2 > 1/\sqrt{2}$, violation of the Bell inequality is observable.

5.3 Transparent Description of the Tautology Problem

Element	Derived? Assumed?	Basis
Single-particle $\mathbb{CP}^1 \simeq S^2$	Close to derived	U(1) from null little group → Hopf bundle
Tensor product $H_1 \otimes H_2$	**Assumed**	External tensor product of fiber bundles is a standard operation
Born rule (2-particle)	**Assumed**	Tensor product extension of single-particle Born rule

Element	Derived? Assumed?	Basis
Dephasing effect of λ	Derived	Natural extension from single-particle model

This verification is a "confirmation that the null fiber model is **consistent with** quantum mechanics" and is not a **derivation** of non-locality.

5.4 Derivability of the Tensor Product Structure

Approach 1: AQFT (most promising). In the Haag–Kastler axioms of algebraic quantum field theory, combining microcausality

$$[\mathcal{A}(O_1), \mathcal{A}(O_2)] = 0 \quad (O_1, O_2 \text{ spacelike separated})$$

with the Schlieder–Roos theorem yields a tensor product decomposition of algebras for spacelike-separated regions: $\mathcal{A}(O_1) \vee \mathcal{A}(O_2) \cong \mathcal{A}(O_1) \bar{\otimes} \mathcal{A}(O_2)$. Combined with the causal structure of the null fiber model, the tensor product is positioned as a consequence of "causal structure + microcausality."

Approach 2: Twistor theory. The decomposition of the second exterior power of twistor space $\mathbb{T} = S \oplus S'$,

$$\Lambda^2(\mathbb{T}) \cong \mathbb{C} \oplus (S \otimes S') \oplus \mathbb{C} \tag{5.1}$$

has a central component $S \otimes S'$ isomorphic to the coordinate space $\mathbb{M}_{\mathbb{C}}$ of complexified Minkowski spacetime, so that the tensor product structure is inherent in the complexification of null geometry.

Downgrading the wall. Through these analyses, the tensor product wall has been downgraded from a "wall that cannot be overcome in principle" to a **"wall with unproven conjectures."** In the logical chain "null geometry → causal structure → microcausality → tensor product," the null-geometric derivation of microcausality is the remaining core challenge.

5.5 Hierarchy of the Axiom System

Layer 1 (null-geometric axioms): Each null direction carries a U(1) fiber, taking the form $F = \varrho\, \omega_{FS}$.

Layer 2 (causal-structure axioms): Microcausality — fiber algebras for spacelike-separated regions commute.

Layer 3 (composite system axioms): The fiber for multi-particle systems is composed via tensor product.

Layer 1 is derived from null geometry. Layer 2 is motivated but assumed as an additional axiom. Layer 3 is close to being justified through AQFT-type arguments, but a complete derivation remains a future task.

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§6 Twistor Compatibility

We confirm that the geometric structure of this framework is compatible with Penrose twistor theory.

6.1 Compatibility Checks

(T1) Agreement of the base space. The twistor line $L_x \simeq \mathbb{CP}^1$ in twistor space $\mathbb{PT}$ and the null Kerr screen $S^2 \simeq \mathbb{CP}^1$ of our framework agree as base spaces.

(T2) Compatibility of fibers. The U(1) gauge transformation by the phase part $\lambda = e^{i\chi}$ of the spinor $\pi_{A'}$ is a structure of the same type as the U(1) connection $A \equiv \Omega_a$ (Part III Eq.(4.1)) in our framework.

(T3) Correspondence of curvatures. The correspondence between the curvature density $\rho_{\mathbb{C}}$ and the Weyl scalar

$$\rho_{\mathbb{C}} = -2\,\Psi_2\big|_{\mathcal{H}} \tag{6.1}$$

means that the curvature information of twistor space is projected as the curvature of a U(1) connection on $\mathbb{CP}^1$.

Twistor-theoretic positioning of $\mathbb{CP}^1$.

$$\begin{array}{rcl} \text{Twistor space } \mathbb{PT} & \supset & L_x \simeq \mathbb{CP}^1 \\ & & \downarrow \text{ (projection/section)} \\ & & \text{This framework: } S^2 \simeq \mathbb{CP}^1 \text{ with } (A, F = \varrho\,\omega_{FS}) \end{array} \tag{6.2}$$

6.2 Summary of Compatibility Assessment

Check item	Twistor side	This framework	Verdict
Base space	$L_x \simeq \mathbb{CP}^1$ (twistor line)	$S^2 \simeq \mathbb{CP}^1$ (cross-section)	Agreement
Fiber	Complex line bundle ($\pi_{A'} \sim \lambda\pi_{A'}$)	U(1) connection (phase part)	Compatible
Curvature	Weyl curvature (non-flatness of twistor space)	$\rho_{\mathbb{C}} = -2\Psi_2$ (Eq.(6.1))	Agreement
Gauge transformation	$\pi_{A'} \to e^{i\chi}\pi_{A'}$	$A \to A + d\chi$	Isomorphic

6.3 Consequences of Compatibility

(i) Twistor-structural peculiarity of Kerr. Kerr spacetime is Petrov type D, with only Ψ_2 being nonzero. The fact that Kerr is an attractor of the gradient flow D1 in our framework (Part III §6) can be interpreted in twistor-structural terms as the process in which "only Ψ_2 survives and the $\ell \geq 2$ radiative modes decay," though this remains at a qualitative level.

(ii) Availability of the existing twistor toolbox. Twistor compatibility makes the following applicable to our framework:

- **Newman–Penrose formalism**: The $\eth$ operator is naturally defined on $\mathbb{CP}^1$, and Δ_{S^2} corresponds to $\eth\bar{\eth}$ (for spin-weight-0 fields).
- **Penrose transform**: Massless fields are represented as cohomology on twistor space. ρ_{EM} (§2) may potentially be positioned as its projection onto $\mathbb{CP}^1$, but details remain unverified.
- **Electric/magnetic decomposition**: The real part K and imaginary part B of $\rho_{\mathbb{C}} = K + iB$ are related to the electric-type and magnetic-type components of the Weyl tensor, respectively, consistent with the CR-type coupling in the evolution equations (Part III §6).

(iii) Connection with tensor product structure. The $S \otimes S'$ in the twistor approach of §5.4 (Eq.(5.1)) is inherent in the algebraic structure of twistor space, and the tensor product can be interpreted as a natural reflection of the exterior algebra of twistor space. However, a complete derivation has not yet been achieved.

6.4 Novelty of This Framework (Differences from Twistor Theory)

Item	Twistor theory	This framework
Role of $\mathbb{CP}^1$	Directional label (parameter space)	Stage for physical fields (domain of ϱ)
Direction of the theory	Spacetime $\to$ twistor space	$\mathbb{CP}^1 \to$ spacetime (effective boundary theory)
Description of dynamics	Complex-geometric conditions (ASD condition)	PDE evolution on $\mathbb{CP}^1$

The greatest novelty of this framework is that it retains the structural elegance of twistor theory while "encapsulating" it in a single complex scalar field $\rho_{\mathbb{C}} = K + iB$.

6.5 Limitations of Compatibility

Caution is needed regarding compatibility in the following situations:

- **Strong gravitational wave regime**: When the effective closure A (Part III §6) breaks down, the full twistor-side structure $(\Psi_0, \Psi_1, \Psi_3, \Psi_4)$ becomes necessary.
- **Non-vacuum**: The correspondence $\rho_{\mathbb{C}} = -2\Psi_2$ holds in the vacuum, $\Lambda = 0$ case. In the presence of matter fields, Ricci correction terms enter.
- **Global structure**: Since this framework deals with local data on $\mathbb{CP}^1$, the global topology of twistor space is not directly reflected.

Summary. This framework is structurally compatible with twistor theory and is positioned as the "boundary version (celestial version)" of the twistor bundle. The compatibility, as a fusion of three lineages—twistor + Newman–Penrose + Bondi–Sachs—supports the reliability of this framework, but the accurate characterization is at the level of "compatible / suggestive."

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§7 Proposal of a Constrained Unified Action (C15)

Remark. From this section onward (§7–§10), the content constitutes a **proposed extension** and is not necessarily derived from the boundary structural unification of Parts I–III. The claim level is speculative.

7.1 Fundamental Variables

The fundamental variables of the unified theory are the following three:

- **Metric** $g_{\mu\nu}$: Geometry of spacetime
- **U(1) connection** A_μ: Carries the Hájíček 1-form, electromagnetic gauge field, and Berry connection in a unified manner
- **Matter spinor** ψ: Fermion field (Dirac field). Rather than claiming "emergence" from geometry alone, it is explicitly introduced as a constituent of the action

The curvature density ϱ is not an independent scalar field but is fixed as a dependent quantity determined from the curvature of the connection:

$$F\big|_{\text{screen}} = \varrho(u)\,\omega_{\text{screen}}, \qquad \omega_{\text{screen}} \simeq \omega_{FS} \tag{7.1}$$

7.2 Unified Action

The unified action on a 4-dimensional spacetime M is defined as follows:

$$S[g, A, \psi] = \int_M \sqrt{-g}\,\Big[\,\frac{1}{16\pi G}\,R \;-\; \frac{1}{4}\,f(\varrho)\,F_{\mu\nu}F^{\mu\nu} \;+\; \bar{\psi}(i\gamma^\mu D_\mu - m(\varrho))\psi \;-\; V(\varrho)\,\Big]\,d^4x \tag{7.2}$$

Here $f(\varrho)$ is a coupling function depending on the curvature density ($f \to 1$ reduces to the standard Maxwell theory), $D_\mu = \partial_\mu + \frac{1}{4}\omega_\mu{}^{ab}\gamma_{ab} - ieA_\mu$ is the covariant derivative containing the spin connection and U(1) gauge field, $m(\varrho)$ is the effective mass, and $V(\varrho)$ is the potential (consistent with the potential W of Part III §3.5 and the gradient flow D1 of §6.1).

To impose ϱ explicitly as a **geometric constraint**, a Lagrange multiplier λ is introduced, and the constrained unified action is defined as

$$S_{\text{tot}}[g, A, \psi, \varrho, \lambda] = S[g, A, \psi] + S_{\text{constraint}} \tag{7.3}$$

The constraint term is

$$S_{\text{constraint}} = \int_M \lambda\,\Big(F\big|_{\text{screen}} - \varrho\,\omega_{\text{screen}}\Big) \tag{7.4}$$

and $\delta_\lambda S_{\text{tot}} = 0$ recovers $F|_{\text{screen}} = \varrho\,\omega_{\text{screen}}$. Therefore ϱ is a geometric dependent quantity determined in the order "field $\to$ curvature $\to$ density," and is not an arbitrarily introduced scalar field.

When the screen is compact, the topological condition

$$\int_{\text{screen}} \varrho\,\omega_{\text{screen}} = 2\pi\,c_1, \qquad c_1 \in \mathbb{Z} \tag{7.5}$$

is automatically guaranteed. This relation unifies quantum Berry curvature ($c_1 = 1$), magnetic monopoles ($c_1 \in \mathbb{Z}$), and horizon rotation ($c_1 = 0$, Part I §3) as a single topological structure.

Chain from the 4D action to the screen ρ equation. Varying the unified action $S_{\rm tot}$ with respect to δg yields the Einstein equations. Decomposing these in the Newman–Penrose formalism and projecting onto the screen $q_{\mu\nu} = g_{\mu\nu} + k_\mu \ell_\nu + \ell_\mu k_\nu$ of the null basis (k, ℓ) ($k^2 = \ell^2 = 0$, $k \cdot \ell = -1$) yields the evolution equation for Ψ_2: $D\Psi_2 = -\eth \Psi_3 + \cdots$ (Part III Eq.(6.4)). Through $\rho_{\mathbb{C}} = -2\Psi_2$, the evolution of ρ is governed by the angular derivative of Ψ_3, and from the geometry, spin-weight, and rotational symmetry of the screen S^2, closure A ($-\eth \Psi_3 \approx c\, \Delta_{S^2} \rho_{\mathbb{C}}$, Part III §6.2) naturally emerges as the minimal local effective term. That is, the chain 4D unified action $\to$ Einstein $\to$ NP decomposition $\to$ screen projection $\to$ ρ effective evolution equation (Part III Eq.(6.7)) is structurally confirmed. The **form** of closure A is justified as the leading term of the screen EFT, and the determination of the closure coefficient c reduces to the bulk matching problem, which is resolved through the QNM consistency condition (Eq.(6.10)) of Part III §6.3.

7.3 Physical Determination of Action Parameters

Determination of the potential $V(\varrho)$. The restriction to the screen of the potential $W(\Psi_2, \bar{\Psi}_2)$ constructed in Part III §3.5 corresponds to $V(\varrho)$. From four physical requirements—(i) the Kerr solution is a critical point, (ii) CR consistency, (iii) Schwarzschild stability, (iv) cross-focusing consistency—the structural form is determined as

$$W(\Psi_2, \bar{\Psi}_2) = \alpha' \, |\Psi_2|^2 + \beta' \, (\mathrm{Re}\, \Psi_2)^2 + V_{\rm cf}(\Psi_2) \tag{7.6}$$

Here the cross-focusing consistency term is

$$V_{\rm cf}(\Psi_2) := \gamma' \left(\mathrm{Re}\, \Psi_2 + \tfrac{1}{4}\, {}^{(2)}R - \tfrac{1}{2}\, |\Omega[\Psi_2]|^2 \right)^2 \tag{7.7}$$

and the stability condition requires $\alpha' + \beta' > 0$.

Determination of flux coefficients via conservation laws. The three coefficients of the flux source term $S_{\rm flux} = \alpha\, |\sigma|^2 + \beta\, T_{\ell\ell} + \gamma\, T_{\ell\phi} + \nabla \cdot V$ of Part III §7 are fixed by three independent physical conservation laws:

$$\alpha = \frac{C_\alpha}{4\pi}, \qquad \beta = \frac{C_\beta}{8\pi}, \qquad \gamma = \frac{C_\gamma}{4\pi} \tag{7.8}$$

- C_α: $C_\alpha = 1$ (standard normalization) from consistency with the Bondi mass-loss formula
- C_β: $\beta > 0$ is proved from the area increase theorem and the DH flux formula
- C_γ: Determined from consistency with the angular momentum flux

Constraints on the coupling function $f(\varrho)$. $f(\varrho)$ is constrained by the condition that the generalized Maxwell equations (§8.1) reproduce the flux equations on the horizon. In the Schwarzschild limit, $f \to 1$. The complete determination of $f(\varrho)$ is linked to the resolution of the closure problem.

Constraints on the mass function $m(\varrho)$. In the Schwarzschild limit, $m(\varrho) \to m_0$ (constant). The linear response to variation $\delta\varrho$, $\delta m = m'(\varrho_0)\, \delta\varrho$, is in principle observable as curvature-dependent fermion scattering on the horizon.

---

§8 Field Equations (Proposed Extension)

8.1 Generalized Maxwell Equations (Variation with Respect to A)

$$\delta_A S = 0 \quad \Longrightarrow \quad \nabla_\mu\big(f(\varrho)\, F^{\mu\nu}\big) = J^\nu, \qquad J^\nu \equiv e\,\bar{\psi}\gamma^\nu\psi \tag{8.1}$$

When $f(\varrho)$ is non-constant, the curvature density acts as a medium, and the null geometry modifies the propagation of the electromagnetic field. This is directly related to the gravitational Faraday effect of Part IV §5.4.

8.2 Einstein Equations (Variation with Respect to g)

$$\delta_g S = 0 \quad \Longrightarrow \quad G_{\mu\nu} = 8\pi G\, T^{(\text{total})}_{\mu\nu} \tag{8.2}$$

$$T^{(\text{total})}_{\mu\nu} = T^{(\text{EM})}_{\mu\nu} + T^{(\psi)}_{\mu\nu} + T^{(\varrho)}_{\mu\nu} \tag{8.3}$$

$$T^{(\text{EM})}_{\mu\nu} = f(\varrho)\Big(F_{\mu\alpha}F_\nu{}^\alpha - \frac{1}{4}\, g_{\mu\nu}\, F_{\alpha\beta}F^{\alpha\beta}\Big) \tag{8.4}$$

Through $f(\varrho)$, ϱ also contributes to the energy-momentum of the electromagnetic field, so that the "correspondence of connection structures" established in the dictionary manifests as a dynamical coupling.

8.3 Dirac Equation (Variation with Respect to ψ)

$$\delta_\psi S = 0 \quad \Longrightarrow \quad (i\gamma^\mu D_\mu - m(\varrho))\psi = 0 \tag{8.5}$$

$$D_\mu = \partial_\mu + \frac{1}{4}\omega_\mu{}^{ab}\gamma_{ab} - ieA_\mu \tag{8.6}$$

The U(1) connection on the screen enters naturally as the minimal coupling of the Dirac field. The consistency of the Dirac structure verified geometrically in §3 is here obtained systematically as a variation of the unified action. The ϱ dependence of $m(\varrho)$ implies that the fermion mass is influenced by the curvature structure of null geometry.

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§9 Limits to Existing Theories (Limit Checks)

We verify that the unified action (7.2) correctly contains the established existing theories.

Electromagnetic limit. In flat spacetime $g_{\mu\nu} \to \eta_{\mu\nu}$ with $f(\varrho) \to 1$, Eq.(8.1) reduces to the standard Maxwell equations

$$\partial_\mu F^{\mu\nu} = J^\nu \tag{9.1}$$

Gravitational limit. With $A_\mu \to 0$, $\psi \to 0$ (elimination of matter fields), Eq.(8.2) becomes the vacuum Einstein equations

$$G_{\mu\nu} = 0 \tag{9.2}$$

On the horizon, the Schwarzschild/Kerr solutions are reproduced through the physical identification $\rho_{\mathbb{C}} = -2\Psi_2|_{\mathcal{H}}$ of Part III §4.

Quantum (topological) limit. When $F = \varrho\,\omega_{FS}$ is maintained on the screen, the integrality of the Chern number

$$\frac{1}{2\pi}\int_{S^2} F = c_1 \in \mathbb{Z} \tag{9.3}$$

holds, and the identification of Berry phase and holonomy is restored. The universality theorem for $c_1 = 0$ of Part I §3 and the geometric quantization of §4 are naturally positioned in this limit.

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§10 Distinctiveness: Curvature Density Fixing and Multipole Spectrum

10.1 Fixing ϱ as a Curvature Density

In conventional theories, the metric g and gauge field A are independent degrees of freedom. In our theory, the constraint (7.1) fixes

$$\varrho \equiv \frac{F\big|_{\text{screen}}}{\omega_{FS}}$$

as a "ratio of curvatures." The screen structure of null geometry becomes a unified variable penetrating all sectors (gravity, electromagnetism, spinor), and $\rho_{\mathbb{C}} = -2\Psi_2|_{\mathcal{H}}$ becomes a dynamical identification backed by an action principle.

10.2 Derivation of the Multipole ℓ Spectrum

We position the exploratory correspondence table of Part IV §4 from the linearization of the constrained unified action (7.3).

Background solution. In the spherically symmetric background (Schwarzschild limit), $F_0 = \varrho_0\,\omega_{FS}$, $\varrho_0 = 1/(4M^2)$.

Perturbation expansion. Expanding $A = A_0 + \delta A$, $\delta A = \sum_{\ell,m} a_{\ell m}(u)\,Y_{\ell m}$, the constraint condition determines $\delta\varrho = \delta F|_{\text{screen}}/\omega_{FS}$ as a dependent quantity.

Quadratic action. The quadratic action on the screen is

$$S^{(2)}[\delta A] = \sum_{\ell,m}\int du\,\Big[\,\frac{1}{2}\,|\dot{a}_{\ell m}|^2 - \frac{1}{2}\,\lambda_\ell\,|a_{\ell m}|^2\,\Big] \tag{10.1}$$

The eigenvalues are $\lambda_\ell = \ell(\ell+1)\,\varrho_0 - \mu_\ell$, where μ_ℓ is a mass term arising from the second derivative of the potential $V(\varrho)$.

Stationarity conditions and physical identification. From the equation of motion for each mode, $\ddot{a}_{\ell m} + \lambda_\ell \, a_{\ell m} = 0$:

ℓ	Eigenvalue λ_ℓ	Physical identification	Dynamical role
0	$-\mu_0$ (curvature of the potential)	Mass, charge (conserved quantities)	Zero mode (Goldstone)
1	$2\varrho_0 - \mu_1$	Angular momentum, magnetic dipole	Rotational degrees of freedom
≥ 2	$\ell(\ell+1)\varrho_0 - \mu_\ell > 0$	Gravitational waves, electromagnetic waves	Radiative modes (exponential decay)

- $\ell = 0$: The spherical average $\langle \varrho \rangle = \varrho_0$ is topologically fixed by the Gauss–Bonnet constraint $\int \varrho \, \omega_{FS} = 2\pi \, c_1$. As a zero mode, it corresponds to mass and charge.
- $\ell = 1$: From the variational principle of Part III §6, $\varrho \propto \cos\theta$ is the Kerr attractor. Neutrally stable due to angular momentum conservation.
- $\ell \geq 2$: With $\lambda_\ell > 0$, oscillatory solutions exist, consistent with the exponential decay $a_{\ell m}(u) \propto e^{-\ell(\ell+1)\varrho_0 u}$ in the gradient flow D1 of Part III §6. $\ell = 2$ is the lowest-order mode of quadrupole radiation.

Thus, the multipole correspondence table of Part IV §4 is positioned not as a "table" but as the "spectrum of equation (10.1)."

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§11 Outlook

11.1 Relationship between Hopf Structure and Spin Structure

The fundamental structure of this framework—U(1) connections on the null Kerr screen S^2—suggests the geometric origin of spin structure through the Hopf bundle.

Attaching a U(1) phase (fiber of the connection A) to each point of S^2, the total space forms the Hopf bundle $\mathrm{U}(1) \hookrightarrow S^3 \twoheadrightarrow S^2$. Since $S^3 \simeq \mathrm{SU}(2)$, it naturally contains the spin group:

$$S^2 \xrightarrow{\text{U(1) connection}} S^3 \simeq \mathrm{SU}(2) \xrightarrow{\text{double cover}} \mathrm{SO}(3) \xrightarrow{\text{Lorentz extension}} \mathrm{Spin}(1,3) \simeq \mathrm{SL}(2,\mathbb{C}) \quad (11.1)$$

Hopf structure recovery condition. In the limit of shear $\sigma = 0$ and expansion $\theta = 0$ (standard NEH conditions), the U(1) bundle on S^2 recovers the structure of the Hopf bundle:

$$\sigma = 0, \ \theta = 0 \quad \Longrightarrow \quad \text{Hopf structure is recovered}$$

General configurations with $\sigma \neq 0$ or $\theta \neq 0$ can be understood as deformations from the Hopf structure.

11.2 Classification of obs/null Degrees of Freedom

This framework suggests a scheme for classifying physical degrees of freedom into "observable degrees of freedom" (obs-DOF) and "null degrees of freedom" (null-DOF).

Observer operator. Readout from the field strength tensor F^{bc}:

$$\mathcal{O}_e[F] := e^a q_{ab} F^{bc} u_c$$

Type of degree of freedom	Number of components	Physical content
obs-DOF	1	Electric field component along the polarizer direction
null-DOF	2	Orthogonal polarization component + U(1) phase (holonomy)

Peculiarity of SU(3). In this classification, SU(3) (strong interaction) alone does not have the $\ell = 2$ free-wave mode in an observable form. The free propagation mode corresponding to $\ell = 2$ of gluons is suppressed by confinement. This suggests a possible geometric manifestation of confinement, but quantitative verification is future work.

11.3 Geometric Reinterpretation of the Newman–Janis Trick

The complex curvature density $\rho_{\mathbb{C}} = K + iB$ of Part III §4 and the tilt model of the null fiber suggest a geometric reinterpretation of the Newman–Janis transformation (the complex coordinate transformation trick from Schwarzschild to Kerr).

Introducing the tilt angle θ of the null fiber and modeling $\rho_{\text{obs}} = \rho_{\text{int}} \cos\theta$, the $r \to r - ia\cos\theta$ appearing in the Kerr metric emerges naturally. This is also consistent with the multipole structure of Kerr, $M_\ell + iS_\ell = M(ia)^\ell$, and the rotation parameter a governs the tilt angle of the null fiber.

This reinterpretation remains at a qualitative level, and rigorous justification in the NP formalism is future work.

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§12 Summary

This part achieved the following.

1. **Natural inclusion of the electromagnetic field (C13)** (§2): We showed that the Dirac quantization condition $eg = n/2$ is equivalent to the stratification $c_1 = n$ of $\mathcal{M}_\varrho$, and the Hopf bundle, Dirac minimum monopole, and spin-1/2 Berry connection were identified as three physical interpretations of the same geometric entity in $\mathcal{M}_\varrho^{(1)}$.
2. **Consistency with Dirac structure (C14)** (§3): We confirmed that the vierbein, spin connection, and Dirac operator are naturally consistent with the null Kerr screen geometry of our framework. This is a consistency verification and not a derivation of the Dirac equation.

3. **Group-theoretical analogy with the weak interaction** (§4): We pointed out the structural analogy with electroweak symmetry from $\mathbb{CP}^1 \simeq \mathrm{SU}(2)/\mathrm{U}(1)$. The unresolved issues of chirality and the Higgs mechanism were explicitly stated, and the limitation to a group-theoretical analogy was made clear.

4. **Consistency verification of non-locality** (§5): We confirmed that the null fiber model is consistent with the non-locality of quantum mechanics. The tensor product wall was downgraded to a "wall with unproven conjectures."

5. **Twistor compatibility** (§6): Compatibility was confirmed across four check items (base space, fiber, curvature, gauge transformation), and the novelty of this framework (physicalization of $\mathbb{CP}^1$, reversal of the theory's direction, PDE-type dynamics) was clarified.

6. **Proposal of a constrained unified action (C15)** (§7–§10): Positioning ϱ on the null Kerr screen as a dependent quantity determined from the curvature of the connection, and imposing the geometric constraint via a Lagrange multiplier, a 4D unified action was presented as a **proposed extension**. Recovery of the limits to existing theories was confirmed, and the positioning of the multipole ℓ spectrum from the action principle was demonstrated.

What Part V established. The electromagnetic field and Dirac structure are naturally included in the ϱ language (resolution of Challenge C), and a 4D unified action has been formulated as a proposed extension (resolution of Challenge D). With this, solutions to all four challenges (A–D) identified in Main Paper §2 have been presented, and the argument for structural unification is closed.

Appendix A: Unified Symbol Table

This is a unified convention used throughout the entire collection of papers.

A.1 Base Geometry

Symbol	Definition	Remarks
ω_{FS}	Fubini–Study 2-form; $\omega_{FS} = \frac{1}{2}\sin\theta\, d\theta \wedge d\phi = \frac{i}{(1+\lvert z\rvert^2)^2} dz \wedge d\bar{z}$	$\int_{S^2} \omega_{FS} = 2\pi$
ϵ_S	Area 2-form of cross-section $S \simeq S^2$; $\epsilon_S = \sin\theta\, d\theta \wedge d\phi$	$\int_{S^2} \epsilon_S = 4\pi$; $\epsilon_S = 2\,\omega_{FS}$
$d\mu_{FS}$	Normalized area measure; $d\mu_{FS} = \frac{1}{2\pi}\omega_{FS} = \frac{1}{4\pi}\epsilon_S$	$\int_{S^2} d\mu_{FS} = 1$

A.2 Curvature Density and Chern Number

Symbol	Definition	Remarks
ϱ	Abstract curvature density variable; $F = \varrho\,\omega_{FS}$	Sector-independent symbol (introduced in Part II §2)
ρ_{rot}	Real rotational curvature density; $\rho_{\mathrm{rot}} := -4\,\mathrm{Im}\,\Psi_2$	Real variable of the gravity sector

Symbol	Definition	Remarks
$\rho_{\mathbb{C}}$	Complex curvature density; $\rho_{\mathbb{C}} = K + iB = -2\Psi_2$	Principal variable from Part III onward
c_1	First Chern number; $c_1 = \frac{1}{2\pi}\int_{S^2} F = \langle \varrho \rangle$	$c_1 \in \mathbb{Z}$
$\mathcal{M}_{\varrho}$	Curvature density space	$= \bigsqcup_n \mathcal{M}_{\varrho}^{(n)}$
$\mathcal{M}_{\varrho}^{(n)}$	Chern stratum ($c_1 = n$)	Affine space
$\rho_{\ell m}$	Spherical harmonic expansion coefficients of ϱ	$d\sigma = \sin\theta\, d\theta\, d\phi$

A.3 Connections and Curvature

Symbol	Definition	Remarks
ω_a	Hájíček 1-form; $\omega_a := -\ell_\mu \nabla_a k^\mu$	Global 1-form under fixed (k, ℓ)
$\Omega_a^{(\mathrm{rot})}$	Rotation 1-form; $\Omega_a^{(\mathrm{rot})} := \omega_a$	Compared in U(1) form via Wick-type identification
A_{Berry}	Berry connection; $A_{\mathrm{Berry}} = \frac{1-\cos\theta}{2}\, d\phi$	On the state space $\mathbb{CP}^1$
A	Electromagnetic U(1) gauge connection	On spacetime
$\kappa(\zeta)$	Curvature density of the Kerr horizon; $\kappa(\zeta) = 2\omega'(\zeta)$	Odd function; $\langle \kappa \rangle = 0$

A.4 Weyl Scalars and NP Formalism

Symbol	Definition	Remarks
Ψ_2	Newman–Penrose Weyl scalar	Convention: ABL [2001]
$P = \mathrm{Re}\,\Psi_2$	Real part	Determined by the cross-focusing equation
$Q = \mathrm{Im}\,\Psi_2$	Imaginary part	$\rho_{\mathrm{rot}} = -4Q$
ρ_{NP}	NP spin coefficient (convergence)	Distinct from curvature density ϱ

A.5 Quantum-Side Symbols

Symbol	Definition	Remarks
$\hat{\rho}$	Quantum density matrix	Avoids collision with curvature density ϱ
γ	Decoherence visibility; $\gamma = \lvert\Gamma\rvert$	$0 \le \gamma \le 1$

A.6 Spherical Harmonic Expansion

Symbol	Definition	Remarks
$Y_{\ell m}$	Spherical harmonics	Orthonormal system with respect to $d\sigma$
C_ℓ	Curvature power spectrum	$C_\ell = \frac{1}{2\ell+1}\sum_m \lvert\rho_{\ell m}\rvert^2$

A.7 Observation-Related Quantities

Symbol	Definition	Remarks
$\omega_{\ell mn}$	QNM frequency	$\omega_R + i\omega_I$
$\Delta\Phi_{\text{grav}}$	Gravitational polarization rotation angle	$\sim 2aM/b^2$

A.8 Physical Identifications

Symbol	Definition	Remarks
K	$K \coloneqq -2\,\mathrm{Re}\,\Psi_2$ (on the horizon)	In the non-rotating limit, $K = \frac{1}{2}\,{}^{(2)}R$
B	Rotational vorticity $B = *d\Omega$	Vorticity of the rotation 1-form

A.9 Effective Dynamics

Symbol	Definition	Remarks
$\partial_u \rho_{\mathbb{C}}$	Null time evolution	u = Bondi retarded time
Δ_{S^2}	Laplace–Beltrami operator on S^2	
S_{flux}	Flux source term	$\alpha\lVert\sigma\rVert^2 + \beta T_{\ell\ell} + \gamma T_{\ell\varphi} + \nabla\cdot V$
D1	Gradient flow dynamics	$\partial_u \varrho = \Delta_{S^2}\varrho + \lambda_1 \cos\theta$

A.10 Unified Principle

Symbol	Definition	Remarks
HVP	Holonomy Variational Principle	Part II §3
$\mathcal{Z}[A]$	Holonomy functional; $\exp(i\oint_C A)$	Part II Eq.(3.1)
$W(\Psi_2, \bar{\Psi}_2)$	Potential of the Ψ_2 boundary theory	Part III §3.5
V_{cf}	Cross-focusing consistency term	Part III §3.5

Appendix B: Research Notes: Hypothesis Map (Supplementary Research Notes)

Remark. This appendix is a hypothesis map organized as exploratory considerations for future research directions, rather than rigorous theorems or derivations. Hypothetical correspondences are marked with *****.

B.1 Space × Harmonic Level Map

ℓ	Null fiber cross-section ($S^2 \simeq \mathbb{CP}^1$)	Gravitational space	Electromagnetic space U(1)	Weak space SU(2)	Strong space SU(3)	Quantum state space
0	Spherically symmetric mode	Schwarzschild mass	Charge (Gauss)	Higgs radius*	Color neutral*	Ground state
1	Tilt mode	Kerr rotation	Dipole field	Weak isospin*	Color current*	Berry connection
2	Quadrupole mode	Gravitational waves	Maxwell waves	W boson waves*	Gluon excitation*	Quantum transition
≥ 3	Higher-order distortion	Higher gravitational waves	Higher electromagnetic multipoles	Weak multipoles*	Color multipoles*	Higher transitions

* = hypothesis

B.2 Spatial Layer Version

Spatial layer	Shape	Role
Null geometry	Light cone	Causal structure
Cross-section space	S^2	Harmonic modes
Phase space	U(1)	Electromagnetic connection
Weak space	SU(2)	Internal rotation
Color space	SU(3)	Color degrees of freedom
State space	Hilbert	Quantum transitions

Appendix C: Torus Representation and Outlook for Internal Geometry (Supplementary Research Notes)

Remark. This appendix is at the level of "interpretation" and "visualization" rather than "derivation," and is positioned as working notes toward future formulation.

C.1 Torus Representation as Projection of Internal Geometry

Among the degrees of freedom in the internal space, the component perpendicular to the gravitational direction is defined as the effective radius ϱ. Due to the presence of the gravitational field,

this structure acquires a tilt, and the observable component of the internal degrees of freedom is projected onto external spacetime as

$$\rho_{\mathrm{obs}} = \varrho \cos\theta \tag{C.1}$$

This bears structural similarity to the appearance of $a = J/M$ as the radius of the ring singularity in the Kerr solution.

Generation differences (electron, muon, tau, etc.) can be interpreted as the same structure of internal geometry observed under different projection conditions. This interpretation is also consistent with the equivalence principle.

Remark (scope of the claim). Quantitative derivation (particle spectrum, mass ratios) requires specifying the geometry of the internal space and dynamically determining the projection conditions, which remain future challenges.

C.2 Deformation from Hopf Structure

In the zero-gravity limit, the internal structure is described as the Hopf fibration $S^3 \to S^2$, which coincides with spinor geometry and the $\mathbb{CP}^1$ structure. The presence of a gravitational field deforms the Hopf structure, which manifests as a geometry of the Clifford torus type $S^1 \times S^1 \subset S^3$ (which equally divides S^3 into two solid tori).

$$\varrho = -2\,\Psi_2\big|_{\mathcal{H}} = K + iB$$

The Weyl correspondence above is the fundamental definition, and in the Schwarzschild solution it can be interpreted as $\varrho = \frac{1}{2}\,{}^{(2)}R$ (half the Gauss curvature).

C.3 Laplacian Eigenvalues and Particle Classification Hypothesis (Research Idea)

Remark. The following is an unverified research hypothesis and constitutes a conceptual memo toward future formulation.

Three-layer classification structure. There is a possibility of classifying particles by the triple (c_1, p, λ) (Chern number, number of poles, Laplacian eigenvalue).

Layer	Quantity	Role
Topological	Chern number c_1	Sector selection (superselection rule)
Geometric	Number of poles p	Multipole structure
Vibrational	Laplacian ID λ	Eigenmode

Torus eigenvalues and sphere eigenvalues. There may be hints for the particle spectrum in the correspondence between the Laplacian eigenvalues $\lambda = n^2 + m^2$ of the torus T^2 and $\lambda = \ell(\ell+1)$ of the sphere S^2. If the internal space is torus-like:

$$\text{Internal ID } (n,m) \xrightarrow{\text{gravitational deformation, screen projection}} \text{Observed ID } (\ell,m)$$

Particles = eigenvibrations of the screen. A hypothesis that identifies particles with spherical harmonic modes of ρ. The eigenmodes of the Laplace–Beltrami eigenvalue problem $\Delta_{S^2}\psi = -\ell(\ell+1)\psi$ on the screen S^2 correspond to physical degrees of freedom. If the mass satisfies a relation such as $m \sim \ell(\ell+1)$, particles are understood as eigenvibrations of space. This is in the same family of ideas as Kaluza–Klein, twistor, and holography.

Connection with SU(3) (unverified). In the $\mathbb{CP}^1 \to \mathbb{CP}^2$ extension (§5.3 item 7), there is a possibility that the color degrees of freedom of quarks (SU(3)) fit within the (c_1, p, λ) classification. This is a subject for future investigation as the nullColor hypothesis.

Chain from the 4D unified action to screen projection. The pathway confirmed in Step 2 (Q17 discussion):

$$S[g, A, \psi, k, \ell, \lambda] \xrightarrow{\delta g} \text{Einstein} \xrightarrow{\text{NP}} D\Psi_2 = -\eth\Psi_3 + \cdots \xrightarrow{\rho=-2\Psi_2} \partial_u \rho \sim c\,\Delta_{S^2}\rho$$

The 4D action variation → NP equations → screen projection reproduces the effective evolution equation (Eq.(6.7)) of Part III. The form of closure A is natural as the leading term of the screen EFT (see Part III §6.2).

Figure 1 candidate. A four-stage schematic diagram: Hopf geometry → null fiber → gravitational deformation → Kerr exposure. In the zero-gravity limit, it coincides with existing geometry (Hopf $S^3 \to S^2$), and for $g \neq 0$, the null tensor surface tilts.

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